# PHYS-E0510 – Advanced Computational Methods in Physics D

Lecture Notes

Sessions 1 to 5

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Each session's title box below names its companion Jupyter notebook and prerequisites.

## How this course is organized

Every session of this course comes in five pieces, and they are meant to be used together rather than one instead of another. The recording is the lecture itself, with the derivations done on the fly and the questions that come up along the way. The slides are the sequence of definitions, figures and equations in the order they were presented, which makes them the quickest way to find your place in a recording or to go over a session before the exam. These lecture notes follow the same sessions in the same order, but write the physics out in connected prose, with the steps a slide compresses into one line carried through by hand, so you can read them instead of watching a session a second time, or next to the slides to fill in what a bullet point skips. The Jupyter notebook of each session, named in the box under the session title and pointed at again in every *Computational companion* box inside the text, is where the physics is actually computed: nothing there is a fixed figure, and you are meant to run the cells, change the parameters and see what happens. Finally, the exercise shorts (https://joselado.github.io/Advanced_Computational_Methods_Physics_2026/exercise_shorts/) are four short silent films per session, one to two minutes each, where we see one idea of the session with every number on screen computed with the same codes as the notebooks.

None of these pieces is complete on its own: the recordings and slides give the narrative and the intuition, the notes give derivations you can check line by line, and the notebooks give the numerics, which build an intuition that none of the others can. A good way to work through a session is to watch the recording once, read the matching notes with the notebook open next to them, and then go back to the notebook and change things until the results stop feeling like something that happened on someone else's screen.

### The computational libraries

All the numerics of the course are done with two Python libraries, rather than with code written from scratch inside the notebooks, and both have to be installed before the notebooks run; the first cell of each notebook says which environment variables they expect.

- `pyqula` (https://github.com/joselado/pyqula) does single-particle and mean-field electronic structure: tight-binding Hamiltonians, band structures, and the self-consistent Hubbard and charge-density-wave calculations of Session 1, whose notebook, `classical_magnetism.ipynb`, is built on it.
- `dmrgpy` (https://github.com/joselado/dmrgpy) does the quantum many-body calculations: exact diagonalization of spin and fermionic Hamiltonians, and the matrix-product-state and DMRG calculations of the later sessions, so the notebooks of Sessions 2 to 5 are built on it.

Both libraries are used here as tools: you do not need to read their source to follow the course, only to call the functions the notebooks show you.

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# PHYS-E0510 – Advanced Computational Methods in Physics D

Session 1: Classical Magnetism and Mean-Field Theory

**Companion notebook:** `classical_magnetism.ipynb` **Prerequisites:** none (second quantization is introduced from scratch)

## Contents and goals

In this session we build the minimal machinery we need to talk about interacting electrons, second quantization, and we use it to answer a very concrete question: when do repulsive electron–electron interactions spontaneously break a symmetry of the Hamiltonian? We will look at two symmetry-broken phases, classical magnetism and charge density waves, and we will treat both of them entirely at the mean-field level. By the end of the session we will see a case where mean-field theory demonstrably gives us the wrong ground state, the Hubbard dimer, and that failure is precisely what motivates the exact-diagonalization and tensor-network methods we develop in the rest of the course.

## Learning outcomes

By the end of this session, you should be able to:

- Write single-particle and many-body Hamiltonians in second-quantized form, and explain why the interaction term makes the many-body problem exponentially hard.
- Set up and run a mean-field self-consistency loop for the Hubbard and CDW models, and interpret the resulting exchange splitting, gap, and order parameter.
- Derive the superexchange coupling $J \sim t^2/U$ from degenerate perturbation theory and use it to predict collinear versus frustrated classical spin order.
- Identify when mean-field theory fails qualitatively, and explain why, using the entangled singlet ground state of the Hubbard dimer as the counterexample.

## 1 Single-particle versus many-body Hamiltonians

Let us start by distinguishing two broad classes of electronic Hamiltonians. The first class is what we call single-particle Hamiltonians: they are bilinear in the field operators,

$$\mathcal{H}_0 = \sum_{ij} t_{ij}\, c_i^\dagger c_j, \tag{1}$$

where $t_{ij}$ is a hopping amplitude that we can think of as the probability amplitude for an electron in orbital $j$ to jump to orbital $i$. Because $\mathcal{H}_0$ is just a matrix acting on a single-particle Hilbert space, every electron here moves essentially independently of the others, and finding the spectrum is nothing more than diagonalizing an $L \times L$ matrix for $L$ orbitals. This is already enough to describe a wide variety of materials: insulators, semiconductors, ordinary metals.

The case we actually care about most, though, is the many-body case. Here we again have the

bilinear term, but we also add a term that is quartic in the fermionic operators,

$$\mathcal{H} = \mathcal{H}_0 + \sum_{ijkl} V_{ijkl}\, c_i^\dagger c_j^\dagger c_k c_l\,, \tag{2}$$

and $V_{ijkl}$ tells us the strength of this interaction. This is the term that is responsible for essentially all of the exotic phenomena we care about in quantum materials: magnetism, charge density waves, superconductivity, and every strongly correlated phase we will meet later in the course. What makes second quantization so useful is that it lets us write both the single-particle and the many-body piece of Eq. (2) in one common language.

The price of the quartic term is the size of the space it acts on. One electron on $L$ orbitals has $L$ states, so $\mathcal{H}_0$ is an $L \times L$ matrix, while any number of electrons on the same $L$ orbitals has $2^L$ occupation configurations, since each orbital is either empty or occupied, and the interacting $\mathcal{H}$ is a $2^L \times 2^L$ matrix. Figure 1 counts both, first for a chain of four sites and then as $L$ grows.

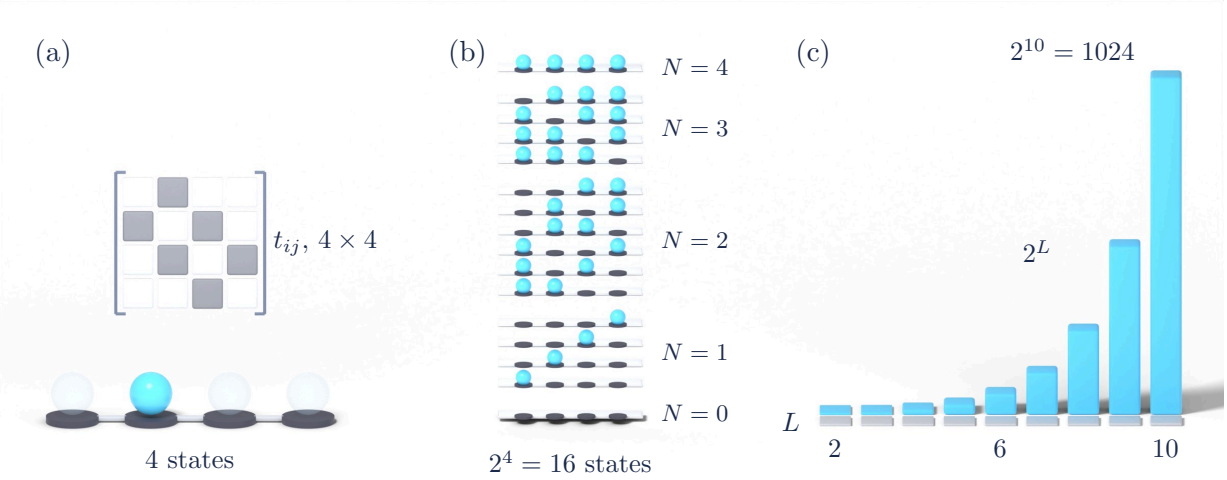



**Figure 1:** The dimension of the single-particle and of the many-body problem. (a) One electron on a chain of four sites can sit on any of the four, drawn here as one electron and three ghosts, so the hopping $t_{ij}$ is a $4 \times 4$ matrix with entries only between neighbors. (b) The same four sites holding any number of electrons, from none to four: each slab is one occupation configuration, a filled ball an occupied site, and the $2^4 = 16$ slabs come in groups of 1, 4, 6, 4 and 1 by electron number $N$. (c) The two dimensions against $L$ on one linear scale, $L$ in front and $2^L$ behind, meaning that the single-particle problem stays flat while the many-body one reaches 1024 already at $L = 10$.

## 2 Second quantization

So let us set up that language. We define the fermionic annihilation and creation operators $c_i$, $c_i^\dagger$ for orbital $i$, and a vacuum state $|\Omega\rangle$ that has no electrons at all, $c_i\,|\Omega\rangle = 0$ for every $i$. You can think of the vacuum as a vector and of the operators as matrices that map one vector to another; a single-particle basis state is simply $c_i^\dagger\,|\Omega\rangle$, and the matrix elements of $\mathcal{H}_0$ in this basis are exactly the hopping amplitudes $t_{ij}$ we started with.

How do we enforce, mathematically, that no site can hold more than one electron of a given

species? We impose the canonical anticommutation relations,

$$\{c_i, c_j^\dagger\} = \delta_{ij}, \qquad \{c_i, c_j\} = \{c_i^\dagger, c_j^\dagger\} = 0. \tag{3}$$

This single algebraic statement does two things at once: swapping two fermions picks up a minus sign, which is exactly antisymmetry of the wavefunction, and acting with $c_i^\dagger$ twice on any state gives zero, which is exactly the Pauli exclusion principle. Throughout the course we will meet three flavors of local degree of freedom built this way:

- **Spinful fermions**: each site has four possible states, $|0\rangle$, $|\uparrow\rangle$, $|\downarrow\rangle$, $|\uparrow\downarrow\rangle$. This is the realistic case, the one you would use for something like a hydrogen atom.
- **Spinless fermions**: each site has only two states, $|0\rangle$ and $|1\rangle$. Formally, a spinful fermion is really nothing more than two decoupled spinless fermions, one per spin flavor.
- **Spin-$\frac{1}{2}$ operators**: again two states, $|\uparrow\rangle$ and $|\downarrow\rangle$, but now the operators are *bosonic*: they commute on different sites instead of anticommuting. As we will see in Section 6, a local spin emerges naturally from a spinful fermion once we project out double occupancy and the empty state. This is the origin of what is called spin-charge separation.

**✘ Computational companion**

Open `classical_magnetism.ipynb`, section "Bandstructure of a tight binding chain" and "Finite size chains." Diagonalize $\mathcal{H}_0$ for a chain of $L$ sites and check for yourself that the number of eigenvalues grows linearly in $L$, while their spacing shrinks: this is the discrete precursor of a continuous band. Then, in "Density of states of a finite system as a function of the system size," plot the density of states, $\rho(\omega) = -\frac{1}{\pi}\,\mathrm{Im}\,\mathrm{Tr}\,[\omega - \mathcal{H}_0 + i0^+]^{-1}$, and think about how it connects to what a tunneling-spectroscopy or ARPES experiment actually measures.

## 3 Worked example: the tight-binding dimer

Before we add any interactions, let us solve the smallest possible chain by hand, since we will keep coming back to it. Take $L = 2$ sites with hopping $t$, and let us include spin from the start, so each site can hold a spin-up and a spin-down electron independently:

$$\mathcal{H}_0 = -t \sum_\sigma \left( c_{0\sigma}^\dagger c_{1\sigma} + c_{1\sigma}^\dagger c_{0\sigma} \right). \tag{4}$$

Spin never enters the hopping, so $\mathcal{H}_0$ splits into two identical $2 \times 2$ spinless problems, one per spin flavor:

$$\mathcal{H}_0^{(\sigma)} = -t \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \implies \varepsilon_\pm = \mp t, \qquad |\pm\rangle = \frac{1}{\sqrt{2}} \left( |0\rangle \pm |1\rangle \right). \tag{5}$$

We call $|+\rangle$ the **bonding** orbital, at energy $-t$, and $|-\rangle$ the **antibonding** orbital, at energy $+t$. Now restore spin: each of these two orbital levels can separately hold a spin-up and a spin-down electron, so we actually have $2 \times 2 = 4$ single-particle eigenstates in total, even though there are only *two* distinct energies, $\pm t$, each doubly degenerate. This is the direct answer to a question you may have already asked yourself while looking at the notebook: why four eigenstates for two sites, and why are some of them degenerate? The degeneracy is counting spin, not orbital content, and it follows from the fact that $\mathcal{H}_0$ commutes with total spin: nothing in $\mathcal{H}_0$ distinguishes $\uparrow$ from $\downarrow$.

If we fill in electrons one at a time starting from $|\Omega\rangle$, we generate the full many-body basis for two spinful sites, exactly as we set up in the previous section. At half filling with two electrons, for instance, there are $\binom{4}{2} = 6$ states in total: four of them are singly occupied ($\uparrow\downarrow$, $\downarrow\uparrow$, $\uparrow\uparrow$, $\downarrow\downarrow$, one electron per site) and two of them are doubly occupied (both electrons on the same site). Keep this counting in mind, since it is exactly what we will use in Section 6 to build the strong-coupling effective spin model.

**✘ Computational companion**

Section "The spectra of a dimer." Diagonalize the two-site, spinful tight-binding Hamiltonian and check that you recover the bonding/antibonding energies $\pm t$ and their spin degeneracy directly from the numerical spectrum.

## 4 Two microscopic interactions, two broken symmetries

Equation (2) is completely general, so in practice we have to pick a concrete form for $V_{ijkl}$ and ask what symmetry-broken state it favors once we go to the thermodynamic limit, $L \to \infty$. We will focus on two of the simplest and most physically motivated choices.

**The Hubbard model (spinful fermions).** Suppose we only keep the interaction between opposite spins sitting on the *same* site:

$$\mathcal{H}_{\text{Hub}} = -\sum_{ij,\sigma} t_{ij}\, c^\dagger_{i\sigma} c_{j\sigma} + U \sum_i n_{i\uparrow} n_{i\downarrow}, \qquad n_{i\sigma} \equiv c^\dagger_{i\sigma} c_{i\sigma}. \tag{6}$$

Physically, this local repulsion is what an onsite electrostatic (Coulomb) interaction reduces to once screening is included, and it is the minimal Hamiltonian we need for **magnetism**. Magnetic order breaks time-reversal and spin-rotational symmetry: the exact Hamiltonian is invariant under any spin rotation, but its ground state is not, and if you apply time reversal to the ground state you get back a distinct state with the opposite magnetization.

**The spinless nearest-neighbor model.** Now drop the spin degree of freedom altogether and instead keep a density–density repulsion between neighboring sites:

$$\mathcal{H}_{\text{CDW}} = -t \sum_i \left( c^\dagger_i c_{i+1} + \text{h.c.} \right) + V \sum_i n_i n_{i+1}, \qquad n_i \equiv c^\dagger_i c_i, \tag{7}$$

which is what you would get from a longer-range electrostatic repulsion once you truncate it to nearest neighbors. This is the minimal Hamiltonian for a **charge density wave (CDW)**: here the broken symmetry is the lattice's *translational* symmetry, and charge piles up on one sublattice while it is depleted on the other.

Note that the two models live on the same chain and differ only in what the interaction penalizes, two electrons on one site or two electrons on neighboring sites, and as a result in what orders, the spin or the charge. Figure 2 puts the two side by side.

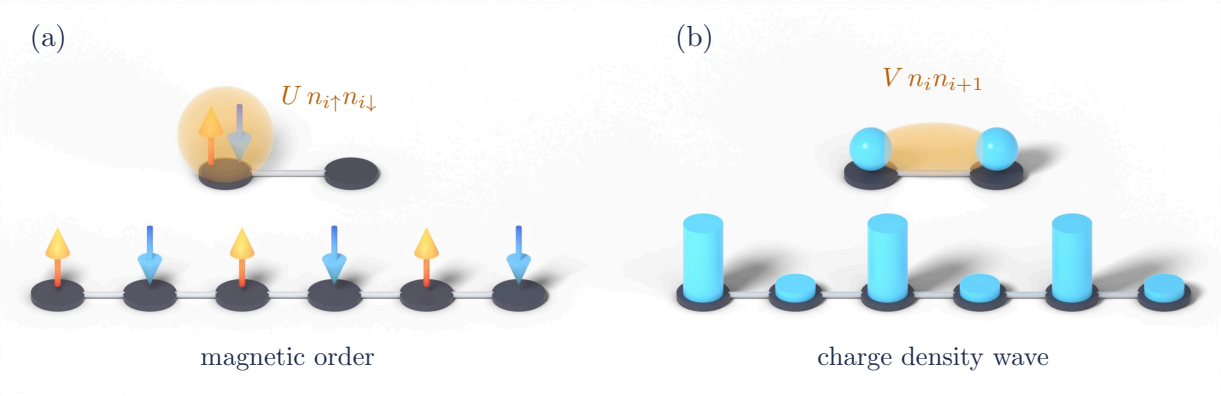



**Figure 2:** The two interactions of this section and the order each one favors. (a) In the Hubbard model of Eq. (6) an up and a down electron sharing a site cost $U$, so at half filling the electrons spread out one per site and their spins align antiparallel from site to site, the antiferromagnet. (b) In the spinless model of Eq. (7) two electrons on neighboring sites cost $V$, so the charge piles up on every other site, drawn here as the height of a column, and leaves the sites in between nearly empty, the charge density wave.

### 4.1 Where do $U$ and $V$ come from physically?

We will treat $U$ and $V$ as given parameters for the rest of the course, but let us pause on where they actually come from, because that also explains why both repulsive and attractive effective interactions are physically reasonable.

- **Direct electrostatics.** Screened Coulomb repulsion between electrons is always present, and it is always repulsive. Its onsite piece is what gives us $U$; its longer-range piece, truncated to nearest neighbors, is what gives us $V$.
- **Quasiparticle-mediated interactions.** If we integrate out some other bosonic degree of freedom that couples to the electron density or spin, lattice vibrations (phonons), collective spin fluctuations (magnons), or collective charge fluctuations (plasmons, excitons), we generate an *effective* quartic interaction of exactly the same form as Eq. (2), even though there was no bare electron–electron term to begin with.

The interaction we actually see is the sum of all these contributions, and depending on which mechanism wins it can come out net repulsive or net *attractive*. In this course we work with repulsive $U, V > 0$, which is the regime relevant for magnetism and CDW order; attractive interactions instead favor Cooper pairing and superconductivity, a case we will not pursue here but which uses exactly the same formal machinery.

## 5 Mean-field theory

Solving Eq. (6) exactly means working in a Hilbert space of dimension $4^L$ (or $2^L$ per spin species), which is exponential in the system size. So the question we want to ask is not how to solve it exactly, but how to solve it approximately: is there some single-particle Hamiltonian $\mathcal{H}_{\mathrm{MF}}$ whose ground state is close to the true ground state of $\mathcal{H}$? Finding that Hamiltonian is the whole art of mean-field theory.

### 5.1 Decoupling the interaction

The simplest way to do this is to write each quartic term as a product of two bilinears and replace one of the bilinears by its expectation value, dropping the fluctuation term that is left over (this is exact only at $U = 0$, but we will see it can still be a very good approximation away from that limit):

$$n_{i\uparrow}n_{i\downarrow} \;\longrightarrow\; \langle n_{i\uparrow}\rangle\, n_{i\downarrow} + n_{i\uparrow}\,\langle n_{i\downarrow}\rangle - \langle n_{i\uparrow}\rangle\,\langle n_{i\downarrow}\rangle\,. \tag{8}$$

If we define the local magnetization as $m_i \equiv \langle n_{i\uparrow}\rangle - \langle n_{i\downarrow}\rangle$, the Hubbard interaction turns into an effective, spin-dependent onsite potential:

$$\mathcal{H}_{\mathrm{MF}} = -\sum_{ij,\sigma} t_{ij}\, c^{\dagger}_{i\sigma}c_{j\sigma} - \frac{U}{2}\sum_{i} m_i\,\sigma_z + \text{const.} \tag{9}$$

where $\sigma_z = \pm 1$ for $\sigma = \uparrow, \downarrow$, so the onsite term reads $-\frac{U}{2}m_i\sigma$ site by site. So spin-up electrons now see a shifted onsite energy $-Um_i/2$, and spin-down electrons see $+Um_i/2$. This splitting, which is generated entirely by the interaction, is what we call the **exchange splitting**. More generally, if we also keep expectation values like $\left\langle c^{\dagger}_{i\uparrow}c_{i\downarrow}\right\rangle$, we let $\mathcal{H}_{\mathrm{MF}}$ describe non-collinear magnetization, with components $m_i^x \equiv 2\,\mathrm{Re}\left\langle c^{\dagger}_{i\uparrow}c_{i\downarrow}\right\rangle$, $m_i^y \equiv 2\,\mathrm{Im}\left\langle c^{\dagger}_{i\uparrow}c_{i\downarrow}\right\rangle$, and $m_i^z \equiv m_i$, so in principle mean-field theory can capture ferromagnets, antiferromagnets, spin spirals, or even skyrmion textures.

**The self-consistency loop in practice.** Concretely, here is what we actually do:

1. Guess an initial value for the local order parameters, e.g. $m_i^{(0)}$ for magnetism or $\langle n_i\rangle^{(0)}$ for a CDW.
2. Build $\mathcal{H}_{\mathrm{MF}}$ from Eq. (9) using this guess, and diagonalize it.
3. Fill the lowest single-particle orbitals up to the target electron number, and recompute $m_i$ (or $\langle n_i\rangle$) from the newly occupied orbitals.
4. Compare the new order parameters to the previous ones. If they have not changed, within some tolerance, we are done: this is the self-consistent solution. Otherwise, mix the old and new guesses and go back to step 2.

Figure 3 draws these four steps as a loop, each one on a chain of four sites. Because this is a nonlinear fixed-point problem, it can converge to different fixed points depending on where you start it, and that is exactly the initial-guess dependence we come back to below.

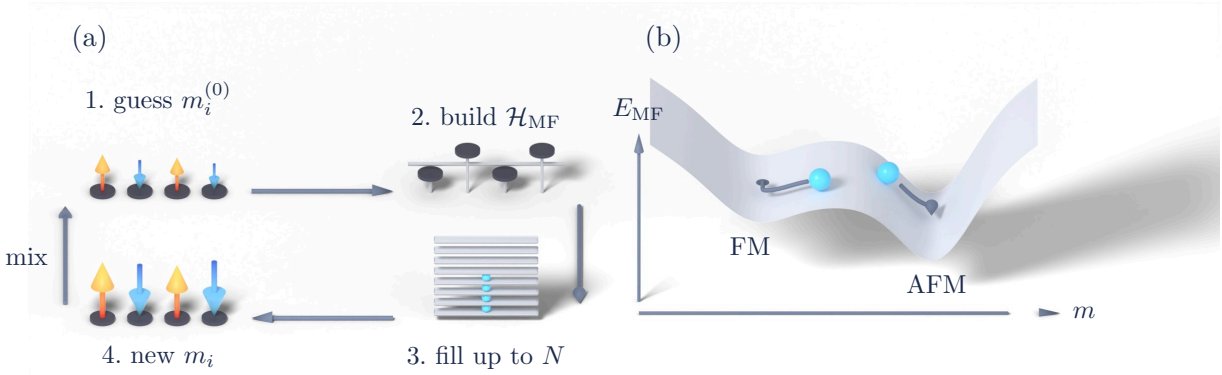



**Figure 3:** The self-consistency loop, which is what every mean-field calculation in this session actually runs. (a) The four steps as four stations, each drawn on a chain of four sites. Steps 2 and 3 are the two halves that alternate: an order parameter fixes a single-particle Hamiltonian through Eq. (9), drawn as the onsite energy of the spin-up electrons lowered where $m_i > 0$ and raised where $m_i < 0$, and diagonalizing that Hamiltonian and filling its lowest orbitals up to the target electron number, the lowest four of the eight single-particle levels of four spinful sites at half filling, fixes a new order parameter. Convergence means the two agree to within a tolerance, and until they do we mix the old and new guesses and go around again. (b) The mean-field energy over the order parameter, with a ferromagnetic and a lower antiferromagnetic valley. Two starting guesses on either side of the ridge roll into different valleys, meaning that the loop converges to a different self-consistent solution depending on where we start it.

## 5.2 Mean-field theory as a variational method

There is an equivalent way to think about all of this, and in a way a more useful one. Suppose we restrict the many-body ground state to be a single Slater determinant, $|\Psi\rangle = c_{a1}^\dagger c_{a2}^\dagger \cdots |\Omega\rangle$, and we minimize $\langle \mathcal{H} \rangle$ over the choice of single-particle orbitals $\{a_n\}$. Since $\mathcal{H}_{\mathrm{MF}}$ depends on $\langle n_{i\sigma} \rangle$, and $\langle n_{i\sigma} \rangle$ itself depends on the ground state of $\mathcal{H}_{\mathrm{MF}}$, minimizing the energy this way is precisely the self-consistency loop we just described: guess $\{m_i\}$, diagonalize $\mathcal{H}_{\mathrm{MF}}$, recompute $\{m_i\}$ from the filled orbitals, and repeat until it converges. So mean-field theory is really the best approximation we can build out of *product states*, and that restriction to product states is exactly what will get us into trouble in Section 9.

**✘ Computational companion**

Section "Magnetic order driven by interactions in a chain." Run the self-consistency loop for a chain at fixed filling and look at the converged spin-up/spin-down bands: you should see one cosine band per spin, split by the exchange field, with only the spin-down band occupied at the filling used in the notebook.

## 5.3 Gap opening at half filling

Let us now force half filling and double the unit cell, so we have two sites with alternating majority spin. At $U = 0$ this just folds the metallic cosine band onto itself. Once $U > 0$, the self-consistent solution opens a gap between the occupied and empty bands, and that gap grows with $U$, eventually becoming linear, $\Delta \sim U$, once $U \gg t$. If you look at the single-particle spectral function instead of the eigenvalues, you see exactly the same story: a continuous metallic density of states at $U = 0$ develops a clean gap that widens with $U$, which is precisely what a spectroscopy

experiment would pick up.

### 5.4 Multiple self-consistent solutions and their energetics

Because the self-consistency loop is a *nonlinear* fixed-point problem, the solution it converges to depends on where we start it. A ferromagnetic initial guess converges to a metallic, ferromagnetic solution; an antiferromagnetic initial guess converges to an insulating, antiferromagnetic solution. Both of these are legitimate stationary points of the mean-field energy, but only one of them is the true, lowest-energy mean-field ground state. If we take a half-filled chain with only nearest-neighbor hopping, it turns out the antiferromagnetic solution is always lower in energy, and the gap between the two scales as

$$E_{\mathrm{FM}} - E_{\mathrm{AFM}} \sim \frac{1}{U} \xrightarrow{U\to\infty} 0. \tag{10}$$

So the two solutions only become degenerate in the strict atomic limit $U \to \infty$; at any finite $U$ they are physically distinct states, one metallic and one insulating. Panel (b) of Figure 3 draws this as an energy landscape with two valleys, and a run that starts on the ferromagnetic side stays in the higher one.

**✘ Computational companion**

Section "The interacting Hubbard dimer." Compare a ferromagnetic-seeded and an antiferromagnetic-seeded self-consistent run on the dimer of Section 3, now with $U > 0$ switched on, and check numerically which solution has the lower variational energy and how the gap between them scales with $U$. Then go back to "Gap opening in the Hubbard model" with a ferromagnetic initialization, to see this initial-guess dependence show up directly in the spectrum.

## 6 The strong-coupling limit: from electrons to spins

As $U/t \to \infty$, something nice happens to the local moment $\langle n_{i\uparrow}\rangle - \langle n_{i\downarrow}\rangle$: it saturates, in magnitude, to 1 on every site. Double occupancy and empty sites become energetically forbidden, and every site ends up carrying exactly one localized electron. This is the regime where the physics is no longer best described by itinerant fermions, but by *spins*, and let us now see in detail why.

### 6.1 The Hubbard dimer and its low-energy manifold

Take our two-site dimer again, with hopping $t$, interaction $U$, and two electrons total (so we are at half filling). We already counted a six-dimensional many-body basis for this problem in Section 3. For $U \gg t$, this basis splits cleanly into two groups:

- a four-dimensional **ground-state manifold** of singly-occupied configurations ($\uparrow\downarrow$, $\downarrow\uparrow$, $\uparrow\uparrow$, $\downarrow\downarrow$ across the two sites), sitting at energy $\sim 0$;
- a **virtual (high-energy) manifold** of doubly occupied configurations, sitting at energy $\sim U$.

Look closely at the ground-state manifold: it is exactly the Hilbert space of two spin-$\frac{1}{2}$ degrees of freedom. In other words, our strongly interacting electronic problem has quietly turned into a spin problem. Panel (c) of Figure 4 draws the six states with their energies.

## 6.2 Superexchange: $J = 4t^2/U$

The only term that connects the two manifolds is the hopping $t$, and symmetry alone (there is no spin-rotation breaking anywhere in $\mathcal{H}_{\text{Hub}}$) already tells us that the low-energy effective Hamiltonian has to be proportional to $\mathbf{S}_1 \cdot \mathbf{S}_2$ at leading order in spin operators. To actually pin down the coefficient, we treat $\mathcal{H}_U = U \sum_i n_{i\uparrow} n_{i\downarrow}$ as the unperturbed Hamiltonian, with the singly and doubly occupied configurations of Section 3 as its eigenstates, and the hopping $V = -t \sum_\sigma (c^\dagger_{0\sigma} c_{1\sigma} + \text{h.c.})$ as the perturbation. The parallel configurations $|\uparrow\uparrow\rangle$ and $|\downarrow\downarrow\rangle$ have no doubly occupied virtual state to hop into at all, since Pauli exclusion blocks it, so they are exact eigenstates already, unshifted at any order: this is our triplet, sitting at $E_{\text{triplet}} = 0$. Panel (a) of Figure 4 draws this blocked hop, and panel (b) the one that is allowed when the spins are antiparallel.

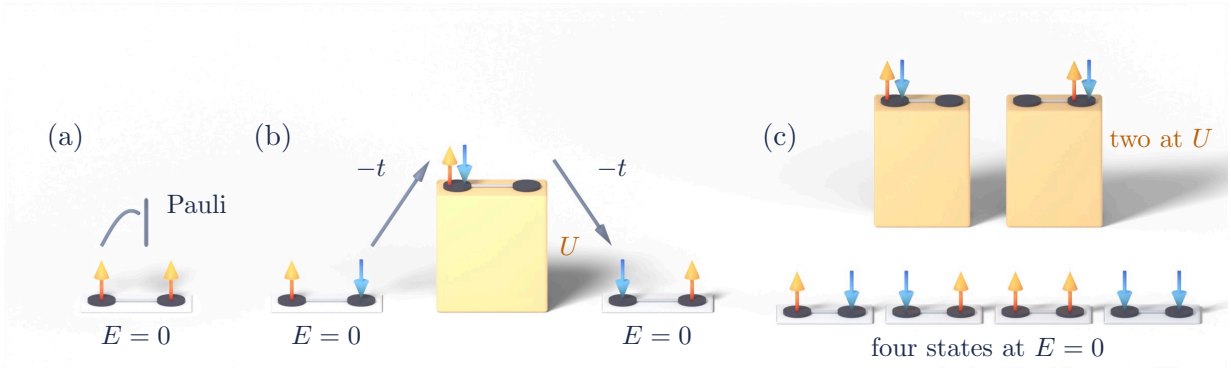



**Figure 4:** The two spin channels of the half-filled Hubbard dimer at strong coupling, which is where the $J = 4t^2/U$ of Eq. (13) comes from. (a) With parallel spins there is no doubly occupied state available to hop into, Pauli exclusion blocks the process outright, and the triplet stays exactly at $E_{\text{triplet}} = 0$ at every order. (b) With antiparallel spins either electron can hop onto its neighbor, passing through a virtual doubly occupied configuration at energy $U$ and landing on the exchanged configuration. Each hop carries a matrix element $-t$ and the virtual state is paid for by an energy denominator $-U$, so second-order perturbation theory lowers the antiparallel channel by an amount of order $t^2/U$ while leaving the parallel one untouched. (c) All six two-electron states of the dimer as levels: the four with one electron per site at $E = 0$, the ground-state manifold, which is the Hilbert space of two spins, and the two doubly occupied states raised by $U$, which the hops of (b) visit only virtually.

The antiparallel configurations are more subtle, since *either* electron can hop, so each one of $|\uparrow\downarrow\rangle = c^\dagger_{0\uparrow} c^\dagger_{1\downarrow} |\Omega\rangle$ and $|\downarrow\uparrow\rangle = c^\dagger_{0\downarrow} c^\dagger_{1\uparrow} |\Omega\rangle$ connects through $V$ to *both* virtual doubly occupied states, $|2,0\rangle = c^\dagger_{0\uparrow} c^\dagger_{0\downarrow} |\Omega\rangle$ and $|0,2\rangle = c^\dagger_{1\uparrow} c^\dagger_{1\downarrow} |\Omega\rangle$, each sitting at energy $U$ above the low-energy manifold, with matrix elements $\langle 2,0| V |\uparrow\downarrow\rangle = -t$ and $\langle 0,2| V |\uparrow\downarrow\rangle = -t$ (and, by the site-exchange symmetry of the problem, $\langle 2,0| V |\downarrow\uparrow\rangle = +t$, $\langle 0,2| V |\downarrow\uparrow\rangle = +t$, the sign flipping because of fermionic anticommutation once the two electrons are reordered onto the same site). It is only the total spin, not the sublattice, that is conserved here, so the clean way to organize the calculation is to work directly in the singlet/triplet basis rather than $\{|\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle\}$. Only the singlet combination $|s\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$ has any overlap with the doubly occupied states at all (the triplet combination's matrix elements to $|2,0\rangle$ and $|0,2\rangle$ cancel by the sign flip above), and by the same site-exchange symmetry only the symmetric combination $|d\rangle = \frac{1}{\sqrt{2}}(|2,0\rangle + |0,2\rangle)$ couples back to it, since the antisymmetric combination $\frac{1}{\sqrt{2}}(|2,0\rangle - |0,2\rangle)$ is odd under the mirror that $|s\rangle$ is even under.

Explicitly,

$$\langle d|\,V\,|s\rangle = \tfrac{1}{\sqrt{2}}\left[\langle 2,0|\,V\,|{\uparrow\downarrow}\rangle - \langle 2,0|\,V\,|{\downarrow\uparrow}\rangle + \langle 0,2|\,V\,|{\uparrow\downarrow}\rangle - \langle 0,2|\,V\,|{\downarrow\uparrow}\rangle\right]\tfrac{1}{\sqrt{2}} = \tfrac{1}{2}\left[-t-t-t-t\right] = -2t, \tag{11}$$

so the singlet sector reduces to an exact $2\times 2$ problem, $|s\rangle$ at energy 0 coupled to $|d\rangle$ at energy $U$ by the matrix element $-2t$, with the orthogonal antisymmetric doubly occupied combination decoupled entirely, remaining an exact eigenstate at energy $U$. Standard non-degenerate second-order (Schrieffer–Wolff, or equivalently Löwdin) perturbation theory on this $2\times 2$ block gives

$$E_{\text{singlet}} \approx 0 + \frac{|\,\langle d|\,V\,|s\rangle\,|^2}{0-U} = -\frac{4t^2}{U}, \tag{12}$$

which one can check is exactly the $U \gg t$ expansion of the closed-form eigenvalue of the full $2\times 2$ matrix, $E_- = \frac{U}{2} - \sqrt{(U/2)^2 + 4t^2} \to -4t^2/U$; a full numerical exact diagonalization of the two-site, two-electron Hubbard model confirms both this asymptotic value and the exact triplet energy $E_{\text{triplet}} = 0$ at every $U/t$, not just at strong coupling. Compare this to the exact spectrum of $J\,\mathbf{S}_1\cdot\mathbf{S}_2$ itself: writing $\mathbf{S}_{\text{tot}} = \mathbf{S}_1 + \mathbf{S}_2$ gives $\mathbf{S}_1\cdot\mathbf{S}_2 = \frac{1}{2}\left[S_{\text{tot}}(S_{\text{tot}}+1) - \frac{3}{4} - \frac{3}{4}\right]$, which evaluates to $-\frac{3}{4}$ for the singlet ($S_{\text{tot}}=0$) and $+\frac{1}{4}$ for the triplet ($S_{\text{tot}}=1$), i.e. $E_{\text{singlet}} = -\frac{3}{4}J$ and $E_{\text{triplet}} = \frac{1}{4}J$, a gap of exactly $J$ between them. Matching this gap to the one we just computed fixes

$$J = E_{\text{triplet}} - E_{\text{singlet}} = \frac{4t^2}{U} \qquad \Longrightarrow \qquad \mathcal{H}_{\text{eff}} = J\,\mathbf{S}_1\cdot\mathbf{S}_2, \tag{13}$$

the **antiferromagnetic superexchange** coupling. Intuitively: virtual hopping into the doubly occupied states, which is Pauli-blocked for parallel spins, lowers the energy of the singlet combination of antiparallel configurations, and that is why the effective low-energy interaction ends up favoring antialignment even though the underlying Hubbard repulsion has no explicit spin dependence at all. The same argument goes through unchanged on an arbitrary lattice with hoppings $t_{ij}$: the half-filled, strongly interacting Hubbard model on *any* lattice maps onto a spin-$\frac{1}{2}$ **Heisenberg model**,

$$\mathcal{H}_{\text{Heis}} = \sum_{ij} J_{ij}\,\mathbf{S}_i\cdot\mathbf{S}_j, \qquad J_{ij} \sim \frac{t_{ij}^2}{U}. \tag{14}$$

with the same numerical prefactor as Eq. (13), $J_{ij} = 4t_{ij}^2/U$, for each bond treated in isolation this way. If instead we have more than one electron per site, as in multi-orbital atoms, we map onto higher-spin Heisenberg models instead: spin-1 for two electrons per site, spin-$\frac{3}{2}$ for three, and so on. Additional weakly interacting orbitals can even flip the *sign* of $J_{ij}$, giving us ferromagnetic rather than antiferromagnetic coupling, which is why real materials are not always antiferromagnets.

**✘ Computational companion**

Section "The strongly interacting limit of the Hubbard model." Compute $E_{\text{FM}} - E_{\text{AFM}}$ as a function of $U$ and check that it follows the $1/U$ scaling we derived above. Then push to smaller $U$ and see where the perturbative assumption $t \ll U$ behind Eq. (13) starts to break down.

## 7 Classical magnetic order and frustration

If we now treat the spins in $\mathcal{H}_{\text{Heis}}$ as classical vectors, an approximation that is in spirit very close to mean-field theory itself, finding the ground state becomes a straightforward energy min-

imization.

**Bipartite lattices.** On the square and honeycomb lattices, we can always split the sites into two sublattices $A, B$ such that every bond connects $A$ to $B$. For antiferromagnetic $J > 0$, the classical ground state is then simply $\mathbf{S}_A = -\mathbf{S}_B$: collinear Néel order.

**Frustration.** Things get more interesting on lattices that are *not* bipartite, like the triangular and kagome lattices, which are built out of corner- or edge-sharing triangles. There, no assignment of two sublattices can satisfy every antiferromagnetic bond at once. The classical ground state instead becomes **non-collinear**: neighboring spins rotate by 120° as you go around each triangle, producing a spiral magnetic order with a multi-site magnetic unit cell. Figure 5 draws the Néel state, a single frustrated triangle and the 120° order side by side.

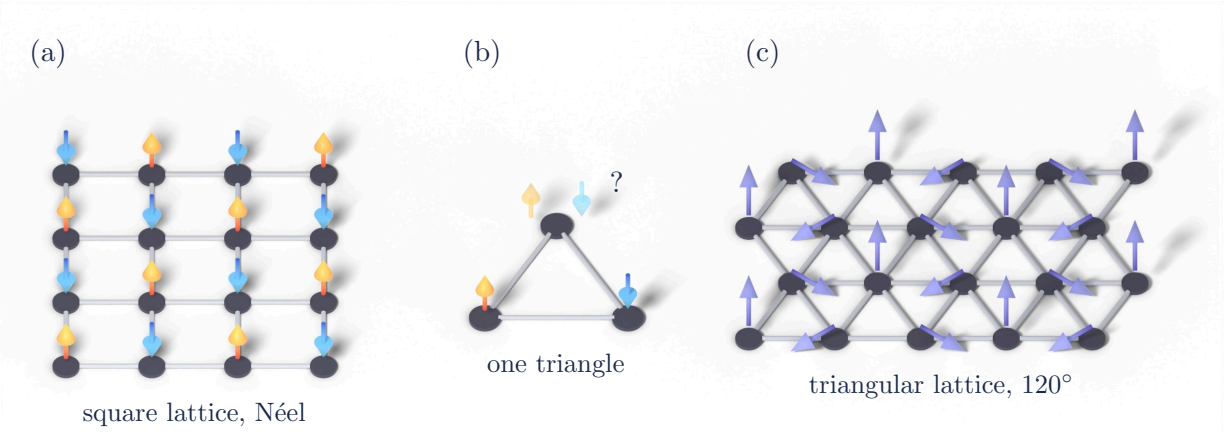



**Figure 5:** Classical ground states of the Heisenberg model of Eq. (14) with antiferromagnetic $J > 0$, obtained by treating every spin as a classical vector. (a) On a bipartite lattice every bond connects the two sublattices, so $\mathbf{S}_A = -\mathbf{S}_B$ satisfies all of them simultaneously and the order is collinear Néel. (b) A single triangle is where this fails: once two spins are antiparallel and satisfy their bond, the third spin cannot be antiparallel to both, drawn as two ghost arrows, so one bond is left unsatisfied regardless of its choice. (c) On the triangular lattice the bonds sit on edge-sharing triangles and no two-sublattice assignment can satisfy all of them, so the minimum-energy configuration is instead the non-collinear 120° spiral, with a magnetic unit cell spanning three sites, drawn in one color since none of the three sublattices is up or down.

**✘ Computational companion**

Section "Non-collinear magnetism in the Hubbard model from competing interactions" and "Non-collinear magnetic order from geometric frustration." Add a second-neighbor hopping to the mean-field chain and see that it, too, can stabilize non-collinear order: competing exchange paths play the same frustrating role as triangular geometry does. Then move to the triangular/kagome supercells and read off the 120° spiral directly from the self-consistent magnetization pattern.

**Anisotropic exchange.** Real materials often break the SU(2) spin-rotation symmetry of $\mathcal{H}_{\text{Heis}}$ through spin–orbit coupling, adding bilinear-in-spin terms that are *not* symmetric under a common rotation of all spins. There are three that come up often enough to have their own names:

- **Antisymmetric (Dzyaloshinskii–Moriya) exchange**, $\mathbf{D}_{ij} \cdot (\mathbf{S}_i \times \mathbf{S}_j)$, which shows up when mirror symmetry is broken and favors spins sitting at 90° to each other, i.e. non-collinear order even without any geometric frustration.
- **Ising anisotropy**, coupling only $S_i^z S_j^z$, which arises from strong spin–orbit coupling and single-ion anisotropy and pushes the spins toward classical behavior, confined to a single axis or plane.
- **Kitaev exchange**, a *bond-dependent* Ising-like coupling where $x$-bonds couple $S^x S^x$, $y$-bonds couple $S^y S^y$, and $z$-bonds couple $S^z S^z$. No single spin configuration can satisfy all three bond types at once on, say, the honeycomb lattice, which is exactly why this term is intrinsically frustrating and a key ingredient behind quantum spin liquid models.

We will not compute with any of these three terms in this course, but they are what you reach for as soon as spin-orbit coupling is strong, and they are behind most of the exotic magnets and quantum spin liquid candidates studied today.

## 8 Charge density waves as a mean-field problem

The very same machinery applies to $\mathcal{H}_{\mathrm{CDW}}$ from Eq. (7): we decouple the density–density term via $\langle n_i \rangle n_{i+1} + n_i \langle n_{i+1} \rangle - \langle n_i \rangle \langle n_{i+1} \rangle$ and iterate to self-consistency, exactly as before. At $V = 0$ the chain is a featureless, half-filled metal with $n_i = \frac{1}{2}$ on every site. As we turn on $V$, we get exactly the same qualitative spectral evolution as in the Hubbard case: a gap opens and grows $\sim V$ at large $V$. What changes is the order parameter, which is now the **charge imbalance**,

$$\delta n \equiv n_{\mathrm{even}} - n_{\mathrm{odd}}, \tag{15}$$

and it grows from 0 at $V = 0$ toward $\delta n \to 1$ at large $V$, meaning the sites end up alternating fully filled and fully empty. The broken symmetry here is translational rather than time-reversal, but notice how closely the mechanism, and even the shape of the spectral evolution, mirrors the magnetic case.

**✘ Computational companion**

Sections "Gap opening in a spinless interacting model" and "Symmetry breaking in the spinless fermionic chain." Track the gap and the charge imbalance $\delta n$ as a function of $V$, then seed the self-consistency loop with an explicit external CDW field and see how it selects which sublattice orders.

## 9 Where mean-field theory breaks down

So far mean-field theory has worked well every time, and that is really because it is a very good technique precisely when the true ground state *does* spontaneously break a symmetry. But nothing guarantees this in advance, and the Hubbard dimer at half filling gives us a clean counterexample.

The exact Hamiltonian $\mathcal{H}_{\mathrm{Hub}}$ for two sites is fully spin-rotation and time-reversal symmetric. Its mean-field decoupling, however, is only self-consistent with a *symmetry-breaking* ansatz once $U$ passes a threshold: below threshold, $\mathcal{H}_{\mathrm{MF}}$ stays non-magnetic, and above it, the converged solution has finite, eventually saturating local moments of opposite sign on the two sites, exactly

the regime we studied in Section 6. The mean-field gap tracks this transition too, staying constant below threshold and then growing linearly above it.

Now compare this to the *exact* many-body ground state, and the story looks quite different. In the strongly interacting limit, Section 6 already told us the dimer maps onto $\mathcal{H}_{\text{eff}} = J\,\mathbf{S}_1 \cdot \mathbf{S}_2$, and we can just diagonalize that by hand: its exact ground state is the singlet

$$|\Psi_{\text{singlet}}\rangle = \frac{1}{\sqrt{2}}\Big( |\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle \Big). \tag{16}$$

This state has $\langle S_1^z \rangle = \langle S_2^z \rangle = 0$. It simply does not break time-reversal symmetry, in direct conflict with what the symmetry-broken mean-field solution told us to expect. The reason is structural, not a numerical accident: $|\Psi_{\text{singlet}}\rangle$ is an entangled superposition, and it cannot be written as any single Slater determinant, no matter what choice of orbitals or spin quantization axis we try. Mean-field theory, built by construction out of product states (Section 5.2), can simply never reach it.

**Why this matters.** The Hubbard dimer is the smallest Hamiltonian you can write down that shows this failure mode, but the failure itself is completely generic: whenever the true many-body ground state is entangled and not adiabatically connected to a product state, mean-field theory gets the physics qualitatively wrong, the wrong symmetry, the wrong order, sometimes an entirely wrong phase. This is exactly the agenda for the rest of the course: *exact diagonalization* (Sessions 2 and 3) solves small many-body Hamiltonians with no product-state restriction at all, and *tensor network* methods (Sessions 4 and 5) let us recover the exact answer at system sizes where exact diagonalization is no longer feasible.

| | **Mean-field theory** | **Exact diagonalization** | **Tensor networks (MPS)** |
|---|---|---|---|
| Ansatz | product state (Slater det.) | unrestricted, full Hilbert space | bounded-entanglement state (MPS) |
| Cost vs. size $L$ | polynomial | exponential, $\sim 2^L$ | polynomial in $L$ and bond dimension |
| Captures entanglement? | no | yes, exactly | yes, up to the bond dimension |
| Typical reach | any $L$ | $L \lesssim 20$ spins, half that for spinful fermions | $L \sim 40$ to 100 and more (1D) |
| Fails when… | ground state is not adiabatically connected to a product state (this session) | never, in principle (only limited by $L$) | entanglement exceeds what the bond dimension can store |

**Table 1:** The three methods used in this course, previewed side by side: their ansatz, cost scaling with system size $L$, whether they capture entanglement, typical reach, and failure mode.

Think of Table 1 as a preview rather than a derivation: we will develop and use each of these methods on its own terms starting next session.

## Closing summary

- Second quantization lets us write single-particle and many-body ($\mathcal{O}(4)$-fermion) Hamiltonians in one language; the interaction term is what drives all correlated physics.
- Mean-field theory decouples the interaction into an effective single-particle problem that we solve self-consistently; equivalently, it is a variational search over Slater determinants.
- The Hubbard model gives us magnetism (broken time-reversal/spin symmetry); the spinless nearest-neighbor model gives us charge density waves (broken translational symmetry), and both come from essentially the same mean-field mechanism.
- In the strongly interacting limit, second-order perturbation theory maps the Hubbard model onto a Heisenberg spin model with $J \sim t^2/U$, the superexchange coupling. Classically this model shows collinear order on bipartite lattices and non-collinear/frustrated order otherwise.
- Mean-field theory can fail qualitatively whenever the true ground state is entangled beyond any product state, exactly as we saw for the Hubbard dimer's singlet ground state. That failure is what motivates exact diagonalization and tensor networks, which we take up next session.

# PHYS-E0510 – Advanced Computational Methods in Physics D

Session 2: Exact Diagonalization of the Heisenberg Spin Chain

**Companion notebook:** `quantum_magnetism.ipynb` **Prerequisites:** Session 1 (second quantization and the superexchange derivation of the Heisenberg model)

## Contents and goals

Last time we ended on a cautionary note: mean-field theory, built entirely out of product states, cannot represent the true ground state of the Hubbard dimer, an entangled singlet that spontaneously breaks nothing at all. Today we stop approximating. Instead of guessing a variational ansatz, we are going to write down the many-body Heisenberg Hamiltonian that emerges from superexchange, represent it as an honest matrix acting on the full $2^L$-dimensional many-body Hilbert space, and diagonalize it exactly. Along the way we will learn to read a many-body spectrum: how the many-body gap tells us whether a spin chain is gapped or gapless, how a field or a coupling constant can be dialed to drive level crossings and even phase transitions, how symmetries let us block-diagonalize the Hamiltonian so that "exact" does not immediately mean "impossible," and how correlators, both static and dynamical, let us extract order and excitations directly from the wave function. We will close by asking whether the excitations of a quantum magnet admit their own single-particle description, in terms of emergent fermionic quasiparticles called spinons. Everything here is numerically exact, no product-state restriction anywhere, and that honesty comes at the price we will keep bumping into all session: the Hilbert space grows as $2^L$, which is exactly the wall that motivates the tensor-network methods we build starting in Session 4.

## Learning outcomes

By the end of this session, you should be able to:

- Build the Heisenberg Hamiltonian as an explicit matrix on the many-body product-state basis and diagonalize it exactly for a finite spin chain.
- Use the finite-size scaling of the many-body gap, power-law versus saturating, to tell a gapless spin chain from a gapped one.
- Exploit conserved quantum numbers, total spin and $S^z_{\text{tot}}$, to block-diagonalize the Hamiltonian and explain level crossings and magnetization curves in a field.
- Compute static and dynamical spin correlators and connect them to an auxiliary-fermion (spinon) mean-field description of the excitations.

## 10 From the Hubbard dimer to a genuine quantum magnet

Let us start by reminding ourselves what a symmetry-broken classical magnet actually looks like at the microscopic level, since it is the natural foil for what comes next. Take the simplest possible magnetic Hamiltonian, an Ising coupling between two spins,

$$\mathcal{H}_{\text{Ising}} = J\, S^z_1 S^z_2, \qquad J > 0. \tag{17}$$

There is no $S^x$ or $S^y$ anywhere, so you can just as well think of $S_i^z = \pm\frac{1}{2}$ as a classical Ising variable. Minimizing the energy is trivial: we want $S_1^z$ and $S_2^z$ to have opposite sign, and there are exactly two ways to do that, $|\uparrow\downarrow\rangle$ and $|\downarrow\uparrow\rangle$. Both are legitimate ground states, both break time-reversal symmetry (flipping every spin sends one into the other), and each one individually has a finite, sharply defined local moment $\langle S_i^z\rangle = \pm\frac{1}{2}$. This is a classical magnet in miniature: a symmetry-broken state with a finite magnetization at every site.

Now let us restore the full quantum mechanics of the spin operators and couple all three components,

$$\mathcal{H}_{\text{Heis}} = J\,\mathbf{S}_1\cdot\mathbf{S}_2 = J\left(S_1^xS_2^x + S_1^yS_2^y + S_1^zS_2^z\right). \tag{18}$$

This is exactly the superexchange Hamiltonian we derived last session from the strong-coupling limit of the Hubbard dimer, with $J = 4t^2/U > 0$. We already know, from Session 1, that its exact ground state is the singlet $|\Psi_{\text{singlet}}\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle)$: a single, non-degenerate state, with $\langle S_i^z\rangle = 0$ on every site, that cannot be written as any product state. This is what we will from now on call a genuine **quantum magnet**: unlike the Ising ground states, this state does not break time-reversal symmetry at all, and it is entangled in an essential way. Our job today is simply to compute this kind of ground state honestly, for chains longer than two sites, and to understand everything we can read off from it and from the excited states that sit above it.

## 11 Exact diagonalization: basis, matrix, and a worked dimer

Generalize Eq. (18) to a chain of $L$ spin-$\frac{1}{2}$'s with nearest-neighbor exchange,

$$\mathcal{H}_{\text{Heis}} = J\sum_{i=1}^{L-1}\mathbf{S}_i\cdot\mathbf{S}_{i+1}, \qquad J>0 \text{ (antiferromagnetic)}. \tag{19}$$

Regardless of how complicated $J_{ij}$ ends up being on some lattice, solving this Hamiltonian exactly always reduces to the same three-step recipe: pick a basis of the many-body Hilbert space, write $\mathcal{H}$ as a matrix in that basis, and diagonalize it. The natural basis is the set of all product states $|\sigma_1\sigma_2\cdots\sigma_L\rangle$ with $\sigma_i\in\{\uparrow,\downarrow\}$: two states for $L=1$, four for $L=2$, eight for $L=3$, and in general $2^L$ states for $L$ sites. This is precisely the exponential growth of the many-body Hilbert space we have already met once, in Session 1's discussion of the Hubbard model; here it shows up again because a spin-$\frac{1}{2}$ chain is nothing but a projected, single-occupied corner of that same fermionic Hilbert space.

To build the matrix we need the matrix elements of $\mathbf{S}_i\cdot\mathbf{S}_j$, and it is most convenient to split it into a diagonal and an off-diagonal piece using the raising and lowering operators $S^\pm = S^x \pm iS^y$,

$$\mathbf{S}_i\cdot\mathbf{S}_j = S_i^zS_j^z + \tfrac{1}{2}\left(S_i^+S_j^- + S_i^-S_j^+\right). \tag{20}$$

The first term is diagonal in our product basis: it just multiplies each basis state by $\pm\frac{1}{4}$ depending on whether $\sigma_i,\sigma_j$ are parallel or antiparallel. The second term is the one that actually moves us around the basis: $S_i^+S_j^-$ flips an antiparallel pair $\downarrow_i\uparrow_j \to \uparrow_i\downarrow_j$ and annihilates everything else, so it only ever connects basis states that differ by swapping one $\uparrow\downarrow$ pair on a bonded pair of sites. This is why the Heisenberg matrix, even though it lives in a $2^L$-dimensional space, is *sparse*: each row has at most $L-1$ off-diagonal entries, one per bond, which is exactly the kind of structure that

iterative solvers like Lanczos exploit to reach much bigger $L$ than a naive dense diagonalization ever could. Figure 6 goes through the three steps for $L = 4$.

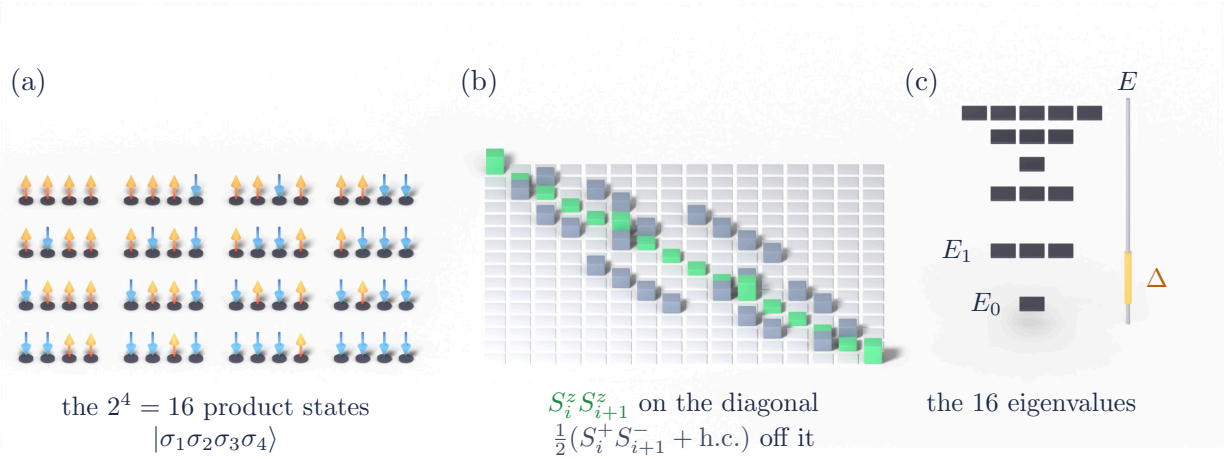



**Figure 6:** The three steps of exact diagonalization for the open Heisenberg chain of Eq. (19) with $L = 4$. (a) The basis, the $2^4 = 16$ product states in the order a code would count them, $|\uparrow\uparrow\uparrow\uparrow\rangle$ at the back left and $|\downarrow\downarrow\downarrow\downarrow\rangle$ at the front right. (b) The Hamiltonian in that basis, one cell per matrix element in the same order, with the height of a bar the size of the element: the $S_i^z S_{i+1}^z$ term of Eq. (20) fills the diagonal and the flip-flop term puts at most $L-1 = 3$ bars in each row, so only 40 of the 256 cells are occupied, which is what is meant by a sparse matrix. (c) The result, the sixteen eigenvalues drawn as levels at their energy, degenerate levels side by side: a singlet ground state at $E_0 \approx -1.62J$, a triplet at $E_1 \approx -0.96J$, and the many-body gap $\Delta = E_1 - E_0$ of Eq. (23) in amber; at the top sit the five members of the fully polarized $S = 2$ multiplet at $\frac{3}{4}J$.

**Worked example: the dimer, twice over.** Let us see this explicitly for $L = 2$. In the basis $\{|\uparrow\uparrow\rangle, |\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle, |\downarrow\downarrow\rangle\}$, the stretched states $|\uparrow\uparrow\rangle$ and $|\downarrow\downarrow\rangle$ have no partner to flip into, so $\mathcal{H}_{\text{Heis}}|\uparrow\uparrow\rangle = \frac{J}{4}|\uparrow\uparrow\rangle$ and likewise for $|\downarrow\downarrow\rangle$. On the remaining two-dimensional subspace $\{|\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle\}$, Eq. (20) gives the $2 \times 2$ block

$$\mathcal{H}_{\text{Heis}}\Big|_{\{\uparrow\downarrow,\downarrow\uparrow\}} = J\begin{pmatrix} -\frac{1}{4} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{4} \end{pmatrix} \Longrightarrow E_\pm = J\left(-\tfrac{1}{4} \pm \tfrac{1}{2}\right), \qquad |\pm\rangle = \tfrac{1}{\sqrt{2}}\left(|\uparrow\downarrow\rangle \pm |\downarrow\uparrow\rangle\right). \tag{21}$$

So $E_- = -\frac{3}{4}J$ for the antisymmetric combination, exactly the singlet we already knew, and $E_+ = \frac{1}{4}J$ for the symmetric combination, degenerate with $|\uparrow\uparrow\rangle$ and $|\downarrow\downarrow\rangle$ at the same energy $\frac{J}{4}$: together these three states form the **triplet**. Diagonalizing a $4 \times 4$ matrix by hand already reproduces the full singlet/triplet structure.

There is a second, quicker way to get the same answer, and we will use it again because it generalizes. Define the total spin $\mathbf{S}_{\text{tot}} = \mathbf{S}_1 + \mathbf{S}_2$, so that $\mathbf{S}_{\text{tot}}^2 = \mathbf{S}_1^2 + \mathbf{S}_2^2 + 2\,\mathbf{S}_1 \cdot \mathbf{S}_2$, and hence

$$\mathcal{H}_{\text{Heis}} = J\,\mathbf{S}_1 \cdot \mathbf{S}_2 = \frac{J}{2}\left[S_{\text{tot}}(S_{\text{tot}}+1) - \tfrac{3}{4} - \tfrac{3}{4}\right], \tag{22}$$

using $\mathbf{S}_i^2 = \frac{3}{4}$ for spin-$\frac{1}{2}$. Since $\mathbf{S}_{\text{tot}}^2$ has eigenvalues $S_{\text{tot}}(S_{\text{tot}}+1)$ with $S_{\text{tot}} = 0$ or $1$ for two spin-$\frac{1}{2}$'s, Eq. (22) immediately gives $E(S_{\text{tot}} = 0) = -\frac{3}{4}J$ and $E(S_{\text{tot}} = 1) = \frac{1}{4}J$, matching Eq. (21) exactly. Notice, too, why the notebook's ground-state energy happens to equal the sum of the separately computed Ising and XY ground energies: on the two-dimensional subspace

$\{|\uparrow\downarrow\rangle, |\downarrow\uparrow\rangle\}$, the Ising piece $S_1^z S_2^z$ is literally proportional to the identity, $-\frac{1}{4}\mathbb{1}$, since *both* basis states carry the same $S_1^z S_2^z$ eigenvalue. Minimizing the sum of a constant and the XY block is the same as minimizing each separately and adding, which is why the two energies happen to add up here. This is special to $L = 2$: for longer chains the Ising term is no longer a multiple of the identity on the subspace the XY term mixes, and the two pieces generically no longer minimize on the same state.

**✘ Computational companion**

Open `quantum_magnetism.ipynb` and work through the "Basics" block: "Creating a spin_chain object," "Creating a Hamiltonian," "Computing the ground state energy," and "Computing expectation values." Build the $L = 2$ Heisenberg Hamiltonian directly from the spin operators, diagonalize it, and check that you recover $E_{\text{singlet}} = -\frac{3}{4}J$ and $E_{\text{triplet}} = \frac{1}{4}J$ numerically. Then replace $S^x$ by $S^y$ in the expectation-value cell and think about why nothing changes.

## 12 The many-body spectrum and the many-body gap

Now let us stop looking at $L = 2$ by hand and let the computer diagonalize Eq. (19) for increasing $L$. The number of eigenvalues you get out is, of course, exactly the dimension of the matrix, $2^L$: two for $L = 2$, sixteen for $L = 4$, sixty-four for $L = 6$, and so on, growing exponentially with system size, exactly like the basis itself. In practice, though, we almost never care about the whole spectrum. Any real experiment at low temperature only probes the states closest to the ground state, so from now on we will systematically focus on the bottom of the many-body spectrum: the ground state $E_0$ and the handful of low-lying excited states just above it.

The single most useful number we can read off from this part of the spectrum is the **many-body gap**,

$$\Delta(L) \equiv E_1(L) - E_0(L), \tag{23}$$

the energy cost of the cheapest excitation above the ground state, marked for $L = 4$ in panel (c) of Figure 6. For the uniform Heisenberg chain, $\Delta(L)$ shrinks steadily as $L$ grows, and it does so following a power law rather than saturating to a finite value: this is the hallmark of a **gapless** system, one whose excitations become arbitrarily cheap in the thermodynamic limit. Where does this power law come from? Qualitatively, from exactly the same free-fermion logic we will use in Section 19: replace each spin operator by an auxiliary (Abrikosov) fermion $f_{i\sigma}$ and mean-field decouple the exchange term, and $\mathcal{H}_{\text{Heis}}$ turns into a free-fermion **spinon** Hamiltonian whose band is gapless, with Fermi points at $k_F = \pm\pi/2$. The cheapest excitations are then particle-hole pairs created right at those Fermi points, which disperse *linearly* in momentum, and a linearly dispersing excitation on a ring of $L$ sites is exactly a one-dimensional particle-in-a-box problem: its levels are spaced by

$$\Delta(L) \;\sim\; \frac{\pi v_s}{L}, \tag{24}$$

with $v_s$ the velocity of the low-energy excitations, i.e. a $1/L$ power law rather than a finite gap. (Numerically, the finite-size data is well fit by $\Delta(L) \approx \pi^2 J/(2L)$; pinning down $v_s$ this precisely is a job for the exact Bethe-ansatz solution, not for mean-field theory, which gets the qualitative power law right but not the precise coefficient.) Keep this "particle in a box" picture in mind: we will meet it again, in a much more literal form, when we look at dynamical correlators in Section 18, and again when we build the explicit spinon band structure behind this argument in Section 19.

**✘ Computational companion**

Section "Many-body energies of the Heisenberg model." Diagonalize the uniform chain for $L = 2, 4, 6, 8$ and plot the full spectrum each time: check that the number of eigenvalues really is $2^L$. Then plot only $E_1(L) - E_0(L)$ and see the gap shrink with $L$; think about what the biggest $L$ is that you could still store the full spectrum for on your own machine.

## 13 The XXZ model: from classical order to quantum entanglement

Section 10 presented the Ising model and the Heisenberg model as two separate Hamiltonians, but it is much more illuminating to see them as the two endpoints of a single family. Take

$$\mathcal{H}_{\mathrm{XXZ}} = \sum_i \left[ J_{xy} \left( S_i^x S_{i+1}^x + S_i^y S_{i+1}^y \right) + J_z \, S_i^z S_{i+1}^z \right], \qquad J_z \equiv 1 \text{ fixed}, \tag{25}$$

and dial $J_{xy}$ from 0 to 1. At $J_{xy} = 0$ we are exactly back at the classical Ising model of Eq. (17), generalized to $L$ sites: in the thermodynamic limit its ground state is doubly degenerate, the Néel-ordered $|{\uparrow}{\downarrow}{\uparrow}{\downarrow}\cdots\rangle$ and its time-reversed partner, each carrying a finite, symmetry-broken staggered moment. At $J_{xy} = 1$ we recover the isotropic Heisenberg model of Eq. (19), whose ground state, as we already know from the dimer, is unique and carries no local moment at all: a purely quantum magnet. So $J_{xy}$ literally interpolates between a classical, symmetry-broken magnet and a quantum, entangled one, and the natural question is: how does the ground state actually get from one description to the other as we sweep $J_{xy}$?

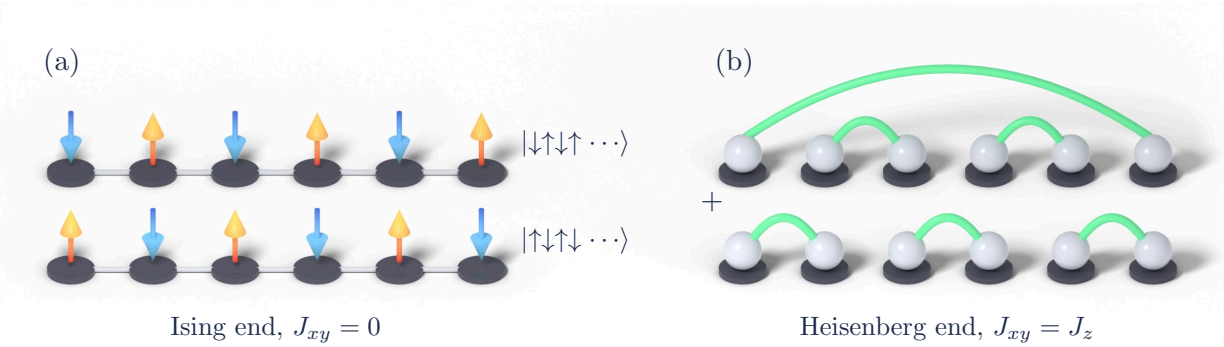



**Figure 7:** The two ends of Eq. (25). (a) The two classical ground states at the Ising endpoint $J_{xy} = 0$, $|{\uparrow}{\downarrow}{\uparrow}{\downarrow}\cdots\rangle$ in front and $|{\downarrow}{\uparrow}{\downarrow}{\uparrow}\cdots\rangle$ behind it. Each is a product state with a definite $S^z$ on every site, each carries a finite staggered moment, and the two are exchanged by time reversal. In the thermodynamic limit they are exactly degenerate and the system picks one of them, which is what symmetry breaking means here. Once $J_{xy}$ is switched on, on a finite chain they are instead split by a tiny amount, and watching that splitting grow as $J_{xy} \to 1$ is how the crossover to the unique, entangled Heisenberg ground state shows up in the spectrum. (b) At the Heisenberg end $J_{xy} = J_z$ there is no moment on any site, so the spins are drawn as spheres without arrows. You can think of the ground state as neighbors paired into singlets, the green bonds, resonating between the two ways of pairing them, drawn in front and behind, where the second one pairs the two ends as on a ring; this is a cartoon of the entanglement rather than the exact state, which also contains singlets between distant sites.

The finite-size spectrum answers this directly. At small $J_{xy}$, the two classical Néel states are not

exactly degenerate on a finite chain (only an infinite chain can spontaneously break a symmetry), but they sit extremely close together, split by a tiny amount that we can literally watch on the screen: this is the finite-size ghost of the classical double degeneracy. As we increase $J_{xy}$ toward 1, this splitting grows continuously, and for any finite $L$ that growth eventually turns into a finite, clearly non-degenerate gap once $J_{xy}$ gets close enough to 1. What changes as we make $L$ bigger is exactly how close to $J_{xy} = 1$ this crossover sits: for larger and larger chains, the two low-lying states stay nearly degenerate over a longer and longer stretch of $J_{xy}$, and the crossover sharpens toward the exact, thermodynamic-limit transition of the XXZ chain, which the Bethe-ansatz solution places precisely at the isotropic, SU(2)-symmetric point $J_{xy} = J_z$. In other words: just by watching how two eigenvalues in a finite-size spectrum move and split as we tune a coupling constant, we can already diagnose a quantum phase transition, well before we ever reach the thermodynamic limit itself. This is a technique we will lean on again and again in the rest of the course.

**✘ Computational companion**

Section "From classical to quantum magnetism." Plot the low-lying spectrum of $\mathcal{H}_{\mathrm{XXZ}}$ as a function of $J_{xy}$ at fixed $L$: identify the near-degenerate pair at small $J_{xy}$ and watch it split as $J_{xy} \to 1$. Then increase $L$ and see how the splitting crossover sharpens and moves, tracking the exact thermodynamic-limit transition at the isotropic point.

## 14 Good quantum numbers: total spin, the Zeeman field, and block-diagonalization

Section 11 used the identity $\mathbf{S}_1 \cdot \mathbf{S}_2 = \frac{1}{2}[\mathbf{S}^2_{\mathrm{tot}} - \mathbf{S}^2_1 - \mathbf{S}^2_2]$ as a shortcut for the dimer, but it is worth asking why it worked: it worked because $\mathbf{S}^2_{\mathrm{tot}}$ and $S^z_{\mathrm{tot}}$ commute with $\mathcal{H}_{\mathrm{Heis}}$. This is not special to two sites. Because Eq. (19) is built entirely out of dot products $\mathbf{S}_i \cdot \mathbf{S}_j$, it is invariant under any global spin rotation, and Noether's theorem in its operator form tells us immediately that the total spin $\mathbf{S}_{\mathrm{tot}} = \sum_i \mathbf{S}_i$ is conserved for *any* $L$ and any choice of exchange couplings $J_{ij}$:

$$\left[\mathcal{H}_{\mathrm{Heis}},\, \mathbf{S}^2_{\mathrm{tot}}\right] = 0, \qquad \left[\mathcal{H}_{\mathrm{Heis}},\, S^z_{\mathrm{tot}}\right] = 0. \tag{26}$$

So every eigenstate of $\mathcal{H}_{\mathrm{Heis}}$ can be labeled by a definite total spin $S$ and a definite projection $m$, exactly the singlet ($S = 0$) and triplet ($S = 1$) labels we used for the dimer, now promoted to good quantum numbers for a chain of any length.

**The Zeeman field does not mix states.** Let us now add a uniform magnetic field,

$$\mathcal{H}(B) = \mathcal{H}_{\mathrm{Heis}} - B\, S^z_{\mathrm{tot}}. \tag{27}$$

This term explicitly singles out the $z$-axis, so it breaks the full SU(2) spin-rotation symmetry down to rotations about $z$ alone. But look closely at what it actually does to our two conserved quantities. It trivially commutes with $S^z_{\mathrm{tot}}$, since it is built out of nothing else. And because $[\mathbf{S}^2_{\mathrm{tot}}, S^z_{\mathrm{tot}}] = 0$ is a completely general identity of angular momentum algebra, $\mathcal{H}(B)$ *also* commutes with $\mathbf{S}^2_{\mathrm{tot}}$. So both quantum numbers survive even in the presence of the field: adding $B$ cannot mix any eigenstates of $\mathcal{H}_{\mathrm{Heis}}$ into each other, it can only shift each one's energy by $-Bm$. This is exactly why the field-dependent spectrum you see in the notebook is a set of perfectly

straight lines rather than curves: each state with a definite $m$ shifts linearly, $E(B) = E(0) - Bm$, and states only bend or avoid one another once you add a term, like a transverse field, that fails to commute with $S^z_{\text{tot}}$.

Applied to the dimer: the singlet has $m = 0$ and stays completely flat, $E_{\text{singlet}}(B) = -\frac{3}{4}J$. The triplet's three members split according to their own $m = -1, 0, +1$: the $m = 0$ member is also completely flat, at $E = \frac{1}{4}J$, while the $m = \pm 1$ members disperse as $E = \frac{1}{4}J \mp B$. Those are precisely the two "flat lines" you can spot in the notebook's zero-field spectrum: they sit at different energies, separated by exactly the many-body gap $\Delta = J$ of Eq. (23), and what distinguishes them is not $m$, which is 0 for both, but the total spin $S$, 0 for one and 1 for the other.

**Block-diagonalizing a long chain.** The real payoff of Eq. (26) is computational. Since $S^z_{\text{tot}} = \sum_i S^z_i$ is conserved, the full $2^L \times 2^L$ Hamiltonian never actually mixes basis states with different numbers of up spins: if $n_\uparrow$ counts the up spins, then $m = n_\uparrow - L/2$ is fixed within each block, and $\mathcal{H}_{\text{Heis}}$ is block-diagonal with blocks of dimension

$$\dim(m) = \binom{L}{L/2 + m}. \tag{28}$$

The largest block sits at $m = 0$ (for even $L$), with $\dim(0) = \binom{L}{L/2} \sim 2^L/\sqrt{\pi L/2}$ by Stirling's approximation: still exponential in $L$, but a factor $\sim \sqrt{L}$ smaller than the full Hilbert space, and, what matters most, it can be diagonalized independently of every other block. If the chain is also periodic, translational invariance gives us a second conserved quantity, the total momentum $k$, and combining $S^z_{\text{tot}}$ sectors with momentum sectors shrinks the largest block by roughly another factor of $L$. Exploiting every available conserved quantity together with the sparse-matrix Lanczos techniques of Section 11 is how exact-diagonalization codes reach the $L \sim 20$ spins that mark the limit of this session; without it, $2^{20} \approx 10^6$ would already be a serious dense-matrix computation, and $L = 30$ would simply be out of reach.

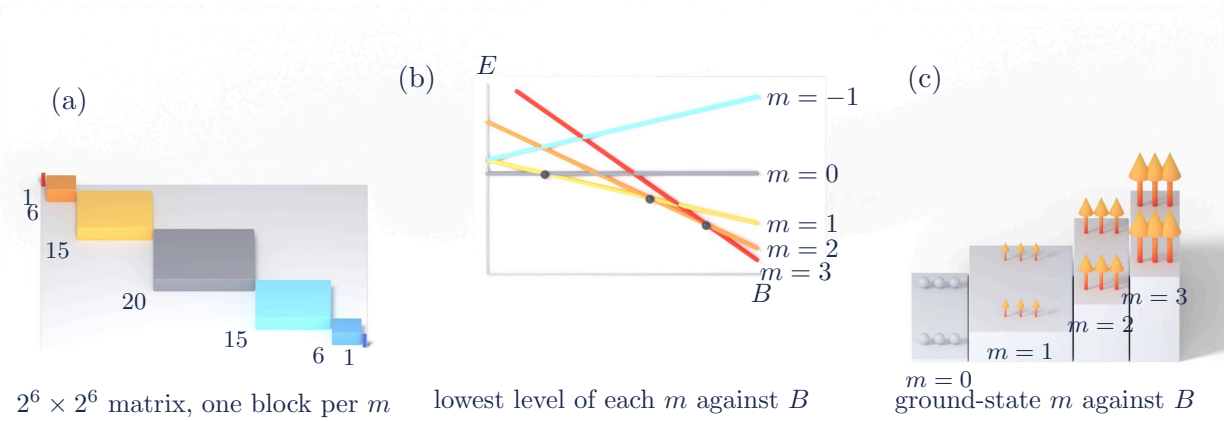



**Figure 8:** What the conservation of $S^z_{\text{tot}}$ buys us, for the open chain of $L = 6$. (a) Block structure of $\mathcal{H}_{\text{Heis}}$ in the $S^z_{\text{tot}}$ basis, with the blocks to scale and $m = +3$ at the back. Because $S^z_{\text{tot}}$ is conserved, the Hamiltonian never connects basis states with different numbers of up spins, so the $2^L \times 2^L$ matrix falls apart into one block per magnetization sector, of dimension $\binom{L}{L/2+m}$ as in Eq. (28), with nothing at all outside them. Each block is diagonalized completely independently of the others, and even the largest one, at $m = 0$, is already smaller than the full space by a factor $\sim \sqrt{L}$. (b) The lowest level of each block, in the color of its block, against the field of Eq. (27), leaving out $m = -2$ and $m = -3$, which rise out of the frame at once: every line is straight, $E_m(B) = E_m(0) - Bm$, and the dark beads mark where the bottom of the fan changes hands, at $B \approx 0.49J$, $1.39J$ and $1.87J$. (c) The ground-state magnetization read off from (b), the staircase of Section 15, with the chain standing on each plateau: spheres with no moment on the singlet plateau, then arrows whose length is the average moment $m/L$ per site, saturated from $B = J[1 + \cos(\pi/L)] \approx 1.87J$ on, the one-magnon edge of the open chain, which tends to the $2J$ of the infinite chain as $L$ grows.

**✘ Computational companion**

Sections "Energies of a Heisenberg model as a function of the external magnetic field" and "Quantum numbers in the Heisenberg model." Plot the dimer spectrum against $B$ and check the linear, non-mixing evolution predicted by Eq. (27); identify which line is the singlet and which is the $m = 0$ triplet member, and confirm that they are exactly $\Delta = J$ apart at $B = 0$. Then think about what field-dependent term would bend the lines instead of keeping them straight.

## 15 Level crossings and magnetization curves

Equation (27) does more than split levels: because different $m$-sectors disperse with different slopes, they can cross, and a crossing at the very bottom of the spectrum is a **level crossing of the ground state**. For the dimer this is easy to pin down exactly: the ground state is the singlet, at $-\frac{3}{4}J$, until the $m = +1$ triplet branch $E = \frac{1}{4}J - B$ drops below it, which happens once

$$\tfrac{1}{4}J - B = -\tfrac{3}{4}J \quad \Longrightarrow \quad B = J. \tag{29}$$

Below $B = J$ the ground state is the singlet, with zero magnetization; above $B = J$ it is the fully polarized triplet member $|\uparrow\uparrow\rangle$, with saturated magnetization $\langle S^z_{\text{tot}} \rangle = 1$. This is the simplest possible example of a field-driven quantum phase transition: a single, sharp level crossing between a sector with $S = 0$ and a sector with $S = 1$.

Longer chains show the same phenomenon, just repeated many times over. As $B$ grows from

zero, the ground state hops successively through sectors of increasing total spin, $S = 0 \to 1 \to 2 \to 3 \to \cdots$, each hop showing up as an abrupt jump in the ground-state magnetization $\langle S^z_{\text{tot}} \rangle$: a magnetization staircase. Panels (b) and (c) of Figure 8 draw both for $L = 6$, the crossings at the bottom of the fan and the staircase they give. Because bigger chains pack their low-lying $S$-sectors closer together in energy, these crossings happen at progressively smaller fields as $L$ grows, and in the thermodynamic limit the staircase smooths out into the continuous magnetization curve of an infinite antiferromagnetic chain. Eventually, once $B$ exceeds roughly the bandwidth of the low-energy spectrum ($B \gtrsim zJ$, with $z$ the coordination number; for a chain, $z = 2$ and the saturation field is exactly $B_{\text{sat}} = 2J$), the ground state saturates completely: every spin points along the field, and this fully polarized state is in fact an *exact* eigenstate of $\mathcal{H}_{\text{Heis}}$ itself, a rare case where the many-body ground state really is a simple product state, since there is nothing left for the exchange term to do once every spin is already maximally aligned. Sweeping $B \to -B$ reproduces the identical staircase with the opposite sign of magnetization, exactly the time-reversal partner of the $B > 0$ curve.

Let us also look directly at the excited states in a small field, before any level crossing has happened. The ground state stays a total-$S = 0$ singlet with zero magnetization on every site. The first excited states, the $S = 1$ triplet members, do carry a finite total magnetization, but that magnetization is not localized anywhere: it is smeared out over the whole chain, one more signature that these are delocalized many-body excitations rather than a single flipped spin sitting at one site. Higher excited states repeat this pattern with different sign structures, several of them related to each other by time reversal, and this whole family of spread-out, roughly particle-in-a-box-like excitations is exactly what the dynamical correlator of Section 18 will let us resolve in detail.

**✘ Computational companion**

Sections "Level crossings in the Heisenberg model," "Many-body wavefunctions in a magnetic field," "Level crossings as a function of the magnetic field," and "Magnetization of moderately large Heisenberg model." Confirm the dimer's crossing at $B = J$ from Eq. (29), then move to $L = 6$ and $L = 10$ chains and track the full magnetization staircase, its saturation field, and its mirror symmetry under $B \to -B$.

## 16 Gapped versus gapless: the dimerized chain

Section 12 told us the uniform Heisenberg chain is gapless, $\Delta(L) \to 0$ as a power law. Let us now build a chain that is not, by alternating the exchange coupling bond by bond,

$$\mathcal{H}_{\text{dimer}} = \sum_i J_i \, \mathbf{S}_i \cdot \mathbf{S}_{i+1}, \qquad J_i = J\left[1 + (-1)^i \delta\right], \tag{30}$$

so odd and even bonds alternate between $J(1 + \delta)$ and $J(1 - \delta)$. At $\delta = 0$ we recover the uniform, gapless chain. At the opposite extreme $\delta \to 1$, the weak bonds vanish entirely and the chain falls apart into $L/2$ completely decoupled dimers, each one exactly the $L = 2$ problem of Section 11: the exact ground state is simply a product of singlets, one per strong bond, and the cheapest excitation is a single dimer flipping from its singlet into its triplet, costing exactly $\Delta_\infty = J(1 + \delta)$, the very same singlet-triplet gap we computed in Eq. (21). Turning the weak bond back on perturbs this picture, letting the singlets talk to each other and giving the triplet excitation some dispersion, but it does not close the gap: for any finite $\delta$, no matter how small,

the dimerized chain remains gapped, with a finite $\Delta_\infty$ that shrinks smoothly to zero only as $\delta \to 0$. Figure 9 puts the two chains side by side.

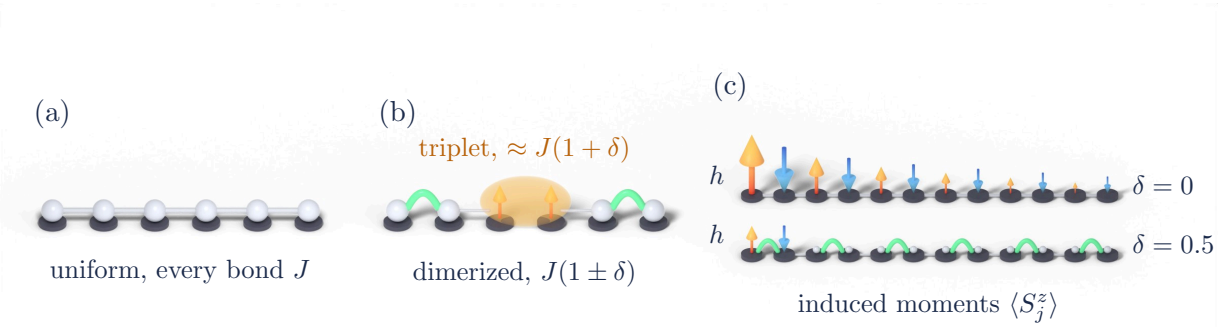



**Figure 9:** The uniform and the dimerized chain of Eq. (30). (a) At $\delta = 0$ every bond is the same and no pair of sites is singled out. (b) At finite $\delta$ the strong bonds, drawn as green singlets, alternate with weak ones, drawn thin, and the cheapest excitation is one strong bond lifted from its singlet into a triplet, here its $m = +1$ member with both spins up, which costs about $J(1 + \delta)$ and is what keeps the chain gapped. (c) A field $h$ pins the first spin of a chain of $L = 12$, and the arrows are the induced moments $\langle S^z_j \rangle$ of the exact ground state, to scale, for $h = 0.2J$: along the uniform chain at the back they alternate in sign and fall off slowly (on the open chain they add up to zero, so the far end compensates), the algebraic decay of Section 17, while along the dimerized chain in front, with $\delta = 0.5$ and its strong bonds drawn as singlets, the first singlet absorbs almost all of the response and the moment is gone by the third dimer, the exponential decay set by $\xi$; moments below 0.02 are drawn as small spheres.

This qualitative difference shows up very cleanly in how $\Delta(L)$ approaches its thermodynamic-limit value. For the gapless uniform chain, we already saw the slow, power-law approach of Eq. (24): even $L \sim 20$ is not enough to convince yourself the gap is really heading to zero. For the dimerized chain, a finite gap sets its own length scale, the correlation length $\xi \sim v_s/\Delta_\infty$, and finite-size corrections to the gap are suppressed exponentially, $\Delta(L) - \Delta_\infty \sim e^{-L/\xi}$, rather than as a power law. That is why, numerically, a dimerized chain's gap already looks converged by $L \sim 12$ to 14, while the uniform chain's gap is still visibly drifting at the same size: the presence or absence of a length scale $\xi$ is precisely what separates fast, exponential finite-size convergence from slow, power-law convergence.

**✘ Computational companion**

Section "Gap of a many-body spin chain." Compute $\Delta(L)$ for the uniform chain ($\delta = 0$) and for a dimerized chain ($\delta > 0$), and compare how quickly each one saturates as you increase $L$; check that the dimerized gap has essentially converged well before the uniform one has.

## 17 Diagnosing the gap: impurities and non-local correlators

Comparing gaps at different $L$ works, but it needs several system sizes side by side. Is there a way to diagnose gapped versus gapless behavior from a *single* calculation? Yes: look at how the system responds in space rather than how its energy responds in size.

**A magnetic impurity.** Pin a small local field on one site, $\mathcal{H} \to \mathcal{H}_{\text{Heis}} - h\, S^z_{i_0}$, and measure the induced magnetization profile $\left\langle S^z_j \right\rangle$ as a function of distance $r = |j - i_0|$ from the impurity. For the gapless uniform chain, this response decays *algebraically*, $\left\langle S^z_j \right\rangle \sim r^{-1}$, the standard hallmark of critical, scale-free correlations in a one-dimensional gapless system. For the gapped dimerized chain, the very same measurement decays *exponentially*, $\left\langle S^z_j \right\rangle \sim e^{-r/\xi}$, with the same correlation length $\xi$ that controlled the finite-size convergence of the gap in Section 16. Panel (c) of Figure 9 draws both profiles for a chain of twelve sites. So the way a localized perturbation "heals" as you move away from it already tells you, from a single system size, whether the underlying chain is gapped or gapless.

**Non-local static correlators.** We can get exactly the same information without introducing any impurity at all, by computing the connected two-point correlator

$$C(i,j) = \langle \mathbf{S}_i \cdot \mathbf{S}_j \rangle - \langle \mathbf{S}_i \rangle \cdot \langle \mathbf{S}_j \rangle \,. \tag{31}$$

In a genuine quantum magnet $\langle \mathbf{S}_i \rangle = 0$ everywhere, so this reduces to the bare correlator $\langle \mathbf{S}_i \cdot \mathbf{S}_j \rangle$; for a classical, product-state magnet, on the other hand, $C(i,j)$ vanishes identically, since a product state trivially factorizes the expectation value. So a nonzero $C(i,j)$ is itself a direct signature of entanglement beyond any mean-field description, echoing exactly the failure mode we uncovered for the Hubbard dimer in Session 1. Formally, this static correlator is tied to the impurity response of the previous paragraph by linear response (a Kubo-formula argument): the induced magnetization at $j$ from a small field at $i$ is, to leading order, proportional to $C(i,j)$ itself. It is therefore no surprise that $C(i,j)$ shows exactly the same dichotomy: power-law decay for the gapless chain, exponential decay, with a rate set by $\delta$, for the dimerized one. The bigger the dimerization, the bigger the gap, and the faster $C(i,j)$ falls off, a clean, single-snapshot diagnostic for how gapped a quantum magnet really is.

**✘ Computational companion**

Sections "Magnetization of a Heisenberg model with a magnetic impurity," "Magnetization of a dimerized Heisenberg model with a magnetic impurity," and "Non-local static correlators." Compare the decay of the induced magnetization for the uniform and dimerized chains, then compute Eq. (31) directly for several values of the dimerization and check that faster decay tracks a bigger gap.

## 18 Dynamical correlators, and a first look at spin-1

So far we have only used static, equal-time expectation values. A **dynamical spin correlator** carries strictly more information, because it resolves excitations by their energy as well as their location. Define

$$S(i,\omega) = \langle 0 |\, S^z_i \, \delta(\omega - (\mathcal{H} - E_0))\, S^z_i \,| 0 \rangle \,. \tag{32}$$

This looks abstract until we insert a resolution of the identity in the exact eigenbasis of $\mathcal{H}$, $\mathbb{1} = \sum_n |n\rangle \langle n|$, which turns Eq. (32) into the Lehmann representation

$$S(i,\omega) = \sum_n |\langle n|\, S^z_i \,|0\rangle|^2 \; \delta(\omega - (E_n - E_0)) \,. \tag{33}$$

Read this as a spectroscopy experiment: it says that if you inject a spin flip at site $i$ with energy $\omega$, you get a nonzero response exactly when there exists an eigenstate $|n\rangle$ at energy $E_n = E_0 + \omega$ that is actually connected to the ground state by a local spin flip, weighted by how big that overlap $\langle n| S_i^z |0\rangle$ is.

For the spin-$\frac{1}{2}$ uniform chain, $S(i,\omega)$ shows a sequence of discrete peaks, spread out over the whole chain, separated by gaps: exactly the particle-in-a-box quantization we anticipated in Eq. (24). As $L$ grows, these peaks soften toward lower $\omega$, tracking the shrinking many-body gap of Section 12, and more of them appear as the low-energy spectrum becomes denser.

**A first look at spin-1.** If we redo this same calculation for a spin-1 Heisenberg chain instead of spin-$\frac{1}{2}$, the profile of $S(i,\omega)$ looks qualitatively different. This is because the two cases sit in different phases: the spin-$\frac{1}{2}$ chain is gapless in the thermodynamic limit, as we have seen throughout this session, while the spin-1 chain is what is called a **topological quantum magnet**, with a full gap in the bulk but gapless, fractionalized spin-$\frac{1}{2}$ excitations pinned to the two ends of an open chain. The underlying reason half-integer and integer spin chains behave so differently is the Lieb–Schultz–Mattis theorem, which forbids a translationally invariant, SU(2)-symmetric chain with half-integer spin per site from having a unique, gapped ground state, while placing no such restriction on integer spin. In our small finite-size dynamical correlators we can already see hints of this: a low-energy excitation that lives mostly at the two edges and gets progressively softer as $L$ grows, sitting well below a much larger, roughly size-independent bulk gap. Resolving this fractionalization cleanly, however, needs system sizes well beyond what exact diagonalization can reach, which is exactly the job tensor networks will do for us in Session 4.

**✘ Computational companion**

Section "Dynamical correlators." Compute $S(i,\omega)$ for the spin-$\frac{1}{2}$ Heisenberg chain, make the system smaller and watch the discrete peaks move, and add a magnetic field to see how the Zeeman shifts of Section 14 show up directly in the correlator's peak positions.

## 19 Quasiparticles for the ground state: Abrikosov fermions and spinon mean-field theory

Everything we have computed so far treats the ground state and its excitations as intrinsically many-body objects, with no single-particle description in sight. But we can ask the same question we asked in Session 1 about the Hubbard model: is there some auxiliary single-particle Hamiltonian whose spectrum reproduces, at least qualitatively, the low-energy excitations of $\mathcal{H}_{\text{Heis}}$? The answer is yes, and it comes from representing the quantum spin operators themselves in terms of auxiliary fermions.

**The Abrikosov (auxiliary) fermion representation.** Introduce a fermion $f_{i\sigma}$ per site per spin flavor and write

$$S_i^\alpha = \frac{1}{2} \sum_{\sigma\sigma'} f_{i\sigma}^\dagger \, \tau_{\sigma\sigma'}^\alpha \, f_{i\sigma'}, \qquad \alpha = x, y, z, \tag{34}$$

with $\tau^\alpha$ the Pauli matrices, subject to the local constraint $\sum_\sigma f^\dagger_{i\sigma} f_{i\sigma} = 1$ on every site: exactly one auxiliary fermion, no more, no fewer. This constraint is not optional bookkeeping, it is what makes the representation exact rather than approximate: an unconstrained spinful fermion per site has four states, empty, singly occupied (↑ or ↓), and doubly occupied, but the constraint removes the empty and doubly occupied configurations and leaves exactly the two singly-occupied states $\{|\uparrow\rangle, |\downarrow\rangle\}$, which is precisely the spin-$\frac{1}{2}$ Hilbert space we started with. These auxiliary fermions are emphatically not electrons; they never hop in or out of the constrained subspace, they simply parameterize the same two-dimensional local Hilbert space in fermionic language. Panel (a) of Figure 10 draws the four states of one site and the two the constraint keeps.

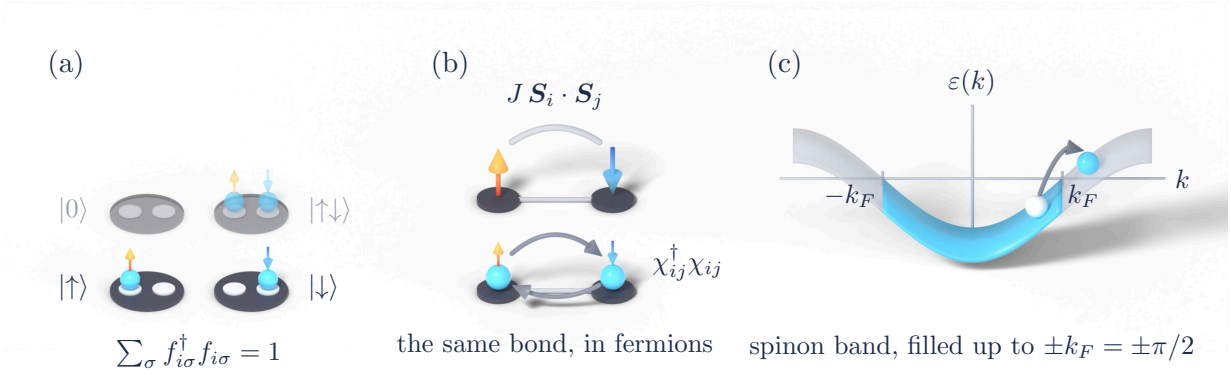



**Figure 10:** Spins as Abrikosov fermions, Eq. (34). (a) One site as two fermion slots, ↑ on the left and ↓ on the right, and its four states: the empty and the doubly occupied ones, ghosted, are removed by the constraint, and the two that remain are exactly the spin-$\frac{1}{2}$. (b) One bond twice: at the back the spin exchange, at the front the same two sites in fermion language, where Eq. (35) trades the exchange for the bond operator, the up fermion hopping across the bond one way and the down fermion the other way, which leaves one fermion per site and the two spins flipped. (c) The spinon band $\varepsilon(k) = -J\chi\cos k$ of the uniform chain, from Eq. (36), filled up to the Fermi points $k_F = \pm\pi/2$, with a particle-hole pair at the right Fermi point, a hole just below $k_F$ and a spinon just above it: this is the excitation that costs arbitrarily little energy, the gapless spinons behind Eq. (24).

Substituting Eq. (34) into $\mathbf{S}_i \cdot \mathbf{S}_j$ and using the constraint on each site, one finds the standard identity

$$\mathbf{S}_i \cdot \mathbf{S}_j = -\frac{1}{2}\,\chi^\dagger_{ij}\chi_{ij} + \frac{1}{4} n_i n_j, \qquad \chi_{ij} \equiv \sum_\sigma f^\dagger_{i\sigma} f_{j\sigma}, \tag{35}$$

where $\chi_{ij}$ is the fermionic **bond operator**, drawn in panel (b) of Figure 10. This turns the quartic spin exchange into a quartic fermion interaction, exactly the kind of term that Session 1 taught us how to handle: decouple it at the mean-field level. Introduce the self-consistent bond amplitude $\chi_{ij} \equiv \left\langle \sum_\sigma f^\dagger_{i\sigma} f_{j\sigma} \right\rangle$ and Lagrange multipliers $\lambda_i$ to enforce the occupancy constraint on average, and Eq. (35) collapses to an effective free-fermion, or **spinon**, Hamiltonian,

$$\mathcal{H}_{\mathrm{MF}} = -\sum_{ij} \frac{J_{ij}\chi_{ij}}{2} \sum_\sigma f^\dagger_{i\sigma} f_{j\sigma} + \mathrm{h.c.} + \sum_i \lambda_i \left(n_i - 1\right). \tag{36}$$

This is precisely the same kind of self-consistency loop as Session 1's Hubbard mean-field theory: guess $\chi_{ij}$, diagonalize the resulting single-particle Hamiltonian, recompute $\chi_{ij}$ from the filled spinon orbitals, and repeat until convergence.

The payoff is that Eq. (36) is just a tight-binding model, and we already know how to read its spectrum. For the uniform chain, the self-consistent $\chi_{ij} \equiv \chi$ is the same on every bond, giving a simple cosine spinon band $\varepsilon(k) = -J\chi \cos k$, gapless Fermi points, and hence gapless spin excitations (panel (c) of Figure 10), matching everything we found directly from exact diagonalization in Section 12. For the dimerized chain of Eq. (30), the bond amplitude inherits the alternation of $J_i$, giving a dimerized, Su-Schrieffer-Heeger-like tight-binding chain for the spinons, whose spectrum opens a gap set by the difference between the strong- and weak-bond hopping amplitudes, mirroring the many-body gap of Section 16. The same construction generalizes to two dimensions, where an analogous mean-field ansatz on, say, the honeycomb lattice produces gapless **Dirac spinons**, the auxiliary-fermion language behind many candidate quantum spin liquids. We will not compute spinon bands in the notebook, but the picture is worth keeping, since it is the language in which most proposals for quantum spin liquids are written.

## Closing summary

- Exact diagonalization means writing the Heisenberg Hamiltonian as a matrix on the full $2^L$ product-state basis and diagonalizing it directly, with no product-state restriction anywhere; the dimer's exact singlet ($-\frac{3}{4}J$) and triplet ($\frac{1}{4}J$) energies are the smallest worked example of this.
- The many-body gap $\Delta(L) = E_1 - E_0$ shrinks with system size; a power-law decay signals a gapless chain (the uniform Heisenberg chain, $\Delta \sim \pi^2 J/2L$), while saturation to a finite value, with exponentially fast finite-size convergence, signals a gapped chain (the dimerized chain).
- The XXZ model interpolates continuously between the classical, symmetry-broken Ising limit and the quantum, entangled Heisenberg limit; the finite-size splitting of the two lowest states already reveals the location of the exact, thermodynamic-limit phase transition.
- Total spin $\mathbf{S}^2_{\text{tot}}$ and $S^z_{\text{tot}}$ are exactly conserved quantities of the Heisenberg Hamiltonian, for any $L$; they block-diagonalize the exponentially large Hilbert space into much smaller sectors, they guarantee that a uniform field only splits levels linearly rather than mixing them, and they are exactly what pushes exact diagonalization out to $L \lesssim 20$ spins.
- A field-driven sequence of level crossings between successive total-spin sectors produces the magnetization staircase of a finite chain; impurity responses and non-local static correlators, decaying algebraically when gapless and exponentially when gapped, are single-snapshot diagnostics for exactly the same physics as the gap scaling.
- Dynamical correlators resolve excitations in both space and energy via the Lehmann representation; comparing spin-$\frac{1}{2}$ and spin-1 chains gives our first hint of the fractionalized, topological physics of the spin-1 chain, developed properly with tensor networks in Session 4.
- Representing spins as constrained Abrikosov fermions and mean-field decoupling the exchange term gives an effective free-fermion, spinon, description of the excitations: gapless spinons for the uniform chain, gapped spinons for the dimerized chain, using exactly the same self-consistency machinery as Session 1's Hubbard mean-field theory.
- All of this is numerically exact, but the underlying cost is still exponential in $L$: even with every symmetry we can exploit, exact diagonalization caps out around $L \sim 20$ spins, which is precisely the wall that motivates the matrix-product-state methods of Session 4.

# PHYS-E0510 – Advanced Computational Methods in Physics D

Session 3: Exact Diagonalization in the Fermionic Basis

**Companion notebook:** `quantum_interacting_fermions.ipynb` **Prerequisites:** Session 1 (second quantization, the Hubbard model, mean-field theory); Session 2 (exact diagonalization, though we rebuild everything we need directly in the fermionic language here)

## Contents and goals

Session 1 built the Hubbard model from second quantization and solved it at the mean-field level, and it ended on a warning: mean-field theory can fail badly, even on the smallest possible example, the Hubbard dimer at half filling. Session 2 picked up right at that failure point and solved the resulting Heisenberg spin model exactly, with no product-state restriction at all, by diagonalizing directly in the spin basis. In this session we go back one step further and diagonalize the *original* fermionic Hamiltonian exactly, working in the fermionic Fock space itself rather than first mapping onto spins. This is not just a change of language: it lets us see things a spin model simply cannot represent, charge physics, a real chemical potential, a genuine insulator-versus-metal distinction, and a numerically exact test of Session 1's mean-field magnetization prediction. We will find that mean-field theory gets the Hubbard chain's magnetization qualitatively wrong, not just quantitatively wrong, which is exactly the kind of failure that motivates doing things exactly whenever we can afford to. We close by watching spin and charge decouple from each other as the interaction grows, a many-body effect called spin-charge separation. Session 5 will pick this exact fermionic program back up and push it to far larger system sizes using tensor networks.

## Learning outcomes

By the end of this session, you should be able to:

- Diagonalize interacting fermionic Hamiltonians directly in Fock space and explain how the Hilbert-space dimension scales with system size and spin.
- Use ground-state degeneracy, a chemical-potential sweep, and the charging energy to identify a charge-density-wave transition and control the ground-state particle number.
- Test Session 1's mean-field magnetization prediction against the exact Hubbard result and explain the disagreement using Lieb's theorem.
- Diagnose spin-charge separation from static and dynamical correlators, and connect the strong-coupling limit to the Heisenberg model of Session 2.

## 20 From spins back to fermions: the many-body Fock space

Let us start by rebuilding the same many-body machinery we used for spins in Session 2, but now directly in terms of fermionic creation and annihilation operators. Recall from Session 1 that a single-particle Hamiltonian acting on $L$ orbitals is just an $L \times L$ matrix, and diagonalizing it costs polynomial time. The moment we add any quartic (interaction) term, that is no longer true: we

have to diagonalize $\mathcal{H}$ in the full many-body Fock space, spanned by all the ways of distributing electrons among the $L$ orbitals, and that space grows exponentially in $L$.

Let us count this explicitly, starting from the smallest case. Take two spinless sites, labeled 0 and 1. The many-body basis has exactly four states,

$$|\Omega\rangle\,, \qquad c_0^\dagger\,|\Omega\rangle\,, \qquad c_1^\dagger\,|\Omega\rangle\,, \qquad c_0^\dagger c_1^\dagger\,|\Omega\rangle\,, \tag{37}$$

the empty state, the two singly-occupied states, and the doubly-occupied state, so $\mathcal{H}$ is a $4\times 4$ matrix in this basis. Add a third site and the counting becomes $1+3+3+1=8$ states (one empty configuration, three with one electron, three with two electrons, one fully filled), which is $2^3$. In general, for $L$ spinless orbitals the Fock space has dimension $2^L$: every orbital is independently either empty or filled, so we get one binary choice per orbital. This is worth contrasting with the single-particle sector alone, spanned by $c_i^\dagger\,|\Omega\rangle$ for $i=1,\ldots,L$, which only has dimension $L$: restricting to one electron is precisely what makes single-particle physics polynomially cheap, and dropping that restriction is precisely what makes many-body physics exponentially expensive. Panel (b) of Figure 11 draws the eight states of three orbitals.

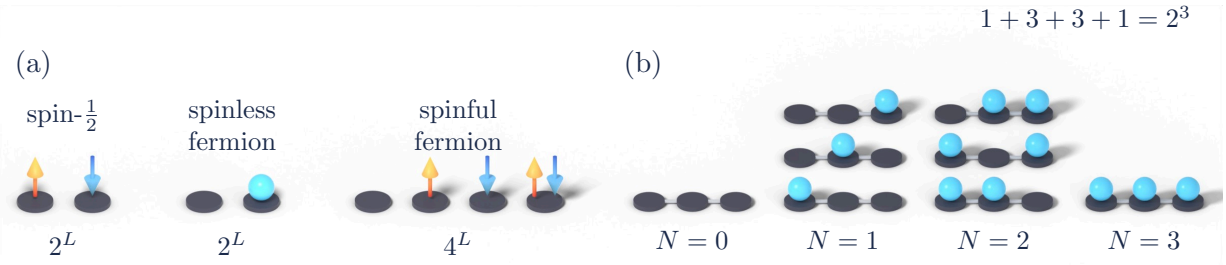



**Figure 11:** The local Hilbert spaces and the many-body basis built from them. (a) The states of a single site: two for a spin-$\frac{1}{2}$, up and down, two for a spinless fermion, empty and occupied, and four for a spinful fermion, empty, up, down and doubly occupied, so a chain of $L$ sites has $2^L$, $2^L$ and $4^L$ many-body states. (b) The complete many-body basis of three spinless orbitals, with a ball on every occupied orbital and the states grouped by their particle number $N$. Reading across, there is one empty configuration, three with a single electron, three with two electrons and one completely filled state, for $1+3+3+1=8=2^3$ basis states in total. Every additional orbital doubles this count, because each orbital is independently either empty or filled, and that doubling is exactly the exponential wall that separates many-body from single-particle physics.

If the fermions carry spin, each site has *four* local states, $|0\rangle$, $|\uparrow\rangle$, $|\downarrow\rangle$, $|\uparrow\downarrow\rangle$, exactly as in Session 1. Equivalently, we can think of a spinful chain as $2L$ independent spinless orbitals (spin becomes just another orbital index), so the Fock space has dimension $2^{2L}=4^L$: twice the exponent of the spinless case, since every site now contributes two binary choices instead of one, which is what panel (a) of Figure 11 puts side by side with the spin and the spinless fermion. Note how this lines up with the spin chains of Session 2: an $L$-site spin-$\frac{1}{2}$ chain also has a $2^L$-dimensional Hilbert space, exactly matching the spinless-fermion count. The reason for this is that a rigorous mapping (Jordan-Wigner, which we will meet properly in Session 5) sends spinless fermions to spin-$\frac{1}{2}$ degrees of freedom one-for-one. For now, the practical upshot is simply this:

- solving an interacting **spinless** fermionic chain costs exactly as much as solving a quantum **spin** chain of the same length, $\sim 2^L$;
- solving an interacting **spinful** fermionic chain, like the Hubbard model, costs the square of that, $\sim 4^L$, so at fixed computational budget we can only reach about half the system size.

Just as in Session 2, once we have $\mathcal{H}$ written as an explicit $2^L \times 2^L$ (or $4^L \times 4^L$) matrix in this Fock basis, the ground state is simply its lowest eigenvector, a set of $2^L$ (or $4^L$) complex coefficients multiplying each basis configuration, and any expectation value we want, $\langle n_i \rangle$, $\langle S_i^z \rangle$, correlators, spectra, follows by ordinary linear algebra on that eigenvector. No approximation, no product-state restriction: this is the entire promise of exact diagonalization, now made good on directly in the fermionic language.

**✘ Computational companion**

Sections "Creating a fermion_chain object," "Creating a Hamiltonian," "Computing the ground state energy," and "Computing expectation values." Build a small fermionic chain object, write down a many-body Hamiltonian directly in terms of the fermionic operators, and extract its ground state energy and a couple of expectation values. This is the fermionic analogue of what you did with spin operators in Session 2, so compare the two interfaces side by side.

## 21 The spinless-fermion model: a charge-density-wave transition, computed exactly

Let us put this machinery to work on the simplest interacting model we have: the spinless nearest-neighbor chain from Session 1,

$$\mathcal{H}_{\mathrm{CDW}} = -t \sum_i \left( c_i^\dagger c_{i+1} + \mathrm{h.c.} \right) + V \sum_i \left( n_i - \tfrac{1}{2} \right) \left( n_{i+1} - \tfrac{1}{2} \right), \qquad n_i \equiv c_i^\dagger c_i . \tag{38}$$

This is exactly Session 1's $\mathcal{H}_{\mathrm{CDW}}$, just with $n_i \to n_i - \frac{1}{2}$ inside the interaction, a purely bookkeeping shift: expanding it out just adds a constant plus an automatic chemical potential of $-V/2$ per site, tuned so that the model sits exactly at half filling without us needing to add and adjust a separate $\mu$ by hand.

Two limits of Eq. (38) are easy to solve outright. At $V = 0$ we just have the free tight-binding chain of Session 1: a single cosine band, filled up to the Fermi points, a metal with a unique, non-degenerate ground state. At $V \to \infty$ we can drop the hopping altogether, and the interaction alone is minimized by the classical alternating pattern $1, 0, 1, 0, \ldots$: exactly one electron every other site. There are exactly *two* such states, related by shifting the pattern by one site, and they are degenerate: this is the charge-density wave, and its defining signature is that the ground state is twofold degenerate, in sharp contrast with the unique metallic ground state at $V = 0$. Figure 12 draws the two limits side by side.

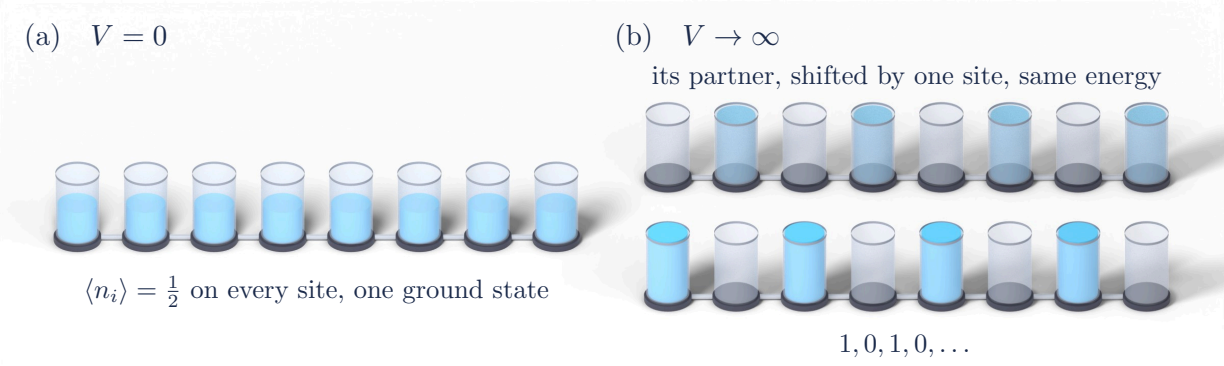



**Figure 12:** The two solvable limits of Eq. (38), with the occupation of every site drawn as the filling of a cup. (a) At $V = 0$ the chain is a free tight-binding metal at half filling, with the same average density $\frac{1}{2}$ on every site and a unique ground state. (b) As $V \to \infty$ the hopping becomes irrelevant and the interaction alone selects the alternating $1, 0, 1, 0, \ldots$ pattern, in front. Shifting that pattern by one site produces a second state of identical energy, drawn behind it as a ghost, and this twofold degeneracy is the defining signature of the charge-density wave.

This is precisely what exact diagonalization lets us watch happen on a finite chain. Diagonalize Eq. (38) at fixed $L$ and plot the low-lying eigenvalues, measured relative to the ground state, as a function of $V$. At small $V$ we see a unique ground state followed by a small gap to a dense set of excited states: this gap is a finite-size artifact, roughly the level spacing $\sim t/L$ of the underlying free-fermion band, and it visibly shrinks as we go from $L = 12$ to $L = 14$, exactly as we would expect if it is destined to close in the thermodynamic limit and leave behind a continuum. As $V$ grows, one particular excited state peels away from that continuum and drops steadily toward the ground state, becoming exactly degenerate with it as $V \to \infty$: this is the CDW partner state, and unlike the metallic gap, this near-degeneracy does *not* close by adding more excited states underneath it; instead a large gap opens above the (now twofold) ground state, growing asymptotically as $\Delta \sim V$. That linear growth has a simple interpretation: at large $V$, the cheapest way to excite the system is to introduce a single domain wall into the alternating pattern, a local defect where the charge order runs into itself, and each domain wall costs an energy of order $V$ regardless of the size of the system, exactly the extensive, size-independent gap we see numerically.

Let us compare this directly with Session 1's mean-field treatment of the very same model. There, the self-consistent gap also opened continuously as $V$ increased and also grew linearly at large $V$, and here exact diagonalization confirms both of those features quantitatively. The reason for this is that the CDW breaks a *discrete* symmetry (translation by one site, a $\mathbb{Z}_2$ symmetry), and discrete symmetry breaking of this kind is allowed even in a one-dimensional quantum system at zero temperature. Keep this in mind, because in Section 23 we will meet a case where mean-field theory tries to break a *continuous* symmetry, and there the story changes completely.

**✘ Computational companion**

Sections "Many-body energies of an interacting fermionic model" and "Towards interaction-induced symmetry breaking." Plot the low-lying many-body spectrum of the spinless chain as a function of $V$ for two system sizes, and identify by eye where the first excited state collapses onto the ground state, this is the CDW transition. Then look at how the large-$V$ gap scales with $V$ and check the linear

behavior we derived above.

## 22 Chemical potential, conserved quantum numbers, and the charging energy

So far we have kept the electron number fixed at half filling. Let us now add a chemical potential explicitly,

$$\mathcal{H}(\mu) = \mathcal{H}_{\mathrm{CDW}} - \mu \sum_i n_i \equiv \mathcal{H}_{\mathrm{CDW}} - \mu N, \tag{39}$$

where $N = \sum_i n_i$ is the total particle number operator. The key structural fact here is that $[\mathcal{H}_{\mathrm{CDW}}, N] = 0$: hopping moves an electron from one site to another without changing how many there are in total, and the interaction only counts electrons, so it commutes with $N$ trivially too. Since $\mathcal{H}_{\mathrm{CDW}}$ and $N$ commute, they share a common eigenbasis, which means adding the $-\mu N$ term does not change a single eigenvector of $\mathcal{H}_{\mathrm{CDW}}$: it only shifts each eigenenergy by $-\mu N$, where $N$ is that eigenstate's own, fixed particle number. Concretely, this means every eigenvalue of $\mathcal{H}(\mu)$ traces out a straight line as a function of $\mu$, with slope $-N$ set entirely by that state's particle number relative to the ground state's.

This has an immediate and useful consequence: because different excited states carry different particle numbers, they have different slopes, and as we sweep $\mu$ these lines cross. Right at a crossing, the identity of the *ground state* itself changes: below the crossing the lowest-energy state has $N$ electrons, above it the lowest-energy state has $N \pm 1$. In other words, $\mu$ is a knob that directly controls the electron number of the ground state, exactly as a gate voltage does in a real quantum dot or nanostructure.

Let us make this quantitative. Define $E(N)$ as the lowest eigenvalue of $\mathcal{H}_{\mathrm{CDW}}$ restricted to the $N$-particle sector. The state with $N$ particles is the ground state of $\mathcal{H}(\mu)$ precisely when $E(N) - \mu N$ is lower than $E(N \pm 1) - \mu(N \pm 1)$, which rearranges into a window,

$$E(N) - E(N-1) \;<\; \mu \;<\; E(N+1) - E(N). \tag{40}$$

The width of this stability window is exactly the **charging energy**,

$$U_c(N) \equiv \big[E(N+1) - E(N)\big] - \big[E(N) - E(N-1)\big], \tag{41}$$

the discrete second derivative of the ground-state energy with respect to particle number. It measures how much harder it gets to add one more electron than it was to add the previous one, and it is exactly the quantity that produces the flat plateaus you see when you plot the ground-state occupation $\langle N \rangle$ against $\mu$: on a plateau, $\mu$ can move freely without changing $N$, because it has not yet exceeded the addition energy in Eq. (40). As $V$ grows, the interaction makes it costlier to double up charge on already-crowded sites, $U_c$ grows with it, and the plateaus in the $\langle N \rangle$-versus-$\mu$ curve widen accordingly, exactly what the notebook shows you when you compare a weakly and a strongly interacting chain. Figure 13 draws the staircase for a ring of eight sites.

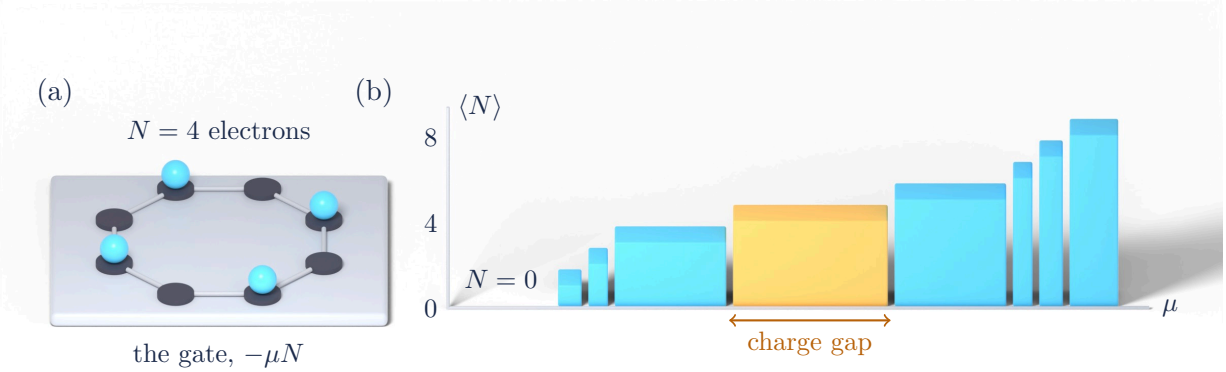



**Figure 13:** The chemical potential as a gate on the electron number. (a) A ring of eight sites standing on a gate, the term $-\mu N$ of Eq. (39), here holding $N = 4$ electrons, half filling. (b) The particle number of the ground state against $\mu$ for the ring at $V = 4t$, computed exactly from Eq. (38), a ring since the ends of an open chain blur the plateaus: every block but the last, which goes on to arbitrarily large $\mu$, is a plateau, the window of Eq. (40) in which $N$ stays put, and its width is the charging energy of Eq. (41). The widest plateau sits at half filling, about $3.8\,t$ against less than $3\,t$ for its neighbors, and that width is the charge gap of the charge-density wave.

**✘ Computational companion**

Sections "Energies of an interacting fermionic model as a function of the chemical potential," "Quantum numbers in the interacting fermionic model," "Level crossings in the interacting fermionic model," and "Electronic density as a function of the chemical potential." Track the particle number of each many-body eigenstate as $\mu$ is swept, watch the ground state switch particle number at level crossings, and confirm that the width of each occupation plateau grows with the interaction strength, exactly the charging-energy statement of Eq. (41).

## 23 The exact Hubbard magnetization: mean-field theory fails again

Now let us bring spin back and repeat this exercise for the Hubbard chain,

$$\mathcal{H}_{\text{Hub}} = -t \sum_{i,\sigma} \left( c^{\dagger}_{i\sigma} c_{i+1\sigma} + \text{h.c.} \right) + U \sum_{i} \left( n_{i\uparrow} - \tfrac{1}{2} \right) \left( n_{i\downarrow} - \tfrac{1}{2} \right), \tag{42}$$

which, just as in Eq. (38), differs from Session 1's Hubbard Hamiltonian only by the same bookkeeping shift, an automatic chemical potential that pins us at half filling.

Now here is the puzzle: if we diagonalize $\mathcal{H}_{\text{Hub}}$ exactly and compute the local magnetization $m_i \equiv \langle n_{i\uparrow} \rangle - \langle n_{i\downarrow} \rangle$ site by site, we find $m_i = 0$ *identically*, on every site, for every $U$, no matter how strong the interaction is. This flatly contradicts what Session 1's mean-field theory told us to expect: there, the self-consistency loop converges, for any $U > 0$ on a bipartite half-filled chain, to a finite staggered moment, $m_i = (-1)^i m$ with $m$ growing toward saturation as $U \to \infty$. Exact diagonalization and mean-field theory are not just quantitatively different here, they disagree about whether the ground state is magnetic at all. Figure 14 puts the two answers side by side, and shows where the antiferromagnetism went in the exact one: into the correlations.

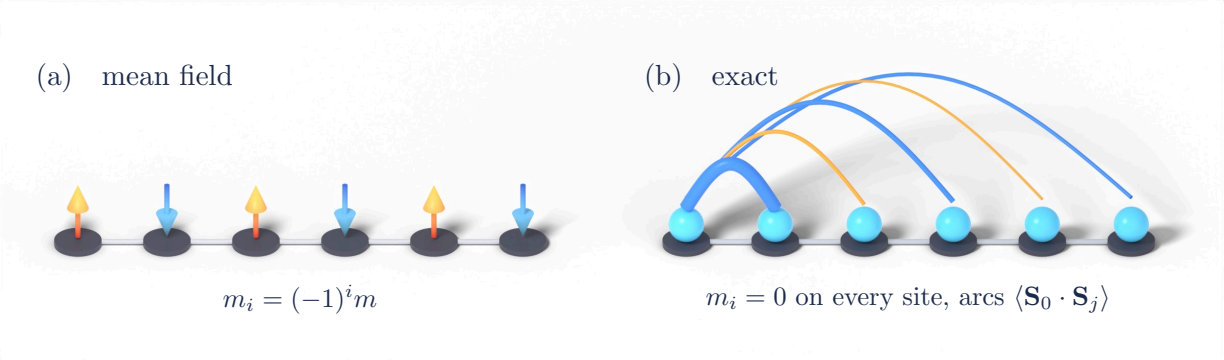



**Figure 14:** Two answers for the half-filled Hubbard chain of Eq. (42). (a) Mean-field theory finds a finite staggered moment $m_i = (-1)^i m$, an arrow on every site. (b) The exact ground state of a chain of six sites at $U = 4t$ has no moment on any site, so every electron is drawn without an arrow, and the antiferromagnetism lives instead in the correlation $\langle \mathbf{S}_0 \cdot \mathbf{S}_j \rangle$ between the first site and site $j$, drawn as an arc whose thickness is its size, blue where it is negative and orange where it is positive, so every arc takes the color of the arrow that panel (a) puts on site $j$. The sign alternates with distance, $-0.54$ for the neighbor and $+0.12$, $-0.17$, $+0.06$, $-0.11$ further out, which is the staggered pattern of panel (a) carried by correlations rather than by moments.

The reason for this is an exact symmetry argument. The Hamiltonian Eq. (42) is invariant under any global spin rotation, so it commutes with the total spin operators, $[\mathcal{H}_{\text{Hub}}, \mathbf{S}^2] = [\mathcal{H}_{\text{Hub}}, S_z^{\text{tot}}] = 0$, and its eigenstates can be labeled by a definite total spin $S$. **Lieb's theorem** pins this down completely for the repulsive Hubbard model on any bipartite lattice at half filling: the ground state is unique and has total spin $S = \frac{1}{2}|N_A - N_B|$, where $N_A, N_B$ are the sizes of the two sublattices. On our chain the sublattices are equal in size, so $S = 0$ exactly, a spin singlet, for every $U$ and every chain length. A rotationally invariant, unique, total-spin-zero state simply cannot have a nonzero expectation value of $S_i^z$ on any individual site, since a spin rotation would map it to an equally valid ground state with the opposite sign of $m_i$, and uniqueness forbids two different ground states. Mean-field theory, by construction, searches only over single Slater determinants, and a Slater determinant with staggered spin *explicitly* breaks spin-rotation symmetry from the very first guess: it can never represent a rotationally invariant, entangled singlet, exactly the same structural failure we met with the Hubbard dimer in Session 1.

**Turning on an explicit symmetry-breaking field.** We can watch this failure mode directly by interpolating between the exact model and the mean-field ansatz. Add an explicit staggered field,

$$\mathcal{H}(h) = \mathcal{H}_{\text{Hub}} + h \sum_i (-1)^i \left(n_{i\uparrow} - n_{i\downarrow}\right), \tag{43}$$

which is precisely the single-particle term that mean-field theory would have generated on its own had it found a self-consistent staggered solution (compare Session 1's $\mathcal{H}_{\text{MF}}$). At $h = 0$ we recover the exact result, $m = 0$. As $h$ increases from zero, $m(h)$ grows *continuously* rather than jumping to a finite value: quantum fluctuations, encoded in the entangled singlet ground state, are actively fighting the ordering field, and only once $h$ is large enough to overwhelm those fluctuations does $m(h)$ approach the mean-field saturation value. This is exactly the standard diagnostic for spontaneous symmetry breaking: the true order parameter is defined as $m \equiv \lim_{h\to 0^+} \lim_{L\to\infty} m(h, L)$, and the order of these two limits matters enormously. At any finite

$L$, symmetry forces $m(h{=}0, L) = 0$ exactly, so we must send $L \to \infty$ *first*. Even having done that, for the one-dimensional Heisenberg antiferromagnet that this model reduces to at large $U$ (Section 24), this limit still gives zero: one dimensional quantum antiferromagnets do not sustain true long-range magnetic order, only quasi-long-range, power-law spin correlations, which is exactly the gaplessness we will meet again in Section 25. This is the reason mean-field theory fails so badly here: it is not just missing quantitative corrections, it is predicting an ordered phase that, in one dimension, does not exist.

| | **CDW (Section 21)** | **Hubbard magnetism (this section)** |
|---|---|---|
| Symmetry broken | discrete, $\mathbb{Z}_2$ (one-site translation) | continuous, SU(2) (spin rotation) |
| Mean-field prediction | finite $\delta n$, growing continuously with $V$ | finite staggered $m$, growing continuously with $U$ |
| Exact 1D result | agrees qualitatively: order onset, gap $\sim V$ | disagrees: $m = 0$ identically, unique spin singlet |

**Table 2:** The two symmetry-breaking scenarios met so far, contrasted: which symmetry is broken, what mean-field theory predicts, and whether the exact one-dimensional result agrees.

Table 2 lines up the two symmetry-breaking scenarios met so far side by side: mean-field theory gets the discrete CDW transition qualitatively right, but gets the continuous Hubbard magnetization badly wrong.

**✘ Computational companion**

Sections "Magnetization of an interacting Hubbard model" and "Magnetization of a Hubbard model with staggered magnetization." First confirm $m_i = 0$ at every site of the exact Hubbard ground state, however large $U$ is. Then switch on the staggered field of Eq. (43) and watch $m$ grow smoothly from zero, and check how the field strength needed to saturate $m$ changes as you increase $U$.

## 24 Recovering the Heisenberg model: an exact worked example

Let us now directly verify, by exact diagonalization, the superexchange result $J \sim t^2/U$ that Session 1 derived by degenerate perturbation theory. Take the Hubbard dimer, $L = 2$, at half filling, and work in the sector with zero total $S_z$, spanned by the four states

$$|1\rangle = c^\dagger_{0\uparrow} c^\dagger_{1\downarrow} |\Omega\rangle\,, \quad |2\rangle = c^\dagger_{0\downarrow} c^\dagger_{1\uparrow} |\Omega\rangle\,, \quad |3\rangle = c^\dagger_{0\uparrow} c^\dagger_{0\downarrow} |\Omega\rangle\,, \quad |4\rangle = c^\dagger_{1\uparrow} c^\dagger_{1\downarrow} |\Omega\rangle\,, \tag{44}$$

the same singly- and doubly-occupied configurations we met in Session 1. Hopping connects a singly occupied state to a doubly occupied one but never $|1\rangle$ directly to $|2\rangle$, or $|3\rangle$ directly to $|4\rangle$, so in this basis $\mathcal{H}_{\text{Hub}}$ (up to the constant $U/2$ shift from Eq. (42): the singly-occupied states sit at $-U/2$ and the doubly-occupied ones at $+U/2$, so adding $U/2$ uniformly reproduces the diagonal below) is

$$\mathcal{H}_{\text{Hub}} \longrightarrow \begin{pmatrix} 0 & 0 & -t & t \\ 0 & 0 & t & -t \\ -t & t & U & 0 \\ t & -t & 0 & U \end{pmatrix} \quad \text{in the basis } \{|1\rangle\,, |2\rangle\,, |3\rangle\,, |4\rangle\}. \tag{45}$$

This block diagonalizes cleanly if we pass to symmetric and antisymmetric combinations, $|a\rangle = \frac{1}{\sqrt{2}}(|1\rangle + |2\rangle)$ (the $S_z = 0$ member of the triplet), $|b\rangle = \frac{1}{\sqrt{2}}(|1\rangle - |2\rangle)$ (the singlet), and likewise $|c\rangle = \frac{1}{\sqrt{2}}(|3\rangle + |4\rangle)$, $|d\rangle = \frac{1}{\sqrt{2}}(|3\rangle - |4\rangle)$ for the doubly-occupied states. A short calculation shows $|a\rangle$ and $|c\rangle$ are already exact eigenstates,

$$\mathcal{H}_{\text{Hub}}\,|a\rangle = 0 \cdot |a\rangle\,, \qquad \mathcal{H}_{\text{Hub}}\,|c\rangle = U\,|c\rangle\,, \tag{46}$$

while $|b\rangle$ and $|d\rangle$ mix into a $2 \times 2$ block,

$$\mathcal{H}_{\text{Hub}}\big|_{\{b,d\}} = \begin{pmatrix} 0 & -2t \\ -2t & U \end{pmatrix} \implies E_{\pm} = \frac{U \pm \sqrt{U^2 + 16t^2}}{2}. \tag{47}$$

The lower eigenvalue, $E_-$, is the exact ground state, and comparing it against the exactly degenerate $S_z = \pm 1$ states $|\uparrow\uparrow\rangle$, $|\downarrow\downarrow\rangle$ (both sitting at energy 0, just like $|a\rangle$, since Pauli exclusion blocks all hopping for parallel spins), we see immediately that the full ground state *is* the total spin singlet ($S = 0$), sitting strictly below the $S = 1$ triplet at energy 0, exactly as Lieb's theorem guaranteed in Section 23. The exact singlet-triplet gap is therefore

$$\Delta_{\text{s-t}} = 0 - E_- = \frac{\sqrt{U^2 + 16t^2} - U}{2} \xrightarrow[U \gg t]{} \frac{4t^2}{U}, \tag{48}$$

which reproduces, from a completely independent, exact calculation, not just the $t^2/U$ scaling but the exact same numerical prefactor that Session 1 obtained perturbatively for the effective coupling $J = 4t^2/U$ of $\mathcal{H}_{\text{eff}} = J\,\mathbf{S}_1 \cdot \mathbf{S}_2$: expanding Eq. (48) to leading order in $t/U$ gives $\Delta_{\text{s-t}} \to 4t^2/U$, digit for digit. This is a satisfying check: two completely different techniques, one perturbative and spin-only, one exact and fermionic, land on exactly the same physics, not just the same scaling.

**Excitations as magnons.** We can probe the excited states further, exactly as in Session 2, by adding a Zeeman field $-h\,S_z^{\text{tot}}$. Since $[\mathcal{H}_{\text{Hub}}, S_z^{tot}] = 0$, this again leaves every eigenvector untouched and only shifts eigenenergies linearly in $h$, by an amount set by each state's own $S_z^{\text{tot}}$. The three states we called the triplet split apart linearly, into $S_z^{\text{tot}} = -1, 0, +1$ branches, while the singlet ground state, having $S_z^{\text{tot}} = 0$, does not move at all. This is precisely the field dependence of the Heisenberg dimer from Session 2, and the physical interpretation is the same either way: the lowest excitations above a singlet ground state are spin-1 objects, magnons, and their Zeeman splitting is a direct, numerically exact fingerprint of the underlying spin physics living inside the fermionic Hubbard model. The same mapping goes through, qualitatively, on longer chains: the exact fermionic spectrum at large $U/t$ reproduces the exact spin spectrum of a Heisenberg chain with $J_{ij} \sim t_{ij}^2/U$, which is exactly what it means to say the two models are equivalent at strong coupling.

**✘ Computational companion**

Section "Connection between the Hubbard and Heisenberg models." Diagonalize the Hubbard model at increasing $U/t$ and compare its low-energy many-body spectrum directly against the Heisenberg spectrum from Session 2's notebook. Check that the states that map onto each other converge as $U$ grows, and see how quickly the mapping breaks down as you push $U$ back down toward $t$.

# 25 Spin-charge separation

We now have all the pieces to understand one of the most striking features of the one-dimensional Hubbard model: at strong coupling, its charge and spin degrees of freedom behave in completely different ways.

## 25.1 The many-body gap is not one thing

Equation (48) already told us the singlet-triplet gap of the dimer shrinks like $1/U$ at large $U$. If we track the same gap as a function of chain length $L$, this trend persists and even sharpens: at $U = 0$ the gap is a purely single-particle, finite-size level spacing, and it closes as a clean power law, $\Delta \sim 1/L$. Turning on $U$ makes the gap *smaller* at fixed $L$, and increasing $U$ further shrinks it more, which looks backwards at first: we expect the Hubbard model to become more insulating, not less gapped, as $U$ grows. The resolution is that this particular gap is the **spin gap**: it is the energy to flip the total spin from $S = 0$ to $S = 1$, and Section 24 already told us this is governed by $J \sim t^2/U$, which indeed shrinks with $U$. In the thermodynamic limit this spin gap vanishes altogether, because the Heisenberg chain that the Hubbard model maps onto at strong coupling is itself gapless. None of this says anything yet about the **charge gap**, the energy to add or remove an electron, and that is a completely different quantity that we need to probe separately.

## 25.2 Nonlocal correlators: telling a metal from an insulator

A clean, ground-state-only way to distinguish a metal from an insulator is to look at how a static correlator decays with distance. Define the nonlocal particle correlator and the nonlocal spin correlator,

$$C_c(i,j) \equiv \left\langle c^\dagger_{i\sigma} c_{j\sigma} \right\rangle, \qquad C_s(i,j) \equiv \langle \mathbf{S}_i \cdot \mathbf{S}_j \rangle, \tag{49}$$

both evaluated in the many-body ground state. Physically, $C_s(i,j)$ measures how strongly the spin at site $i$ is correlated with the spin at site $j$. Even at strong coupling, where the model reduces to the gapless Heisenberg chain, $C_s(r)$ decays as a power law rather than exponentially, precisely because the spin sector has no gap to suppress long-range propagation. What does grow with $U$ is the *amplitude* of this correlator, not its decay rate: at $U = 0$ each site's spin is only weakly and loosely correlated with its neighbors (itinerant electrons barely notice each other's spin), while at large $U$ every site carries an increasingly well formed, singly-occupied local moment, strongly antiferromagnetically correlated with its neighbors, approaching the ideal Heisenberg chain's correlations.

$C_c(i,j)$, by contrast, measures the amplitude to inject an electron at site $i$ and later find it at site $j$: in a metal this decays only as a power law, $C_c(r) \sim \sin(k_F r)/r$, reflecting the fact that gapless charge excitations let the particle propagate arbitrarily far at essentially no energy cost. In an insulator, by contrast, any propagation has to borrow energy of order the charge gap $\Delta_c$, and the correlator instead decays exponentially, $C_c(r) \sim e^{-r/\xi_c}$, with a correlation length $\xi_c$ set by $\Delta_c$ (roughly $\xi_c \sim v/\Delta_c$ for some characteristic velocity $v$). This is exactly the signature we see in the Hubbard chain: at $U = 0$, $C_c(r)$ decays slowly, a power law, while at large $U$ (say $U = 10\,t$) it collapses to zero within a couple of sites, an unmistakable exponential decay confirming the charge sector is gapped and insulating, exactly consistent with the linearly growing charge scale

we will see again below. So interactions act in opposite directions on the two channels: they strongly suppress charge fluctuations while they strengthen spin correlations, even though the associated energy scale $J \sim t^2/U$ of those very same spin fluctuations is shrinking.

**✘ Computational companion**

Sections "Gap of an interacting Hubbard model," "Non-local static correlators for the spin response," and "Non-local static correlators for the particle-particle response." Plot the singlet-triplet gap against $L$ for a few values of $U$ and confirm the $1/U$ trend derived above. Then plot $C_c(1,j)$ and $C_s(1,j)$ against $j$ for weak and strong $U$, and check which one turns from power-law to exponential decay as $U$ grows, that is your direct numerical evidence for a charge gap coexisting with a gapless spin sector.

## 25.3 Dynamical correlators and the two energy scales

Static correlators tell us about decay lengths; dynamical correlators tell us about the actual excitation energies involved, and are the many-body generalization of the single-particle density of states from Session 1. Define the dynamical particle and spin correlators via their Lehmann (spectral) representations,

$$S_c(\omega) = \sum_n \left| \langle n | \, c^\dagger_{i\sigma} \, |0\rangle \right|^2 \delta\big(\omega - (E_n - E_0)\big), \qquad S_s(\omega) = \sum_n \left| \langle n | \, S^+_i \, |0\rangle \right|^2 \delta\big(\omega - (E_n - E_0)\big), \quad (50)$$

where the sums run over all many-body eigenstates $|n\rangle$ of $\mathcal{H}_{\text{Hub}}$. $S_c(\omega)$ picks out the excitation energies reachable by adding a single electron, exactly the charge sector; $S_s(\omega)$ picks out excitation energies reachable by flipping a single spin, exactly the spin sector. At a large but fixed interaction, say $U = 60t$, the two look dramatically different: $S_c$ has essentially no weight until a large threshold energy, and that threshold keeps drifting upward as $U$ increases; $S_s$ has weight starting at a much lower energy, and its overall lineshape closely tracks the Heisenberg-chain spin structure factor from Session 2, exactly what we expect from the strong-coupling mapping.

Scanning the peak position of each correlator as a function of $U$ makes the contrast completely explicit: the leading charge peak moves to higher energy *linearly* in $U$ (it costs $\sim U/2$ just to overcome the local repulsion when adding a second electron to an already singly-occupied site), while the leading spin peak moves to *lower* energy, decreasing $\sim 1/U$ as we derived in Section 24. Charge and spin excitations are governed by opposite trends in the very same parameter $U$, growing apart in energy without bound as $U$ increases. This is what we mean by **spin-charge separation**: in the strongly correlated one-dimensional Hubbard chain, charge transport is exponentially suppressed, essentially frozen out in the thermodynamic limit, while spin transport remains gapless and, in this precise sense, easy, even though both live on the very same lattice of electrons. Figure 15 draws what the two excitations are and how far apart their energies sit.

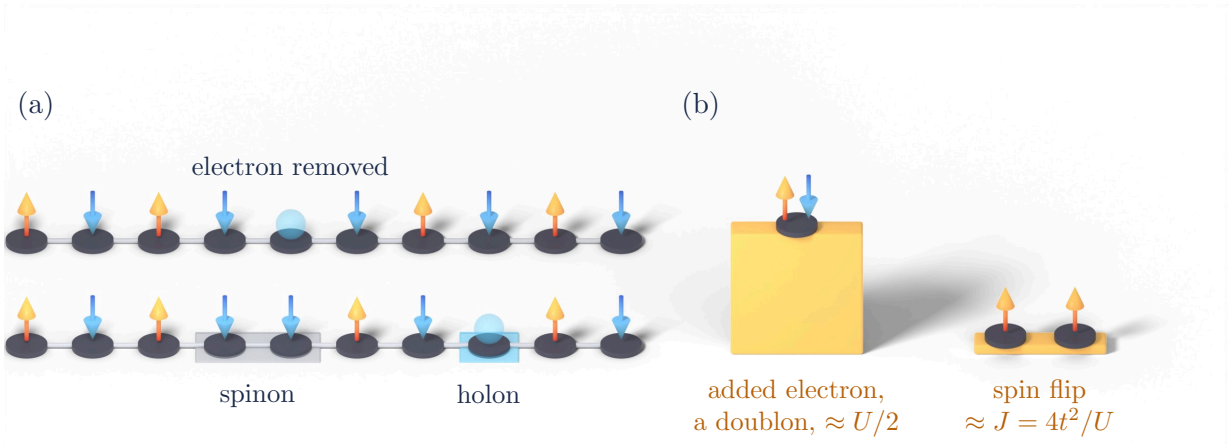



**Figure 15:** Spin-charge separation in the half-filled Hubbard chain of Eq. (42). (a) Behind, an electron removed from the antiferromagnetic chain; in front, the same chain after the hole has hopped three sites to the right. The hole is now an empty site without spin, the holon, carrying the charge, and it has left behind two parallel neighbors where it started, the spinon, carrying the spin, so one removed electron has become two objects that move apart. (b) The two energy scales at $U = 10t$, drawn to scale as two towers: adding an electron onto a singly occupied site makes a doubly occupied one, a doublon, which is what $S_c(\omega)$ of Eq. (50) reaches and costs about $U/2$, while flipping a spin against its neighbor, what $S_s(\omega)$ reaches, costs about $J = 4t^2/U = 0.4\,t$.

**✘ Computational companion**

Sections "Dynamical particle-particle correlators in the Hubbard model," "Dynamical spin-spin correlators in the Hubbard model," and "Interaction dependence of the dynamical excitations in the Hubbard model." Compute both dynamical correlators at a large, fixed $U$ and compare their lineshapes, then track how the leading peak of each one moves as you scan $U$, confirming the linear-growth-versus-$1/U$-decay contrast that defines spin-charge separation.

## Closing summary

- Exact diagonalization directly in the fermionic Fock space costs $2^L$ (spinless) or $4^L$ (spinful) in Hilbert space dimension, exactly matching the $2^L$ cost of the equivalent spin problem from Session 2, and it comes with no product-state restriction whatsoever.
- The spinless nearest-neighbor chain shows an exact charge-density-wave transition: a unique metallic ground state at $V = 0$ becomes a twofold-degenerate, gapped insulator as $V \to \infty$, with a large-$V$ gap growing $\sim V$, in good qualitative and quantitative agreement with Session 1's mean-field treatment, since the broken symmetry here is only discrete.
- A conserved chemical potential term lets us control the ground-state particle number directly; the width of the resulting occupation plateaus is the charging energy, the discrete second derivative of the ground-state energy with respect to $N$.
- The exact Hubbard-model magnetization is *zero* at every site, for every $U$, in sharp and structural disagreement with Session 1's mean-field prediction of finite staggered order; Lieb's theorem guarantees a unique, rotationally invariant spin singlet ground state at half filling, something no single Slater determinant can ever represent. This is the clean, numerically exact confirmation of mean-field theory's failure flagged at the end of Session 1.

- An exact diagonalization of the Hubbard dimer reproduces the $t^2/U$ superexchange scaling of Session 1 from first principles, and adding a Zeeman field shows the low-energy excitations are magnons, exactly matching the Heisenberg dimer of Session 2.
- At strong coupling the Hubbard chain exhibits spin-charge separation: the charge sector is gapped, with a gap growing linearly in $U$ and exponentially decaying nonlocal correlators, while the spin sector stays gapless, with a spin gap shrinking as $1/U$ and power-law correlators, exactly the physics of the Heisenberg chain it maps onto.
- All of this was limited by the $2^L$ and $4^L$ wall, to $L \sim 20$ sites for spinless fermions and about half that for the Hubbard chain. Session 5 returns to this same fermionic program, via the Jordan-Wigner mapping and tensor-network methods, and pushes it to system sizes where the thermodynamic limit becomes directly visible.

# PHYS-E0510 – Advanced Computational Methods in Physics D

Session 4: Matrix Product States for Spin Chains

**Companion notebook:** `mps_quantum_magnets.ipynb` **Prerequisites:** exact diagonalization of the Heisenberg model (Session 2)

## Contents and goals

In the last two sessions we solved quantum many-body problems exactly, first the Heisenberg chain in the spin basis and then the spinless and Hubbard chains in the fermionic basis, and it worked: gaps, level crossings, correlators, all of it came out of the numerics with no approximation at all. But both times we ran straight into the same wall. The Hilbert space of $L$ spin-$\frac{1}{2}$'s has dimension $2^L$ (and $4^L$ for spinful fermions), and that number simply refuses to be tamed: by around $L \sim 20$ spins, even a very good computer has run out of memory, and we are nowhere near the thermodynamic limit where the interesting physics, quantum phase transitions, fractionalization, the emergence of gaps, actually lives. So the question for today is very concrete: is there a way to represent the many-body wavefunction that does not cost us exponentially much, at least for the states we actually care about? The answer is yes, and it goes by the name of matrix product states (MPS), the simplest member of the broader family of tensor network methods. We will see why this representation is efficient, what makes it work so well specifically in one dimension, how to tell numerically whether we trust it, and then we will spend the second half of the session simply redoing Session 2's Heisenberg-model story, but now at 40 to 100 spins and more, large enough to see thermodynamic-limit physics we could only guess at before.

## Learning outcomes

By the end of this session, you should be able to:

- Explain why the matrix product state ansatz compresses an exponentially large wavefunction into a cost that is linear in $L$ and quadratic in the bond dimension $\chi$.
- Use entanglement entropy and the area law to justify why matrix product states work well in one dimension, and where they are expected to fail (2D, volume-law states).
- Assess the convergence of a variational MPS calculation using energy fluctuations, and use entanglement entropy as a diagnostic for a quantum phase transition.
- Run MPS calculations at 40 to 100 spins and more to distinguish the gapless spin-$\frac{1}{2}$ Heisenberg chain from the gapped, fractionalized spin-1 Haldane phase.

## 26 From an exponential wavefunction to a matrix product state

### 26.1 Tensor networks as data compression

Let us set the stage with an analogy before we write down any equations. Imagine you have a high-resolution photograph and you want to store it using much less memory. You do not throw away information at random; you exploit the fact that real photographs are highly structured,

nearby pixels are correlated with each other, and a compressed representation (think of a JPEG) can reproduce something visually indistinguishable from the original using a small fraction of the raw pixel count. Tensor networks do exactly the same thing, but for quantum many-body wavefunctions. The "raw pixels" are the $2^L$ complex coefficients of the wavefunction in the computational basis; the "compression algorithm" is a particular way of factorizing those coefficients into small pieces, and the parameter that controls how aggressive the compression is will turn out to be something called the **bond dimension**. The whole content of this section is making that analogy precise.

## 26.2 The formal matrix product state ansatz

For a chain of $L$ spins, each with $d$ local states ($d = 2$ for spin-$\frac{1}{2}$, $d = 3$ for spin-1, and so on), an arbitrary wavefunction is

$$|\Psi\rangle = \sum_{s_1,\ldots,s_L} C(s_1,\ldots,s_L)\; |s_1\rangle \otimes \cdots \otimes |s_L\rangle\,, \tag{51}$$

and specifying $|\Psi\rangle$ exactly means specifying all $d^L$ numbers $C(s_1,\ldots,s_L)$ independently. The matrix product state ansatz makes a very specific proposal for how to generate these coefficients without storing them one by one: for every site $n$ and every local state $s_n$, introduce a matrix $M_n^{s_n}$, and write

$$C(s_1,\ldots,s_L) = M_1^{s_1} M_2^{s_2} \cdots M_L^{s_L}. \tag{52}$$

For open boundary conditions, $M_1^{s_1}$ is a $1 \times \chi_1$ row vector and $M_L^{s_L}$ is a $\chi_{L-1} \times 1$ column vector, so the whole chain of matrix products in Eq. (52) collapses down to a single number, exactly as it must for $C$ to be a scalar coefficient. The internal matrices $M_n^{s_n}$ are $\chi_{n-1} \times \chi_n$, and the integer $\chi$ (we will mostly take it uniform along the chain) is the **bond dimension**: it is the one knob that controls how much this ansatz can represent, and it is going to be the central character of this whole session. Figure 16 draws the two ways of storing the same state next to each other.

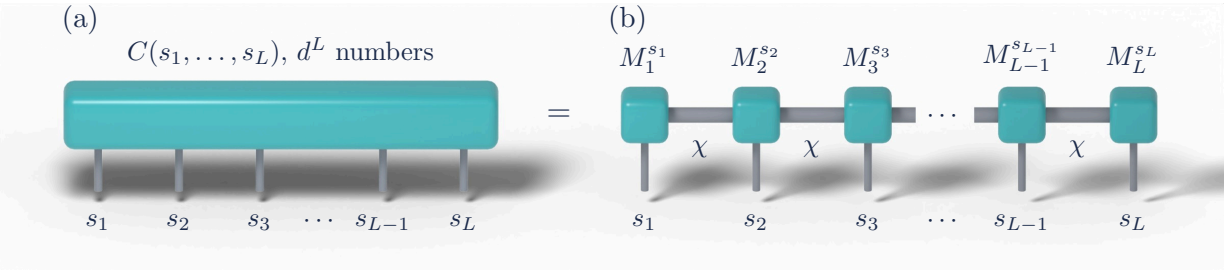



**Figure 16:** The state of Eq. (51) stored in two ways. (a) The coefficients $C(s_1,\ldots,s_L)$ as one tensor with $L$ physical legs, one per site, which means $d^L$ numbers to store. (b) The matrix product state of Eq. (52), the same legs hanging from a train of small tensors. Each site carries one tensor $M_n^{s_n}$, with a vertical *physical* leg running over the $d$ local states of that site and horizontal *bond* legs of dimension $\chi$ joining it to its neighbors. Joining two legs means summing over the shared index, so contracting the horizontal legs from left to right is precisely the matrix product of Eq. (52). The two end tensors carry a single bond leg each, a row vector and a column vector, which is why the whole contraction collapses to one number. What changes between the panels is only where the numbers live: one block that grows as $d^L$, or $L$ small tensors of $d\chi^2$ numbers each, Eq. (53).

Let us count parameters, because this is really the whole point. Storing $C(s_1,\ldots,s_L)$ directly

costs us $d^L$ numbers, exponential in $L$. Storing the matrices $\{M_n^{s_n}\}$ instead costs us

$$N_{\mathrm{MPS}} \sim L\, d\, \chi^2, \tag{53}$$

which is *linear* in the system size $L$ and only quadratic in the bond dimension $\chi$. So if we can get away with representing the states we care about at some fixed, moderate $\chi$, we have turned an exponential problem into a polynomial one. This raises the natural question: is it actually true that the states we care about, ground states and low-lying excited states of local Hamiltonians, can be captured this way with a $\chi$ that stays modest as $L$ grows? We will answer this properly once we introduce entanglement entropy in Section 29, but the short answer, which is really the whole reason this course has a Session 4, is: yes, in one dimension.

### 26.3 Cost: exact diagonalization versus matrix product states

Let us first look at what this buys us in practice, before we justify it theoretically. If we solve the Heisenberg chain $\mathcal{H}_{\mathrm{Heis}} = J\sum_n \mathbf{S}_n \cdot \mathbf{S}_{n+1}$ by exact diagonalization, the time to find the ground state grows exponentially with $L$, simply because we are manipulating vectors and matrices of size $2^L$. If instead we solve it variationally over the matrix product state ansatz of Eq. (52), at fixed bond dimension the cost grows close to linearly in $L$. The two curves cross somewhere around $L \sim 14$ to 16: below that, exact diagonalization is actually faster (there is no point using a variational method when the exact answer is cheap), but above it, matrix product states win, and they keep winning by an ever-growing margin. Concretely, systems of 50, 70, even 80 spins become tractable on a laptop, in seconds, while exact diagonalization would need more memory than exists on the largest supercomputers once you pass $L \sim 40$ to 50. That gap in scaling, exponential versus polynomial, is what the rest of the session builds on.

**✘ Computational companion**

Sections "Energy of the Heisenberg model as a function of the system size" and "Near linear scaling with system size with matrix product states." Plot the ground-state energy per site against $L$ and see it converge toward a thermodynamic-limit value; check it against the exact Bethe-ansatz result $e_0 = J\left(\frac{1}{4} - \ln 2\right) \approx -0.4431J$ as an independent convergence check. Then time exact diagonalization against the matrix product state solver as you increase $L$, and locate for yourself the crossover system size beyond which the tensor network method becomes the only option.

## 27 How good is the variational state? Convergence diagnostics

Matrix product states are a variational ansatz: we are not solving the Schrödinger equation exactly, we are searching for the best approximation to the ground state (or an excited state) within the family of states described by Eq. (52) at some fixed $\chi$. This immediately raises a practical question we did not have to worry about with exact diagonalization: how do we know, without knowing the exact answer, whether our variational state is actually good?

### 27.1 Energy fluctuations

Suppose our solver has converged to some normalized state $|\Omega\rangle$, with energy $E_\Omega = \langle\Omega|\mathcal{H}|\Omega\rangle$. If $|\Omega\rangle$ happened to be an exact eigenstate of $\mathcal{H}$, then $\mathcal{H}\,|\Omega\rangle = E_\Omega\,|\Omega\rangle$ and the variance of the energy,

$$(\Delta E)^2 \equiv \langle\Omega|\mathcal{H}^2|\Omega\rangle - \langle\Omega|\mathcal{H}|\Omega\rangle^2\,, \tag{54}$$

would vanish identically. So a natural diagnostic is: compute $(\Delta E)^2$ using only the variational state itself, no knowledge of the true ground state required, and check how close it is to zero. Let us make this quantitative. Write $|\Omega\rangle$ as mostly the true ground state $|0\rangle$ with a small admixture $\varepsilon$ of the first excited state $|1\rangle$,

$$|\Omega\rangle = \sqrt{1-\varepsilon^2}\,|0\rangle + \varepsilon\,|1\rangle\,, \qquad \varepsilon \ll 1. \tag{55}$$

A short calculation (expand $\langle\Omega|\mathcal{H}|\Omega\rangle$ and $\langle\Omega|\mathcal{H}^2|\Omega\rangle$ to $\mathcal{O}(\varepsilon^2)$ using $\mathcal{H}\,|0\rangle = E_0\,|0\rangle$, $\mathcal{H}\,|1\rangle = E_1\,|1\rangle$) gives

$$E_\Omega - E_0 = \varepsilon^2(E_1 - E_0), \qquad (\Delta E)^2 = \varepsilon^2(E_1 - E_0)^2 + \mathcal{O}(\varepsilon^4)\,. \tag{56}$$

Two things follow immediately, and both are exactly what makes this diagnostic so useful. First, $(\Delta E)^2 \to 0$ if and only if $\varepsilon \to 0$, i.e. if and only if $|\Omega\rangle$ converges onto a true eigenstate: the fluctuation is a perfectly good stand-in for the (unknown) distance to the exact answer. Second, comparing the two relations in Eq. (56) tells us that $(\Delta E)^2 \approx (E_\Omega - E_0)(E_1 - E_0)$, so the energy fluctuation is simultaneously telling us both how contaminated our state is by an excited state *and* roughly what the gap to that excited state is. In practice we watch $(\Delta E)^2$ as we increase the bond dimension $\chi$: it should decrease monotonically and eventually flatten out near machine precision once $\chi$ is large enough for the physics at hand.

### 27.2 Bond dimension and excited states

Everything we just said for the ground state applies unchanged to any targeted eigenstate: an excited-state solver also converges toward some $|\Omega_k\rangle \to |k\rangle$ as $\chi$ increases, and we can monitor the same energy fluctuation, Eq. (54), evaluated on $|\Omega_k\rangle$. What changes is *how fast* this convergence happens. Higher excited states are, generically, more entangled than the ground state (they explore a larger portion of the Hilbert space, and, as we will see in Section 29, entanglement is exactly what a finite bond dimension struggles to capture), so they need larger $\chi$ before their energy fluctuation collapses toward zero. This is a completely generic feature of tensor network calculations: the ground state converges first and fastest, and each successive excited state needs a bit more bond dimension to catch up.

**✘ Computational companion**

Sections "The role of the bond dimension" and "The role of bond dimension in excited states." Plot the ground-state energy and its fluctuation, Eq. (54), against $\chi$ for the Heisenberg chain, and watch the fluctuation collapse as $\chi$ grows. Then repeat for several excited states at once and confirm that higher-lying states need larger $\chi$ to converge to the same fluctuation tolerance.

## 28 Low-energy excitations and level crossings

Before we build up the full entanglement-entropy machinery, let us see what matrix product states buy us physically, using the same first excited state we just discussed as a diagnostic.

Take the Heisenberg chain and compute the matrix element $\langle 0|\,S^x_n\,|1\rangle$ between the ground state and the first excited state, as a function of the site $n$ where we act with the spin operator. Physically, this tells us the amplitude for creating the lowest-energy spin excitation at site $n$. For the spin-$\frac{1}{2}$ chain, this profile looks like the ground-state wavefunction of a particle in a box:

maximal amplitude at the center of the chain, decaying smoothly toward both edges. This is the particle-in-a-box picture we already met in the dynamical correlators of Session 2: the lowest spin excitation is a wave spread over the whole chain, and on an *open* chain the boundaries force it into a standing wave that vanishes at both edges. This is exactly the behavior of a delocalized, bulk spin excitation, and it is our first hint that the low-energy physics of the spin-$\frac{1}{2}$ chain lives in the bulk. We will come back to a sharp contrast with this profile, for spin-1, once we have enough system size to make the comparison convincing (Section 33).

The other natural thing to track is how the ground state itself changes as we turn on a magnetic field, $\mathcal{H} = J\sum_n \mathbf{S}_n\cdot\mathbf{S}_{n+1} - B\sum_n S^z_n$. Since $\sum_n S^z_n$ commutes with the exchange term, applying $B$ never mixes eigenstates of $\mathcal{H}|_{B=0}$; it simply relabels their energies by $-B\times(\text{total } S^z)$, and different total-$S^z$ sectors cross each other as $B$ increases. For $L = 2$ there is exactly one such crossing, from the $S = 0$ singlet to the $S = 1$ triplet; for $L = 4$ there are two, $S = 0 \to S = 1 \to S = 2$; and as we grow $L$, more and more crossings appear, packed closer and closer together in $B$. This is a very direct, numerically clean way of seeing that the spin-$\frac{1}{2}$ Heisenberg chain is gapless in the thermodynamic limit: the staircase of crossings becomes a smooth ramp as $L \to \infty$, meaning there exist excitations arbitrarily close in energy to the ground state at any nonzero $B$.

**✘ Computational companion**

Sections "Low energy excitations of a Heisenberg model" and "Level crossings as a function of the magnetic field with matrix product states." Reproduce the particle-in-a-box matrix-element profile, and separately track the ground-state total spin as a function of $B$ for $L = 2, 4, 10, 30$: watch the discrete staircase of Session 2 smooth out into what will look like a continuous ramp once $L$ is large enough.

# 29 Entanglement entropy: why matrix product states work in one dimension

We have used the matrix product state ansatz for two sections now without ever justifying it. Let us finally do that properly, because the justification is also what tells us, precisely, when this whole approach is going to fail.

## 29.1 Separable versus entangled states

Take a bipartition of our system into two regions, $A$ and $B$ (think of $A$ as the left half of the chain and $B$ as the right half). We say a state $|\Psi\rangle$ is **separable** if it factorizes as a simple tensor product,

$$|\Psi\rangle = |\phi\rangle_A \otimes |\chi\rangle_B \,, \tag{57}$$

for some states $|\phi\rangle_A$, $|\chi\rangle_B$ living purely on $A$ and purely on $B$. A basis state like $|\uparrow\downarrow\rangle$ is trivially of this form, $|\uparrow\rangle_A \otimes |\downarrow\rangle_B$; so, less obviously but just as directly, is a state where every spin points along $x$, since $|{\rightarrow}\rangle\otimes|{\rightarrow}\rangle = \frac{1}{2}(|\uparrow\rangle+|\downarrow\rangle)\otimes(|\uparrow\rangle+|\downarrow\rangle)$ is still a product. Every classical configuration, in fact every state you can write down without any correlation between $A$ and $B$, is separable in this sense. But now take the two-site singlet we met already in Session 1,

$$|\Psi_{\text{singlet}}\rangle = \frac{1}{\sqrt{2}}\Big(\,|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle\,\Big). \tag{58}$$

No choice of $|\phi\rangle_A$, $|\chi\rangle_B$ reproduces this: try to write it as a product and you immediately find you need two different terms with opposite relative sign, which a single product cannot generate. We call such a state **entangled**. This is precisely the distinction that separates classical states from quantum ones, and we need a way to quantify *how* entangled a state is, not just whether it is or is not.

## 29.2 The reduced density matrix and the von Neumann entropy

Given the full density matrix $\rho = |\Psi\rangle\langle\Psi|$, define the **reduced density matrix** of subsystem $A$ by tracing out everything in $B$,

$$\rho_A \equiv \mathrm{Tr}_B\, \rho. \tag{59}$$

The **entanglement entropy** across the cut is then just the ordinary von Neumann entropy of $\rho_A$,

$$S_A \equiv -\,\mathrm{Tr}\left[\rho_A \ln \rho_A\right]. \tag{60}$$

If $|\Psi\rangle$ is separable, $\rho_A = |\phi\rangle\langle\phi|$ is a rank-one projector, its only nonzero eigenvalue is 1, and $S_A = -1\ln 1 = 0$. If $|\Psi\rangle$ is the singlet, a direct computation of $\rho_A$ (trace out site $B$) gives $\rho_A = \frac{1}{2}\mathbb{1}$, with two equal eigenvalues $\frac{1}{2}, \frac{1}{2}$, so $S_A = -2 \times \frac{1}{2}\ln\frac{1}{2} = \ln 2$, the maximal possible entropy for a single spin-$\frac{1}{2}$. So $S_A$ really does interpolate smoothly between "no entanglement" and "maximal entanglement," exactly as we want.

## 29.3 Schmidt decomposition: the bond dimension bounds the entanglement

Now let us connect this directly to the matrix product state ansatz. Cut the chain at bond $n$, between sites $n$ and $n+1$. Any state can be written, via a singular value decomposition of its coefficient tensor reshaped as a matrix between the left block $(1,\ldots,n)$ and the right block $(n+1,\ldots,L)$, in the **Schmidt form**

$$|\Psi\rangle = \sum_{\alpha=1}^{r} \lambda_\alpha\, |\alpha\rangle_A \otimes |\alpha\rangle_B\,, \qquad \sum_\alpha \lambda_\alpha^2 = 1, \tag{61}$$

with $|\alpha\rangle_A$, $|\alpha\rangle_B$ orthonormal on each side and $r$ the Schmidt rank. For a generic state, $r$ can be as large as $d^{\min(n,L-n)}$, exponential in the block size. But for a matrix product state built as in Eq. (52), the sum over the bond index at position $n$ runs only over $\chi_n$ values by construction, so the coefficient tensor manifestly factorizes through a $\chi_n$-dimensional index: the Schmidt rank at that cut can be at most $\chi_n$. Plugging the Schmidt form into Eq. (60) gives $\rho_A = \sum_\alpha \lambda_\alpha^2 |\alpha\rangle\langle\alpha|$, so

$$S_A = -\sum_{\alpha=1}^{r} \lambda_\alpha^2 \ln \lambda_\alpha^2 \;\le\; \ln r \;\le\; \ln \chi_n, \tag{62}$$

with equality in the last step exactly when all $\chi_n$ Schmidt weights are equal, $\lambda_\alpha^2 = 1/\chi_n$. This is the single most important equation in this session: *the bond dimension caps the entanglement entropy the ansatz can represent.* If the state we are after has entanglement entropy $S_A$ across some cut, we need at least $\chi \gtrsim e^{S_A}$ there to capture it. So the entire question of whether matrix product states are a good idea for a given problem reduces to: how does $S_A$ behave as we grow the system? Figure 17 follows what crosses a cut, from two spins to a whole chain, a square lattice and a random state.

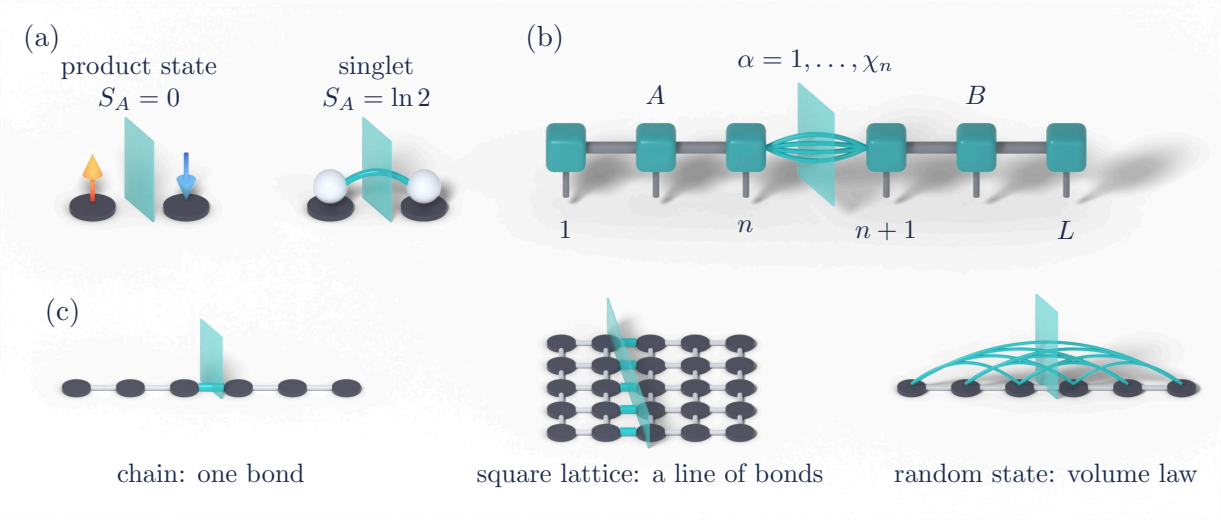



**Figure 17:** What crosses a cut. (a) The two cases of the section on separable and entangled states: nothing crosses the cut of the product state $|\uparrow\rangle_A \otimes |\downarrow\rangle_B$, while the singlet, two spins with no moment of their own, is held together by one strand across it, and $S_A = \ln 2$. (b) The Schmidt decomposition of Eq. (61) read directly off a matrix product state. Cutting the chain at bond $n$ separates it into blocks $A$ and $B$ and exposes the single bond index $\alpha$ that joined them, drawn opened into its $\chi_n$ values. Since that index takes only $\chi_n$ values by construction, the sum in Eq. (61) can run over at most $\chi_n$ terms, the Schmidt rank obeys $r \leq \chi_n$, and the entropy across the cut obeys the bound of Eq. (62). Everything the two halves of the chain know about each other has to pass through that one index. (c) What a cut severs in a gapped ground state: one bond of a chain, a whole line of bonds of a square lattice, and, for a random state, strands everywhere, so that the entropy grows with the volume of the block, the volume law.

## 29.4 The area law, and why one dimension is cheap

Here is the key physical fact. For the ground state of a *gapped*, local, one-dimensional Hamiltonian, correlations decay exponentially beyond some correlation length $\xi$, and it turns out (this is what is called the area law, proven for gapped one-dimensional systems) that $S_A$ *saturates* to an $L$-independent constant as soon as the block $A$ is a few $\xi$ away from both its boundary and the far end of the chain. Only the physics living within $\sim \xi$ of the cut contributes to the entanglement; everything deep in the interior of $A$ or $B$ is irrelevant to it. For a *gapless*, critical chain there is no finite $\xi$, and conformal field theory tells us $S_A$ instead grows, but only logarithmically, $S_A \sim \frac{c}{3} \ln n$ for a cut in the bulk of an infinite or periodic chain (open boundaries, which we adopt throughout this session and motivate in Section 30, roughly halve this to $S_A \sim \frac{c}{6} \ln n$). Either way, this is enormously cheaper than the generic case: a uniformly random state in the Hilbert space (this is Page's theorem) has $S_A$ growing linearly with the size of the smaller block, a *volume law*, which would force $\chi$ to grow exponentially with system size and defeat the whole purpose of the ansatz. Ground states of local 1D Hamiltonians are a vanishingly special corner of the full Hilbert space, and it is precisely because they live in that corner, low or at most logarithmically growing entanglement, that a bond dimension $\chi$ that stays modest (or grows only polynomially) as $L \to \infty$ can capture them accurately. Panel (c) of Figure 17 puts the three cases side by side.

A very clean way to see the saturation directly is to solve an exactly dimerized chain, where sites are grouped into decoupled pairs, $\mathcal{H} = J \sum_k \mathbf{S}_{2k-1} \cdot \mathbf{S}_{2k}$ with no coupling between pairs. The exact ground state is simply a product of singlets, $|\Psi\rangle = \bigotimes_k |\Psi_{\text{singlet}}\rangle_{2k-1,2k}$. Now cut this

state anywhere: every cut either falls *inside* a singlet, contributing $\ln 2$ (from Eq. (60) above), or falls *between* singlets, contributing nothing at all, since the two halves factorize exactly there. However large a block $A$ we choose, as long as its boundary sits between dimers, $S_A$ is either 0 or $\ln 2$ per boundary point, completely independent of the size of $A$. This is the area law in its most transparent form: entanglement is a property of the *boundary* of the cut, not of its volume, and in one dimension a boundary is just a point (or two, for an interior block), so it costs $\mathcal{O}(1)$ no matter how big the system gets.

**✘ Computational companion**

Sections "Entanglement entropy of the ground state," "Size dependence of the entanglement entropy," and "Bond dimension dependence of the entanglement entropy." Compute $S_A$ at every bond of a Heisenberg chain for a few values of $\chi$, and check that increasing $\chi$ beyond some point stops changing the profile at all: that saturation is a direct, numerical confirmation of the bound in Eq. (62). Then track the entanglement entropy at the central bond as you grow $L$, and confirm it grows only very slowly, consistent with the (logarithmic, for the gapless Heisenberg chain) area-law scaling described above.

**Why two dimensions is expensive.** The same bond-counting argument, run in two dimensions, gives the opposite conclusion. Tile a 2D lattice into decoupled clusters (say, plaquettes of four spins each, fully coupled inside and disconnected between clusters) and repeat the exercise: a rectangular block $A$ now cuts through a number of inter-cluster bonds proportional to the *perimeter* of $A$, not to a fixed number of points. Doubling the linear size of $A$ roughly doubles the number of boundary clusters, and hence doubles $S_A$. Since a matrix product state has to linearize the 2D lattice into an effective 1D chain (snaking through the rows), the bond crossing the "middle" of that snake has to carry the entanglement of an entire row, so $\chi$ needs to grow *exponentially* with the linear dimension of the system to keep the truncation error fixed. Matrix product states can still be used in two dimensions (this is how some of the best available numerics for the 2D Hubbard model or frustrated triangular and kagome-lattice antiferromagnets are obtained), but only by working on narrow cylinders and paying for a much larger $\chi$ than a comparable 1D calculation would need. We will not do two-dimensional calculations in this course; the point of the argument is to explain why everything else in this session is specific to one dimension.

**Where matrix product states struggle even in 1D.** Bounded, area-law entanglement is a property of *ground states* of *gapped or mildly critical, local* Hamiltonians. It fails, and matrix product states fail along with it, in a few situations that we will not compute in this course but that you should have in mind: entanglement grows *linearly* in time under quantum quench dynamics (so time evolution with matrix product states is only trustworthy up to some maximal time set by $\chi$), highly excited states deep in the many-body spectrum typically violate the area law entirely, and long-range interactions effectively reintroduce higher-dimensional connectivity even in a nominally 1D chain.

# 30 Boundary conditions

One more practical lesson, before we move to the physics: the choice between open and periodic boundary conditions matters a great deal for tensor networks, even though it barely matters for exact diagonalization (there, periodic boundary conditions just add one more coupling term to the Hamiltonian matrix). With open boundary conditions, a bulk cut severs exactly *one* bond of the chain, so the area-law argument above applies directly. With periodic boundary conditions, the chain is a ring, and any bipartition into two contiguous arcs severs the ring at *two* points rather than one, since the two arcs are each other's neighbors on both ends. Both cuts contribute to the entanglement entropy across that partition, so periodic boundary conditions roughly *double* $S_A$ compared to open boundary conditions on the same chain. Doubling the entropy means, by Eq. (62), that we need roughly the *square* of the bond dimension to reach the same accuracy. This is why virtually every practical matrix product state calculation, including everything else in this session, uses open boundary conditions: periodic chains are simply much more expensive for the same physical answer. Figure 18 draws the two cases side by side.

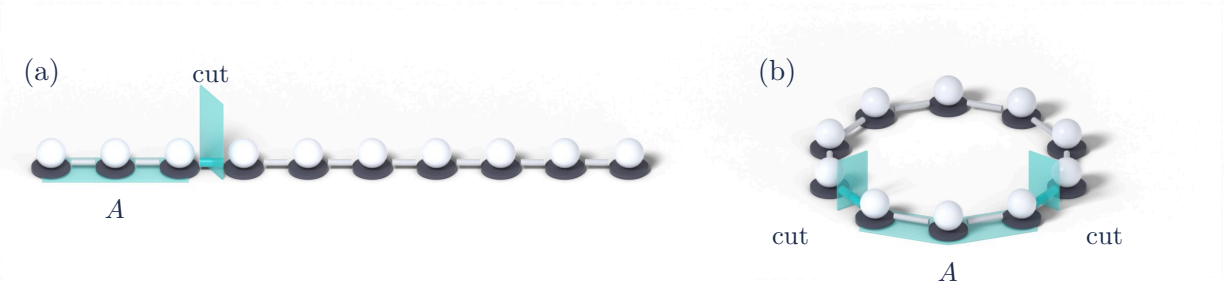



**Figure 18:** Why open boundaries are so much cheaper. The teal tray marks the block $A$ whose entanglement entropy we compute, and each pane a cut, with the bond it severs drawn in teal. (a) On an open chain, a contiguous block borders the rest of the system at a single point, so one bond is severed. (b) On a ring the same block is adjacent to the rest at both of its ends, so any bipartition severs two bonds and each one contributes, roughly doubling $S_A$. Through the bound of Eq. (62), a doubled entropy costs us the *square* of the bond dimension for the same accuracy.

**✘ Computational companion**

Section "Open and closed boundary conditions." Compute the central-bond entanglement entropy of the Heisenberg chain with both open and periodic boundary conditions as a function of $L$, and check numerically that the periodic curve sits roughly at twice the open one.

# 31 Entanglement entropy as a diagnostic for quantum phase transitions

## 31.1 The transverse-field Ising model

Entanglement entropy is not just a bookkeeping device for tensor network cost, it is a physical diagnostic in its own right, and nowhere is this cleaner than at a quantum phase transition.

Consider the transverse-field Ising chain,

$$\mathcal{H}_{\mathrm{TFI}} = -J \sum_n S^z_n S^z_{n+1} - B_x \sum_n S^x_n. \tag{63}$$

This model has two exactly solvable limits. At $B_x = 0$, only the $S^z S^z$ term survives, and even though $S^z_n$ is a quantum operator, the Hamiltonian is diagonal in the $S^z$ basis: the ground state is the classical, fully polarized ferromagnet, $|\uparrow\uparrow\cdots\uparrow\rangle$ (or its degenerate partner $|\downarrow\downarrow\cdots\downarrow\rangle$). At $B_x \to \infty$, we can instead drop the exchange term entirely, and the ground state is again classical, this time fully polarized along $x$, $|\rightarrow\rightarrow\cdots\rightarrow\rangle$. Both limits are separable product states, so by Eq. (60) their entanglement entropy is exactly zero. Somewhere in between, $\mathcal{H}_{\mathrm{TFI}}$ must interpolate between these two very different classical orders, and it does so through a quantum phase transition. (The exact solution of this model comes from mapping spins onto free fermions via a Jordan-Wigner transformation, a construction we will meet properly in Session 5; it places the critical point, for the normalization used here, at $B_x = J/2$.) Figure 19 draws the chain at the two ends of the field and at the transition.

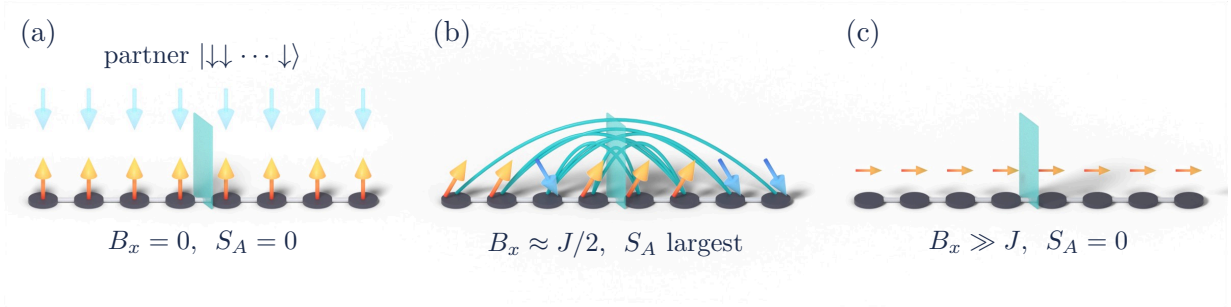



**Figure 19:** The transverse-field Ising chain of Eq. (63) at three fields, each cut at its central bond. (a) At $B_x = 0$ every arrow points along $z$, and the all-down state standing behind as a ghost is its degenerate partner; either one is a product state, so nothing crosses the cut. (b) Near the critical field $B_x = J/2$ the arrows tilt toward $x$ and the chain breaks into domains of every size, and this is where the most strands cross the cut, the peak of $S_A$ found in the subsection on the entanglement peak and the order parameter. (c) For $B_x \gg J$ every arrow points along $x$, again a product state with nothing across the cut. What to look at is that both ends are classical and only the middle is not.

## 31.2 The entanglement peak and the order parameter

Since both classical limits have zero entanglement entropy, and the state in between them is quantum (a superposition that cannot be reduced to a single classical spin configuration), $S_A$ has to rise away from zero somewhere in the middle and come back down, and the natural guess is that it peaks exactly at the phase transition. This is precisely what the numerics show: computing $S_A$ at the central bond of a long chain as a function of $B_x/J$, we see it vanish at both $B_x = 0$ and $B_x \to \infty$, with a pronounced maximum sitting right at the critical point. For a chain of $\sim 100$ spins, close enough to the thermodynamic limit, that peak sits at $B_x/J = 0.5$, the exact value quoted above.

We can cross-check this against a completely independent, and more familiar, diagnostic: the order parameter itself, $\langle S^z_n \rangle$. For $B_x < 0.5$, the ground state retains a finite $\langle S^z_n \rangle$, the ferromagnetic order parameter of the Ising phase; for $B_x > 0.5$, quantum fluctuations wash it out completely and $\langle S^z_n \rangle \to 0$, leaving only $\langle S^x_n \rangle \neq 0$. Both diagnostics agree on where the transition sits, which is reassuring, but they are not equally forgiving numerically. The order-parameter curve for a finite,

20-spin chain locates the transition noticeably away from 0.5: it takes going to $L \sim 100$ before the finite-size estimate settles down at the exact critical value. Exact diagonalization simply cannot reach the system sizes needed to resolve this cleanly; matrix product states can.

**✘ Computational companion**

Sections "The entanglement entropy as a signal for phase transitions" and "Phase transition from the order parameter." Sweep $B_x$ across the transition and plot both the central-bond entanglement entropy and $\langle S^z_n \rangle$; compare the sharpness of the two signals, and repeat for a smaller system to see the finite-size shift in the apparent transition point directly.

## 32 The Heisenberg chain at thermodynamic-limit scale

We now have both the tool (matrix product states) and the diagnostics (energy fluctuation, bond dimension convergence) we need to trust it. Let us put both to work on exactly the kind of questions Session 2 asked about the Heisenberg chain, $\mathcal{H}_{\text{Heis}} = J \sum_n \mathbf{S}_n \cdot \mathbf{S}_{n+1} - B \sum_n S^z_n$, but now at system sizes that were simply out of reach before.

### 32.1 Magnetization: from zero field to a magnetic impurity

At small field $B$, deep inside the gapless phase, the magnetization $\langle S^z_n \rangle$ of a uniform Heisenberg chain remains zero at every site: the field is too weak to overcome the antiferromagnetic exchange, and translational symmetry keeps every site equivalent. This is already a nontrivial check, since it holds even though we have explicitly broken time-reversal symmetry with $B \neq 0$; the ground state is simply not yet in the regime where that symmetry breaking shows up locally.

Things change qualitatively once we replace the uniform field by a single **magnetic impurity**: keep $B = 0$ everywhere except at one site, $n = 0$, where we apply a local field. The system responds by developing a nonzero, spatially varying magnetization $\langle S^z_n \rangle$ around the impurity, and *how* that response decays with distance from the impurity turns out to be one of the sharpest diagnostics we have for whether the chain is gapped or gapless. For the uniform, gapless Heisenberg chain, the induced magnetization decays as a slow power law,

$$\langle S^z_n \rangle \sim \frac{1}{|n|}, \qquad |n| \gg 1, \tag{64}$$

never quite vanishing, however far we go from the impurity. This is the hallmark of gaplessness: the chain has excitations at arbitrarily low energy, so a local perturbation can propagate its influence arbitrarily far with only algebraic loss.

### 32.2 A dimerized chain: correlation length from an impurity response

Now dimerize the exchange coupling, alternating a strong bond $J_1$ and a weak bond $J_2$ along the chain, $\mathcal{H} = \sum_n J_n \, \mathbf{S}_n \cdot \mathbf{S}_{n+1}$ with $J_n$ alternating between $J_1$ and $J_2$. As soon as $J_1 \neq J_2$, the chain develops a gap, and repeating the impurity experiment now gives an exponentially decaying response instead,

$$\langle S^z_n \rangle \sim e^{-|n|/\xi}, \tag{65}$$

with a correlation length $\xi$ that shrinks as the dimerization strengthens. This is a direct numerical demonstration of exactly the physics we anticipated in Section 29: a finite gap sets a finite correlation length, and it is that same $\xi$ that controls both the exponential decay of local response *and* the saturation of the entanglement entropy across any cut. Reaching a clean exponential fit, distinguishing it convincingly from the power law of the gapless case, needs chains of order 100 spins, an order of magnitude beyond anything exact diagonalization can offer.

### 32.3 Excited-state magnetization profiles and the many-body gap

Beyond the ground state, we can also target a handful of low excited states directly and look at their magnetization profiles. In a Heisenberg chain with a small field, the ground state is the usual $S = 0$ singlet with zero magnetization everywhere, but the first excited state develops a standing-wave pattern in $\langle S^z_n \rangle$, finite magnetization in some regions of the chain and the opposite sign in others. The second excited state returns to zero magnetization everywhere (it shares the ground state's time-reversal symmetry), while the third excited state reproduces the standing wave of the first, related to it by time reversal. This alternation, symmetric, antisymmetric, symmetric, antisymmetric, is exactly what we saw in the small chains of Session 2, now visibly stable at much larger $L$.

Finally, we can track the many-body gap itself, $E_1 - E_0$, as a function of $L$, for both the uniform and the dimerized chain. The uniform Heisenberg chain's gap closes as $L \to \infty$ (consistent with the gaplessness we diagnosed via Eq. (64)), while the dimerized chain's gap saturates to a finite value, the same value that sets the correlation length $\xi$ in Eq. (65). Seeing both trends converge cleanly, rather than being cut off by finite-size effects, again needs the large-$L$ reach that only matrix product states give us here.

**✘ Computational companion**

Sections "Magnetization of moderately large Heisenberg model," "Magnetization of a large Heisenberg model with a magnetic impurity," "Magnetization of a dimerized Heisenberg model with a magnetic impurity," "Magnetization of many-body excitations in large quantum chains," and "Gap of a many-body spin chain." Reproduce the power-law versus exponential contrast around a magnetic impurity, look at the standing-wave excited-state profiles, and extract the many-body gap scaling with $L$ for both the pristine and dimerized chains.

## 33 Resolving fractionalization: spin-$\frac{1}{2}$ versus spin-1 at scale

We are now in a position to settle, cleanly, a question that Session 2 could only gesture at with small chains: what actually distinguishes the spin-$\frac{1}{2}$ and spin-1 Heisenberg models?

### 33.1 Non-local static correlators

A convenient, purely static probe of gappedness is the non-local correlator $\langle S^z_0 S^z_n \rangle$, the linear response at site $n$ to a perturbation applied at site 0. We already saw the impurity version of this idea in Eqs. (64) and (65); the same dichotomy shows up directly in the correlator itself, power-law decay $\sim 1/n$ for a gapless chain, exponential decay $\sim e^{-n/\xi}$ for a gapped one. Applying this to the dimerized model of the previous section at increasing dimerization strength confirms

the trend directly: stronger dimerization means a shorter $\xi$ and a faster exponential falloff, while the undimerized limit recovers the slow power law.

Now apply exactly the same correlator to the spin-$\frac{1}{2}$ and spin-1 Heisenberg models, *without* any dimerization, just plain nearest-neighbor exchange, $\mathcal{H} = J\sum_n \mathbf{S}_n \cdot \mathbf{S}_{n+1}$ for $S = \frac{1}{2}$ and $S = 1$ respectively. Here is where system size really matters. If we only had access to exact diagonalization, capping out around $L \sim 15$ to 16 for spin-1, we might come away thinking the spin-1 correlator is actually the *larger* of the two at those distances. But push to $L \sim 40$ to 50 spins, and the picture flips completely: the spin-$\frac{1}{2}$ correlator keeps decaying as a slow power law all the way out, while the spin-1 correlator, once $n$ exceeds its (much shorter) correlation length, drops off exponentially. The crossover only becomes visible once the system is comparable to, or larger than, that correlation length, which is exactly why small chains give a misleading impression: spin-1 is gapped, spin-$\frac{1}{2}$ is not, but you need $L$ large enough to see it.

## 33.2 Fractionalization in the spin-1 chain

Here is the sharpest version of the contrast. Take an open spin-1 chain and apply a small uniform field, $B \sim 0.05J$. Rather than spreading uniformly, the induced magnetization $\langle S^z_n \rangle$ localizes almost entirely at the two edges of the chain, with the bulk staying at essentially zero. Summing the magnetization over each edge region gives $-\frac{1}{2}$ on the left and $+\frac{1}{2}$ on the right: each edge behaves as if it carried an emergent, free spin-$\frac{1}{2}$ degree of freedom, even though every microscopic spin in the chain is spin-1. This is **fractionalization**: the elementary excitations of the system are not integer multiples of the microscopic spin, but a genuine fraction of it. The cleanest exactly solvable model with this same edge physics is the **AKLT point**, a spin-1 chain built from projected valence bonds that shares the Heisenberg chain's phase and makes the emergent spin-$\frac{1}{2}$ edge states exactly computable rather than just numerically observed; Figure 20 draws how it is built and where the edge spins come from. At $L = 50$ the two edge moments are still visibly "leaking" into the bulk; only by $L = 100$ do we see them cleanly localized at the two ends with essentially nothing in between, confirming that in the thermodynamic limit the spin-1 Heisenberg chain has a fully gapped bulk and a pair of free spin-$\frac{1}{2}$ edge modes. (This is the Haldane phase, and it is one of the simplest examples of a symmetry-protected topological phase in one dimension.)

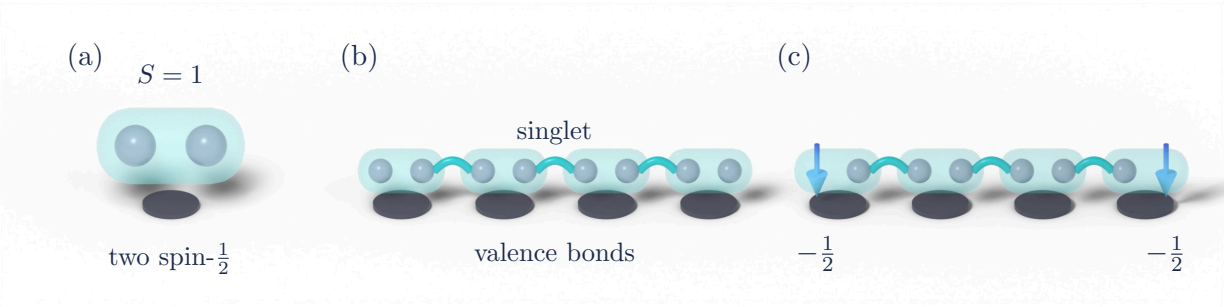



**Figure 20:** The AKLT construction of the spin-1 chain. (a) Each spin-1 is written as two spin-$\frac{1}{2}$ inside one capsule, the projector that keeps only their symmetric, total spin 1 states. (b) The right half of each site forms a singlet with the left half of the next site, so every bond carries one valence bond and the bulk has no free moment. (c) The outer halves of the two end sites have no partner, and they are the free spin-$\frac{1}{2}$ at the edges; a small uniform field polarizes both along it, so each edge region carries a moment of $-\frac{1}{2}$, half of what a spin-1 can carry, while the bulk stays at zero.

We can see the same physics again, even more directly, in the **dynamical** correlator $\langle S_n^+(t) S_n^-(0)\rangle$ as a function of both position $n$ and energy $\omega$, the same kind of object we used in Session 2 to read off excitation spectra. In the bulk of the spin-1 chain, creating a spin excitation costs a finite energy, exactly the bulk gap we just discussed. But right at the two edges, the dynamical correlator has spectral weight sitting at essentially *zero* energy: you can flip an edge spin for free. That zero-energy edge response is the direct spectral fingerprint of the free spin-$\frac{1}{2}$ edge modes, and, again, it only becomes sharp and unambiguous once the chain is long enough, $L \sim 40$ spin-1's or more, that the two edges are decoupled from each other and from any residual bulk contamination.

**✘ Computational companion**

Sections "Non-local static correlators in a large system," "Fractionalization in the $S = 1$ Heisenberg model," and "Non-local correlator in the $S = 1/2$ and $S = 1$ Heisenberg models." Plot $\langle S_0^z S_n^z\rangle$ on a log scale for the spin-$\frac{1}{2}$ and spin-1 chains and watch the power-law/exponential contrast only become clean once $L$ is large enough; then localize the spin-1 edge magnetization at $L = 50$ versus $L = 100$, and finally look at the dynamical correlator to see the zero-energy edge excitation directly. Note that the notebook attaches an explicit spin-$\frac{1}{2}$ to each end of the spin-1 chain for this last comparison, precisely so that the emergent edge modes have somewhere to combine into a conventional, non-fractional total spin.

## Closing summary

- Matrix product states rewrite the exponentially many wavefunction coefficients $C(s_1, \ldots, s_L)$ as a chain of small matrices, Eq. (52); the number of parameters grows only linearly in $L$ and quadratically in the bond dimension $\chi$, Eq. (53).
- This ansatz is only as good as the entanglement of the state it targets: the Schmidt decomposition shows the bond dimension bounds the entanglement entropy across any cut, $S_A \leq \ln \chi$, Eq. (62).
- Ground states of gapped, local, one-dimensional Hamiltonians obey an area law, entanglement entropy that saturates (or at most grows logarithmically at criticality) rather than growing with system size; this is why a modest, $L$-independent $\chi$ suffices in 1D, and why the same ansatz becomes exponentially expensive in two dimensions.
- Energy fluctuations, Eq. (54), give us a way to certify convergence without knowing the exact answer; open boundary conditions roughly halve the entanglement entropy (and hence the required $\chi$) compared to periodic ones.
- The entanglement entropy is itself a physical diagnostic: it peaks exactly at the quantum phase transition of the transverse-field Ising model, cleanly resolved once $L$ is close to the thermodynamic limit.
- Redoing Session 2's Heisenberg-chain program at 40 to 100 spins and more turns qualitative hints into sharp, converged results: power-law versus exponential correlators, a closing versus saturating many-body gap, and a clean separation between the gapless spin-$\frac{1}{2}$ chain and the gapped, fractionalized spin-1 chain with its emergent spin-$\frac{1}{2}$ edge modes.

Matrix product states solved our $2^L$ problem for spin chains, but everything we did here was written in the spin language of Session 2. Next session we go back to the fermions of Session 3, connect them to spins through the Jordan-Wigner mapping we only alluded to in Section 31.1, and lay out the general tensor-network algorithm, the actual variational sweep that produces the

matrices $M_n^{s_n}$ we have been taking for granted throughout this session.

# PHYS-E0510 – Advanced Computational Methods in Physics D

Session 5: Matrix Product States for Fermionic Models

**Companion notebook:** `mps_many_body_fermionic.ipynb` **Prerequisites:** Sessions 3 and 4 (exact diagonalization directly in the fermionic basis; matrix product states for spin chains)

## Contents and goals

Let us recall where we left things. In Session 3 we redid our many-body program directly in the fermionic basis, with no product-state restriction at all, and it worked: we recovered the exact spinless charge-density-wave transition and spin-charge separation in the Hubbard model. But we hit exactly the same wall we always hit with exact diagonalization: the Hilbert space grows as $2^L$ for spinless fermions and $4^L$ for the Hubbard chain, which kept the Hubbard chain to about half the length we could reach for spins. In Session 4 we built the fix, matrix product states, but we only ever applied it to *spin* Hamiltonians. Today we close the loop: we are going to solve interacting *fermionic* Hamiltonians with tensor networks, at system sizes of a hundred sites or more, by first mapping them onto spin Hamiltonians through the Jordan-Wigner transformation and then reusing every piece of MPS machinery we already built. Along the way we will redo the Session 3 spinless and Hubbard results at a scale where the physics is unambiguous, meet two new diagnostics, pairing correlators and the correlation entropy, and finish with the general tensor-network algorithm that has been quietly running underneath everything since Session 4.

## Learning outcomes

By the end of this session, you should be able to:

- Map a fermionic Hamiltonian onto an equivalent spin Hamiltonian via the Jordan-Wigner transformation, and explain why nearest-neighbor hopping needs no explicit string.
- Use MPS at system sizes of $\sim 100$ sites to locate a quantum phase transition that exact diagonalization could not resolve sharply.
- Compute pairing correlators, including their enhancement toward Cooper-pair-like behavior under attractive interactions, and use the correlation entropy to judge how far a ground state is from a mean-field, single-Slater-determinant description.
- Describe, at a diagrammatic level, how the DMRG sweep optimizes an MPS tensor by tensor, and place MPS/DMRG within the broader family of tensor-network methods (PEPS, MERA).

## 34 From spins to fermions: the Jordan-Wigner mapping

The question we want to answer first is almost philosophical: are spin models and fermionic models secretly the same thing? On the face of it they look like different kinds of objects. A fermionic Hamiltonian is built from operators $c_i^\dagger, c_i$ that anticommute on different sites, $\{c_i, c_j^\dagger\} = \delta_{ij}$, while a spin-$\frac{1}{2}$ Hamiltonian is built from operators $S_i^x, S_i^y, S_i^z$ that *commute* on different sites and only satisfy a nontrivial algebra on the same site. Tensor networks, as we built them in Session 4, are

fundamentally bosonic objects: an MPS is just a product of matrices, one per site, and nothing in that construction knows about fermionic minus signs. So if we want to use MPS to solve fermionic Hamiltonians, and that is exactly what we want to do today, we first need a dictionary that turns a fermionic Hamiltonian into an equivalent spin Hamiltonian.

## 34.1 Matching the local Hilbert spaces

The first step is easy. A single fermionic site has exactly two states, empty $|0\rangle$ and occupied $|1\rangle = c^\dagger |0\rangle$, because $c^\dagger c^\dagger = 0$ forbids double occupancy. A single spin-$\frac{1}{2}$ also has exactly two states, $|\downarrow\rangle$ and $|\uparrow\rangle$. So let us just identify them site by site,

$$|0\rangle \leftrightarrow |\downarrow\rangle\,, \qquad |1\rangle \leftrightarrow |\uparrow\rangle\,, \tag{66}$$

and correspondingly identify the operators that move between them: $c^\dagger$ takes empty to occupied, exactly like $S^+$ takes down to up, so we would like to say $c^\dagger \leftrightarrow S^+$, $c \leftrightarrow S^-$, and, since $n = c^\dagger c$ is 0 or 1 while $S^z$ is $\mp\frac{1}{2}$,

$$S^z_i = n_i - \tfrac{1}{2}. \tag{67}$$

So far this looks like a clean bijection between the two Hilbert spaces. But notice the sentence we just glossed over: fermionic operators on different sites *anticommute*, while spin operators on different sites *commute*. That mismatch is not a technicality, it means the naive identification $S^+_i \stackrel{?}{=} c^\dagger_i$ cannot possibly be right as it stands. Panel (a) of Figure 21 draws the matching of the two local spaces, and the other two panels the repair we will now construct.

## 34.2 The string operator

Let us see the mismatch explicitly, since it is the whole reason this section exists. Take the naive guess $\widetilde{S}^+_i \equiv c^\dagger_i$, $\widetilde{S}^-_j \equiv c_j$ for $i \neq j$. Then

$$\widetilde{S}^+_i \widetilde{S}^-_j = c^\dagger_i c_j, \qquad \widetilde{S}^-_j \widetilde{S}^+_i = c_j c^\dagger_i = -c^\dagger_i c_j, \tag{68}$$

where the last step is just the fermionic anticommutation relation for $i \neq j$. So the naive guess gives us two operators that *anticommute*, $\widetilde{S}^+_i \widetilde{S}^-_j = -\widetilde{S}^-_j \widetilde{S}^+_i$, whereas actual spin operators on different sites, being independent two-level systems glued together with an ordinary tensor product, absolutely must *commute*, $S^+_i S^-_j = S^-_j S^+_i$. This is precisely the anticommutation-versus-commutation mismatch we flagged above, and it means we need to dress the naive guess with something that fixes the sign.

The fix is the **Jordan-Wigner string**. Order the sites along the chain, and define, for every site $i$, the operator

$$K_i \equiv \prod_{j<i} (1 - 2n_j)\,, \tag{69}$$

the parity of the number of fermions sitting to the left of site $i$: $K_i = +1$ if an even number of sites to the left are occupied, $K_i = -1$ if odd. Since $K_i$ is built entirely out of number operators $n_j = c^\dagger_j c_j$, which are quadratic in fermionic operators, $K_i$ itself commutes with any fermionic operator on a different site (two sign flips cancel), even though it looks like a long, complicated string of anticommuting objects. It is also Hermitian and unitary, $K_i^2 = 1$. With this in hand, the correct dictionary is

$$S^+_i = c^\dagger_i K_i, \qquad S^-_i = c_i K_i, \qquad S^z_i = n_i - \tfrac{1}{2}. \tag{70}$$

One can check, by direct computation using $K_i^2 = 1$ and the fact that $K_i$ commutes with operators on other sites, that Eq. (70) reproduces exactly the correct spin algebra: $[S_i^\alpha, S_j^\beta] = 0$ for $i \neq j$, and the usual $\mathfrak{su}(2)$ algebra on the same site. The string is not optional decoration, it is the one piece of bookkeeping that makes a fermionic Hilbert space and a spin Hilbert space, which are literally the same vector space, carry consistent operator algebras.

### 34.3 Why nearest-neighbor hopping barely notices the string

The reason this mapping is actually useful in practice, rather than just a mathematical curiosity, is what happens for nearest-neighbor terms. Take $i < j$ and work out the general spin bilinear $S_i^+ S_j^-$ in terms of fermions. Writing $K_j = K_i(1-2n_i)\prod_{i<k<j}(1-2n_k)$ and using $K_i^2 = 1$ together with the operator identity $c_i^\dagger(1-2n_i) = c_i^\dagger$ (which holds because $c_i^\dagger n_i = 0$: we cannot create a particle on a site that is already occupied), a short calculation gives

$$S_i^+ S_j^- = c_i^\dagger c_j \prod_{i<k<j} (1-2n_k)\,, \qquad i < j. \tag{71}$$

So in general, hopping an electron from $j$ to $i$ really does require dragging a whole string of $S^z$ operators across every intervening site. But look at what happens for the special case $j = i+1$: the product in Eq. (71) runs over an empty range and is simply 1, so

$$S_i^+ S_{i+1}^- = c_i^\dagger c_{i+1}. \tag{72}$$

The string cancels completely. One can verify directly, using the same identities, that $S_{i+1}^- S_i^+$ gives the very same operator, confirming that these two spin operators commute as they must. So *nearest-neighbor* hopping maps onto a completely local, string-free spin term, while longer-range hopping does not. This is exactly why the Jordan-Wigner trick is such a good match for one-dimensional, nearest-neighbor tensor-network methods: the models we actually build MPS Hamiltonians for never need the string explicitly, it is only lurking in the background, guaranteeing consistency, but never entering the actual Hamiltonian we hand to the solver.

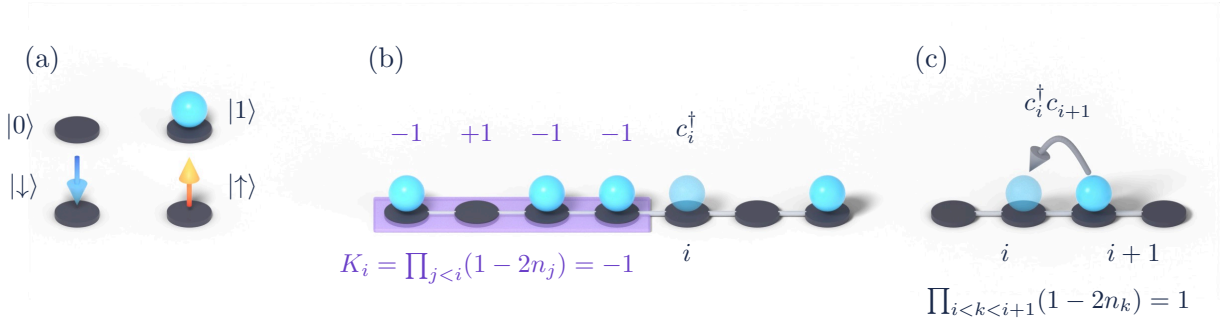



**Figure 21:** The Jordan-Wigner string, and why it usually vanishes from sight. (a) The local Hilbert spaces matched site by site: a spinless fermionic site, behind, is either empty or occupied, and a spin-$\frac{1}{2}$, in front, points either down or up, so $|0\rangle \leftrightarrow |\downarrow\rangle$ and $|1\rangle \leftrightarrow |\uparrow\rangle$ is a clean bijection. What the picture cannot show is the mismatch the rest of the section repairs: operators on different fermionic sites anticommute, while spin operators on different sites commute. (b) The dictionary of Eq. (70) attaches to $c_i^\dagger$, which puts an electron on the ghosted site, the parity factor $K_i = \prod_{j<i}(1-2n_j)$: every site to the left of $i$, shaded, contributes $-1$ if it is occupied and $+1$ if it is empty, so for the three electrons to its left $K_i = -1$. That trailing string is what repairs the mismatch. (c) On a nearest-neighbor bond the product in Eq. (71) runs over the empty range $i < k < i+1$, so the string is 1 and the spin bilinear reduces to the bare hop of the electron from $i+1$ to $i$. This is why the models we build matrix product state Hamiltonians for never meet the string explicitly.

## 34.4 The practical recipe

Putting this together, the scheme we use for the rest of this session is: (1) start from an interacting fermionic Hamiltonian; (2) rewrite it, site by site and bond by bond, as an equivalent spin Hamiltonian using Eq. (70), which for nearest-neighbor models involves no explicit strings at all; (3) solve this *artificial*, unphysical-looking spin Hamiltonian with the MPS machinery from Session 4; (4) map expectation values back to the fermionic language using the same dictionary in reverse. From a practical standpoint, once this is set up, we can simply think of it as a many-body solver for interacting fermions, and never worry about the string again, exactly the way `dmrgpy`'s `fermionchain` class handles it internally for every calculation in the companion notebook, which uses open boundary conditions throughout (Section 34's string cancellation for nearest-neighbor hopping relies on a fixed left-to-right site ordering; on a periodic chain the bond that closes the ring is not nearest-neighbor in that ordering and does pick up an explicit, filling-dependent string, which is one more reason to stick to open chains).

For a Hubbard-type model with spin, each site carries two fermionic flavors, $c_{i\uparrow}^\dagger$ and $c_{i\downarrow}^\dagger$, so the recipe above needs one more choice: an explicit ordering of all $2L$ orbitals along a single one-dimensional string before Eq. (70) can be applied. The natural option is to interleave them site by site, $\uparrow_1, \downarrow_1, \uparrow_2, \downarrow_2, \ldots$, so that same-site operators stay adjacent; the alternative is to order all up-spins first and all down-spins after, $\uparrow_1, \ldots, \uparrow_L, \downarrow_1, \ldots, \downarrow_L$, which keeps each spin sector's own hopping local but pushes the onsite $U$ term into a long string. Either ordering is exact, and `dmrgpy` picks one for us internally, so keep in mind that this choice exists the next time you see a spinful Jordan-Wigner string written down elsewhere.

# 35 Large-scale replay: the spinless interacting chain

Let us put the machinery to work on a model we already know intimately from Session 3: spinless fermions with nearest-neighbor hopping and nearest-neighbor repulsion,

$$\mathcal{H}_{\mathrm{CDW}} = -t \sum_i \left( c_i^\dagger c_{i+1} + \mathrm{h.c.} \right) + V \sum_i n_i n_{i+1}. \tag{73}$$

We already know its two extreme limits by heart: at $V = 0$ it is a free-fermion metal, and as $V \to \infty$ it becomes a charge density wave, one electron on every other site. What exact diagonalization could never tell us cleanly, because $L \lesssim 16$ is simply too small to see a sharp thermodynamic phase transition, is exactly where between these two limits the transition happens, and MPS is going to answer that question for us.

## 35.1 A pleasant surprise: this is the XXZ chain

Before the numerics, let us look at the Jordan-Wigner image of Eq. (73), because it reconnects us to a model from Session 2. Using Eq. (70) and (71) for adjacent sites, the hopping term becomes $-t(S_i^+ S_{i+1}^- + \mathrm{h.c.}) = -2t(S_i^x S_{i+1}^x + S_i^y S_{i+1}^y)$, while the density term is diagonal and trivial to translate, $n_i n_{i+1} = S_i^z S_{i+1}^z + \frac{1}{2}(S_i^z + S_{i+1}^z) + \frac{1}{4}$. Summed over the whole chain, the linear term $\frac{1}{2}\sum_i (S_i^z + S_{i+1}^z)$ is, up to the two uncancelled boundary sites, just $V S_{\mathrm{tot}}^z$, a uniform field that commutes with the bulk hopping and simply shifts the energy at fixed filling (fixed $S_{\mathrm{tot}}^z$); the $O(1)$ residual boundary field on the first and last site does not commute with the bulk hopping, but it is a boundary effect that does not affect the bulk transition, so we drop it here, leaving

$$\mathcal{H}_{\mathrm{CDW}} \longleftrightarrow \sum_i \left[ -2t \left( S_i^x S_{i+1}^x + S_i^y S_{i+1}^y \right) + V S_i^z S_{i+1}^z \right] + \mathrm{const.} \tag{74}$$

This is exactly the **XXZ chain** with anisotropy $\Delta \equiv V/(2t)$. Its exact Bethe-ansatz solution places the transition between a gapless, easy-plane phase and a gapped, Néel-ordered phase precisely at the isotropic Heisenberg point $\Delta = 1$, i.e. at $V = 2t$. Keep that number in mind, we are about to rediscover it numerically. Panel (a) of Figure 22 draws the dictionary on a single bond: the hop of an electron is the flip-flop of an up and a down spin, and two neighboring electrons are two neighboring up spins.

## 35.2 Charge order from a pinning field

Because a translationally invariant Hamiltonian cannot spontaneously break translational symmetry in a finite calculation (the ground state stays a symmetric superposition of the two CDW patterns), the trick we use, both here and back in Session 1's mean-field discussion, is to add an infinitesimal symmetry-breaking seed and see whether the bulk amplifies it or screens it out. Take a chain of $L = 100$ sites, add a tiny pinning term that favors slightly less charge on the first site and slightly more on the last, and look at the resulting charge profile $\langle n_i \rangle$.

For strong interaction, $V = 40t$, the pinning is all it takes: the whole chain locks into the charge-ordered pattern, alternating high and low density from edge to edge. This tells us unambiguously that we are inside the symmetry-broken phase. For moderate interaction, $V = t$, the story is completely different: the charge does oscillate near the pinned edge, but the amplitude of the oscillation decays as you move into the bulk, and by the middle of the chain the density has

relaxed back to the uniform value $\langle n_i \rangle = \frac{1}{2}$. The bulk, in other words, actively wants to screen out the perturbation, which is exactly the signature of a phase that does *not* break translational symmetry, even though $V = t$ is a sizeable interaction by any measure. Panel (b) of Figure 22 shows the two profiles side by side.

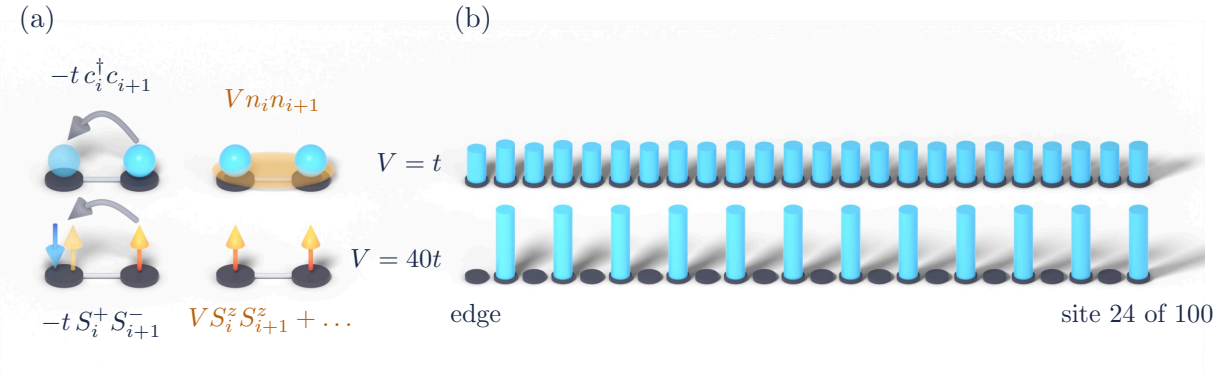



**Figure 22:** The spinless chain as the XXZ chain, and charge order from a pinning field. (a) The dictionary of Eq. (74) on one bond, fermions behind and spins in front: the hop of an electron from $i+1$ to $i$ is the flip-flop $S_i^+ S_{i+1}^-$, the ghosts marking where the electron and the up spin end up, and the repulsion of two neighboring electrons, the energy cost in amber, is the Ising term $V S_i^z S_{i+1}^z$ together with the uniform field that we drop at fixed filling. (b) The density $\langle n_i \rangle$ on the first 24 of the $L = 100$ sites, as columns whose height is the density, computed with the density matrix renormalization group at $\chi = 20$ and the pinning $m(n_1 - n_L)$, $m = 0.1t$, of the notebook, starting the sweeps from the charge pattern the pinning favors. For $V = t$, behind, the pinned edge leaves an oscillation of a few percent that dies out into the bulk, meaning that the bulk screens the seed, while for $V = 40t$, in front, every site is full or empty from the edge on, meaning that the bulk amplifies it.

**✘ Computational companion**

Section "Charge expectation value for weak and strong interactions." Plot $\langle n_i \rangle$ across the $L = 100$ chain for $V = 40t$ and $V = t$ and confirm the two behaviors described above: saturated order versus a decaying edge oscillation. Section "Charge density in the presence of an impurity" lets you repeat essentially the same diagnostic with the perturbation placed inside the chain instead of at the edge, and moving it to the middle is a good way to convince yourself the effect is a bulk property and not an artifact of the boundary.

## 35.3 The phase transition at $V_c$

We can turn the qualitative picture above into a quantitative one by sweeping $V$ continuously and tracking the bulk charge imbalance (or, equivalently, the site density deep in the chain, away from the pinning) as a function of $V$. Below a critical value the bulk density stays flat at $\langle n \rangle = \frac{1}{2}$ no matter how hard we pin the edges; above it, the bulk itself develops a finite charge imbalance, and the pinning field is only needed to select which of the two equivalent sublattices orders. Numerically this happens at $V_c \approx 2t$, in agreement with the exact XXZ result we derived in Eq. (74). This is a thermodynamic-limit quantum phase transition, a nonanalyticity in the ground state as a function of $V$ at zero temperature, and it is one that exact diagonalization at the $L \sim 12$ to 14 of Session 3 simply could not resolve sharply: only at the $L \sim 100$ scale that MPS gives us for free does the transition become crisp.

**✘ Computational companion**

Section "Phase transition in a repulsive fermionic model." Sweep $V$ and locate where the bulk density departs from 0.5; check that this matches $V_c \approx 2t$. Then repeat the whole calculation with a much smaller chain, $L = 10$, and see how much blurrier the transition becomes, this is a direct, hands-on illustration of why we needed to leave exact diagonalization behind.

# 36 Correlators as a diagnostic of the phases

Pinning fields and density profiles are intuitive, but they rely on us adding an explicit symmetry breaking term. A cleaner, field-free diagnostic comes from nonlocal correlation functions, and it generalizes something we already used for the Heisenberg model in Session 2.

## 36.1 The nonlocal particle-particle correlator

Define the single-particle correlator between the first site and site $n$,

$$C(n) \equiv \left\langle c_1^\dagger c_n \right\rangle . \tag{75}$$

Physically this is exactly what a Kubo-formula response calculation would give you if you pinned site 1 infinitesimally and measured the induced amplitude at site $n$, so it packages the same physics as Section 35's pinning-field experiment into a single, field-free object. In a gapless, metallic phase, $C(n)$ decays only as a power law, $C(n) \sim n^{-\alpha}$, because there are excitations at arbitrarily low energy that keep correlations alive at arbitrarily long distance; this is the same phenomenology we met for the gapless quantum magnet in Session 2. In a gapped, insulating phase, by contrast, $C(n)$ decays exponentially,

$$C(n) \sim e^{-n/\xi}, \tag{76}$$

where $\xi$ is the **correlation length**, set by the energy gap: you need a finite amount of energy to create an excitation, and that finite energy cost is what kills correlations beyond a finite range.

For the model of Eq. (73), $V = 0$ and $V = t$ both give power-law decay on a log-log plot, a straight line, while $V = 6t$, deep in the charge-ordered phase, gives a straight line on a *log-linear* plot instead: exponential decay. This is the same $V = 0/1$ metal versus $V = 6$ insulator contrast we already anticipated qualitatively in Section 35, now made completely quantitative.

**✘ Computational companion**

Section "Non-local static correlators for the particle-particle response of a large interacting fermionic model." Plot $\ln|C(n)|$ against $n$ for a few values of $V$ and confirm the metal gives a slowly-varying (power-law) curve while the CDW gives a straight line (exponential decay, whose slope is $-1/\xi$).

## 36.2 Pairing correlators and the Cooper-pair susceptibility

Everything so far involved moving a single electron. Let us now ask a different question: how much does the ground state want to create or destroy a *pair* of electrons at once? Define the local bond-pair annihilation operator $\Delta_i \equiv c_i c_{i+1}$ and its correlator,

$$P(i,j) \equiv \left\langle \Delta_j^\dagger \Delta_i \right\rangle = \left\langle c_{j+1}^\dagger c_j^\dagger \, c_{i+1} c_i \right\rangle . \tag{77}$$

This is a new object, with no analogue in Session 3, where we never asked two-particle questions like this one. It is a four-point (two-particle) correlator: it measures how correlated the fluctuation "destroy a pair near $i$" is with the fluctuation "create a pair near $j$," and by the same Kubo-formula logic as before, it is exactly what governs the linear response of the system to an external pairing field, i.e. its **Cooper-pair susceptibility**.

For the repulsive model of Eq. (73), $P(i,j)$ decays faster than its noninteracting baseline: repulsion actively discourages two electrons from ever being found close together, so it *quenches* pairing fluctuations. If instead we flip the sign of the interaction and make it attractive, $V<0$, the correlator decays more slowly than the noninteracting case, meaning the system is more willing to host a fluctuating pair. This is precisely the microscopic seed of Cooper pairing and superconductivity, the same attractive channel we mentioned but did not pursue back in Session 1: repulsive interactions suppress pair correlations, attractive interactions enhance them.

**✘ Computational companion**

Section "Non-local pairing fluctuations of an attractive fermionic model." Compare $P(i,j)$ for $V=0$, a repulsive $V>0$, and an attractive $V<0$, and confirm that repulsion makes the pairing correlator decay faster than in the free case while attraction makes it decay slower.

# 37 The Hubbard model at scale: spin-charge separation revisited

Let us now restore spin and go back to the Hubbard model,

$$\mathcal{H}_{\text{Hub}} = -t\sum_{i\sigma}\left(c^\dagger_{i\sigma}c_{i+1,\sigma}+\text{h.c.}\right)+U\sum_i n_{i\uparrow}n_{i\downarrow}. \tag{78}$$

In Session 3 we already saw, from exact diagonalization on small chains, that at strong $U$ this model exhibits **spin-charge separation**: even though the individual electron cannot be split, the collective low-energy excitations of the many-body ground state can. Charge excitations are gapped, since it costs a finite energy $\sim U$ to move a charge across the Mott-insulating background, while spin excitations are gapless: at half filling and strong $U$, the Hubbard model maps onto the Heisenberg antiferromagnet from Session 1's superexchange derivation, and the Heisenberg model is a gapless quantum magnet with fractional spin-$\frac{1}{2}$ spinon excitations, as we studied directly in Session 2. What large system sizes give us here is the ability to see this contrast not as a hint in a handful of energy levels, but as two cleanly, exponentially different decay laws in real space.

## 37.1 Charge gap from the particle-particle correlator

Exactly as in Section 36, we track $C(n)=\left\langle c^\dagger_{1\sigma}c_{n\sigma}\right\rangle$. At $U=0$ the chain is a free-fermion metal and $C(n)$ decays as a slow power law. Turning on $U$ makes the decay progressively faster, and once $U$ is large enough to open the Mott gap, $C(n)$ decays exponentially, with a correlation length that shrinks as $U$ grows: exactly the charge-gap signature we are after.

**✘ Computational companion**

Section "Non-local static correlators for the particle-particle response of a large Hubbard model" and, side by side, "Spin and charge correlator in the Hubbard model." Plot $\ln|C(n)|$ for increasing $U$ and confirm the crossover from power-law to exponential decay, this is the charge gap made visible in real space.

### 37.2 Gapless spin excitations from the spin-spin correlator

Now build the analogous correlator for spin instead of charge, $\langle \mathbf{S}_1 \cdot \mathbf{S}_n \rangle$. Since $S_i^z = \frac{1}{2}(n_{i\uparrow} - n_{i\downarrow})$ and $S_i^+ = c_{i\uparrow}^\dagger c_{i\downarrow}$, this is a *biquadratic* object in the fermionic operators, a product of two fermion bilinears rather than a single one, but it is computed the same way once expressed in the underlying MPS. The result is the mirror image of the charge correlator: at $U = 0$ the spin correlator decays quickly, but as $U$ grows the spin correlator becomes progressively *more* long-ranged, the opposite trend from the charge correlator. This is spin-charge separation stated as two decay rates: repulsion gaps the charge sector while it keeps, and even enhances, spin correlations, consistent with mapping onto a gapless Heisenberg spin chain at strong $U$.

**✘ Computational companion**

Section "Spin-Spin non-local correlator of the Hubbard model." Plot $\ln|\langle \mathbf{S}_1 \cdot \mathbf{S}_n \rangle|$ for increasing $U$ and confirm the opposite trend to the charge correlator: the decay gets *slower* as $U$ grows. Try flipping to attractive $U$ as well, and think about why the roles of the spin and charge sectors should swap in that case.

### 37.3 Pairing correlators in the Hubbard model

Finally, define the onsite singlet pair operator $\Delta_i^{\text{onsite}} \equiv c_{i\downarrow} c_{i\uparrow}$ (a different pairing channel than Section 36's nearest-neighbor bond operator $\Delta_i$) and its correlator $\left\langle (\Delta_j^{\text{onsite}})^\dagger \Delta_i^{\text{onsite}} \right\rangle$, the direct Hubbard-model analogue of the pairing susceptibility we built in Section 36. The qualitative conclusion carries over unchanged: repulsive $U$ suppresses the pairing correlator relative to the noninteracting baseline, while attractive $U$ enhances it. Between the two Hubbard results of this section, we have now seen the full statement promised at the top: repulsion gaps the charge sector, leaves (or enhances) the spin sector gapless, and simultaneously quenches pairing, exactly opposite to what an attractive interaction would do.

**✘ Computational companion**

Section "Pairing correlators in the Hubbard model." Repeat the repulsive-versus-attractive comparison from the spinless case, now for the onsite singlet pairing correlator, and check that the same qualitative story holds.

## 38 The correlation entropy: when is mean-field theory valid?

Throughout Sessions 4 and 5 we have quietly assumed that we *need* the full many-body machinery and could never get away with mean-field theory. But is that actually always true? We now have a tool that lets us answer this question directly from the exact ground state, without ever having

to run a separate mean-field calculation and compare energies by hand.

## 38.1 Definition via the correlation matrix

Take the same nonlocal correlator we have been using all session, $C_{ij} \equiv \left\langle c_i^\dagger c_j \right\rangle$, but now think of it not as a single number for fixed $i = 1$, but as an $L \times L$ Hermitian matrix, the **correlation matrix**. Diagonalize it, $C = \sum_k \nu_k \left|k\right\rangle \left\langle k\right|$, with real eigenvalues $\nu_k$.

Suppose first that the ground state is a Slater determinant, $\left|\Psi\right\rangle = \prod_{n\in\mathrm{occ}} c_{a_n}^\dagger \left|\Omega\right\rangle$, exactly the kind of state mean-field theory builds. In the single-particle basis that diagonalizes the mean-field orbitals, $C_{ij}$ is manifestly diagonal, with a 1 for every occupied orbital and a 0 for every empty one, so $C^2 = C$: the correlation matrix of a Slater determinant is a *projector*, and a projector has eigenvalues that are exactly 0 or 1 (since $\nu_k^2 = \nu_k$ forces $\nu_k \in \{0, 1\}$). This gives us a clean litmus test: if every eigenvalue of $C$ is exactly 0 or 1, the state in front of us is, or is equivalent to, a single Slater determinant, and mean-field theory can reach it exactly.

For a correlated ground state, $C^2 \neq C$ in general, and the eigenvalues $\nu_k$ spread out continuously over $[0, 1]$ instead of collapsing onto the two endpoints. We can quantify how far we are from the product-state limit with an entropy built exactly the way we would build a binary Shannon entropy for each "occupation probability" $\nu_k$,

$$S_{\mathrm{corr}} = -\sum_k \Big[\nu_k \ln \nu_k + (1-\nu_k)\ln(1-\nu_k)\Big]. \tag{79}$$

Every term in the sum vanishes when $\nu_k \in \{0, 1\}$ and is strictly positive otherwise, so $S_{\mathrm{corr}} = 0$ if and only if the ground state is exactly a Slater determinant. The bigger $S_{\mathrm{corr}}$ is, the further the true ground state has drifted from anything a single mean-field calculation could ever produce, no matter how it was initialized. Figure 23 draws the eigenvalues $\nu_k$ as the fillings of cups, for a Slater determinant and for a correlated ground state.

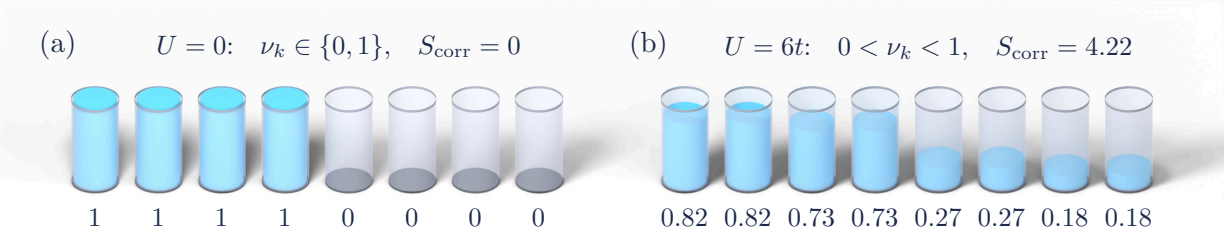



**Figure 23:** The eigenvalues of the correlation matrix, one cup per natural orbital $\left|k\right\rangle$, filled to $\nu_k$. (a) A Slater determinant, the four-site Hubbard chain of the notebook at $U = 0$ and half filling: every cup is full or empty, so every term of Eq. (79) vanishes and $S_{\mathrm{corr}} = 0$. (b) The same chain at $U = 6t$, from exact diagonalization: the eight eigenvalues come in spin pairs, 0.82, 0.73, 0.27 and 0.18, no cup is full or empty, and $S_{\mathrm{corr}} = 4.22$, meaning that no single Slater determinant reproduces this ground state. What $S_{\mathrm{corr}}$ counts is exactly how far the cups are from being full or empty.

## 38.2 Applied to the spinless chain

Let us apply Eq. (79) to the model of Eq. (73). We already know both of its extreme limits are secretly product states: at $V = 0$ the ground state is the filled Fermi sea, a Slater determinant

by construction, and at $V \to \infty$ the ground state is the perfect charge-density-wave pattern, one electron here, none there, alternating, which is also exactly a single Slater determinant (just in real space instead of momentum space). So $S_{\text{corr}} \to 0$ at both ends. In between, and in particular right around the critical point $V \approx 2t$ we found in Section 35, the correlation entropy rises to a clear peak: this is exactly the regime where the ground state is most entangled, most resistant to any single-Slater-determinant description, and it coincides with the quantum phase transition itself, since critical points are generically where correlations, and hence entanglement, are strongest.

**✘ Computational companion**

Section "The correlation entropy of an interacting fermionic model." Plot $S_{\text{corr}}$ against $V$ and confirm it vanishes at both $V = 0$ and large $V$, peaking near $V \approx 2t$. Then shrink the system and see how the peak height and position respond, this connects directly back to the finite-size sensitivity we already saw for the phase transition itself.

## 38.3 Driving a quantum magnet toward its mean-field solution

We can make this diagnostic even more concrete by engineering a controlled crossover between a strongly correlated state and a mean-field one, in a single Hamiltonian. Take the Hubbard model and add exactly the staggered mean-field magnetic term that Session 1's self-consistency loop would have generated on its own, the same staggered field we switched on in Session 3,

$$\mathcal{H} = \mathcal{H}_{\text{Hub}} + J_z \sum_i (-1)^i S_i^z. \tag{80}$$

At $J_z = 0$ this is just the ordinary Hubbard model, which at strong $U$ and half filling behaves like a gapless quantum antiferromagnet, a spin-liquid-like state in the sense that the local magnetization vanishes everywhere by symmetry, even though spin-spin correlations are long-ranged. As we turn on $J_z$, we are explicitly biasing the ground state toward the broken-symmetry, staggered-magnetization state that mean-field theory would give us directly. And indeed, as $J_z$ grows, the correlation entropy of Eq. (79) systematically drops: the ground state is being pushed, by hand, toward exactly the kind of product-like, symmetry-broken state that a self-consistent mean-field calculation would converge to on its own. This closes the circle back to Session 1: in Session 3 we watched the magnetization grow under this same field, and now we can also see, quantitatively, how much entanglement is thrown away in the product-state simplification that mean-field theory always assumes, and so when it is or is not a good approximation.

**✘ Computational companion**

Section "From a quantum magnet to a symmetry broken mean field magnet." Sweep the staggered field $J_z$ and watch both the local staggered magnetization grow and the correlation entropy fall. Then increase $U$ and check how the correlation entropy at $J_z = 0$ responds, and think about why stronger $U$ should push the system further from (or closer to) a mean-field description.

**✘ Computational companion**

The notebook also contains three sections that go beyond these notes and develop the same tools further: "Energy of interacting spinless fermions as a function of the system size" revisits how finite-size scaling of the ground-state energy lets you extrapolate to the thermodynamic limit; "Magnetic correlations in a dimerized Hubbard model" asks how alternating strong/weak bonds reshape the

spin correlator we built in Section 37; and "Correlation entropy of the Anderson model" applies the correlation entropy of Section 38 to a single correlated impurity coupled to a metallic host, tracking how the entropy is distributed in space around the impurity. They are natural extensions to try once you have worked through the sections above.

# 39 The general tensor-network algorithm

We have now used matrix product states extensively, across two full sessions, without ever looking at how the ground-state search is actually carried out. Let us close by doing exactly that.

## 39.1 Diagrammatic notation

It helps a lot to draw tensors rather than write out their indices. A scalar is a dot with no legs, a vector is a dot with one leg, a matrix is a dot with two legs, and a general rank-three tensor is a dot with three legs, one leg per index. Contracting an index between two tensors, i.e. summing over it, is drawn by joining the corresponding legs into a single line; matrix-vector multiplication, for instance, is just a two-legged dot joined to a one-legged dot along one shared leg, leaving one free leg, exactly a vector again, as it should be. An MPS is then nothing more than a one-dimensional chain of such dots: each site carries one *physical* leg, of dimension 2 for a spin-$\frac{1}{2}$ (or, after Jordan-Wigner, for a spinless-fermion site), and two *bond* legs of dimension $\chi$ connecting it to its left and right neighbors,

$$|\Psi\rangle = \sum_{\{s_i\}} M_1^{s_1} M_2^{s_2} \cdots M_L^{s_L} \; |s_1 s_2 \cdots s_L\rangle \,, \tag{81}$$

where each $M_i^{s_i}$ is a $\chi \times \chi$ matrix (a row and a column vector at the two ends) and the matrix product running along the chain is exactly the string of joined bond legs in the diagram. The bond legs carry no physical meaning of their own, they are simply the bookkeeping device that lets a single tensor encode entanglement between the two halves of the chain it separates, capped at $S_A \leq \ln \chi$, exactly the entanglement-entropy bound we derived in Session 4.

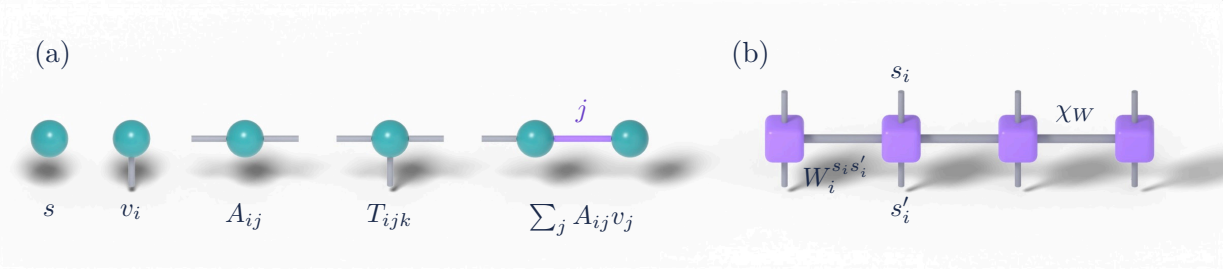



**Figure 24:** Diagrammatic notation for tensors, and the matrix product operator. (a) A dot with one leg per index: a scalar $s$ has no legs, a vector $v_i$ has one, a matrix $A_{ij}$ has two and a rank-three tensor $T_{ijk}$ has three. Summing over an index shared by two tensors is drawn by joining the corresponding legs, here the leg $j$ of a matrix and a vector, so what is left has one free leg and is a vector again, exactly as matrix-vector multiplication should be. (b) A matrix product operator, the operator counterpart of the matrix product state we used throughout Session 4. Each site tensor $W_i^{s_i s_i'}$ carries two physical legs, one up for the row index $s_i$ and one down for the column index $s_i'$, on top of the two internal bond legs, here of dimension $\chi_W$, and the tensors at the ends carry a single bond leg. For a nearest-neighbor Hamiltonian those internal bonds only need to carry a handful of states, which is enough to encode $\mathcal{H}$ exactly rather than approximately.

## 39.2 Matrix product operators

Hamiltonians need the same treatment. A **matrix product operator (MPO)** is the operator version of an MPS: instead of one physical leg per site we now have two, one for the row index and one for the column index of the local operator, plus the same two internal bond legs as before,

$$\mathcal{H} = \sum_{\{s_i, s_i'\}} W_1^{s_1 s_1'} W_2^{s_2 s_2'} \cdots W_L^{s_L s_L'} \, |s_1 \cdots s_L\rangle \langle s_1' \cdots s_L'| \,. \tag{82}$$

For a nearest-neighbor Hamiltonian like $\mathcal{H}_{\text{Heis}} = J \sum_i \mathbf{S}_i \cdot \mathbf{S}_{i+1}$, this internal bond dimension, which we call $\chi_W$, can be made very small, just large enough to carry "nothing has happened yet," "an $S^x$, $S^y$, or $S^z$ was just placed and is waiting for its partner on the next site," and "the interaction has already been closed," so a handful of internal states per bond is enough to represent the exact Hamiltonian, not an approximation to it. Panel (b) of Figure 24 draws this chain of operator tensors.

## 39.3 Applying $\mathcal{H}$ to a state: contraction and compression

The operation we need over and over is $\mathcal{H}|\Psi\rangle$. Diagrammatically this is simply joining the physical legs of the MPO to the physical legs of the MPS, one site at a time, and merging the two bond legs (one from the MPS, one from the MPO) that now run in parallel between neighboring sites into a single, wider bond. The result is again an MPS, but with bond dimension $\chi\chi_W$, the product of the two bond dimensions we started with. This is the practical cost of contraction: every time we act with $\mathcal{H}$, the bond dimension grows, and if we did this repeatedly without intervention, it would grow exponentially and defeat the whole purpose of using an MPS in the first place. The fix is exactly the tool we already used for state compression in Session 4: a singular value decomposition on every bond, keeping only the largest $\chi$ singular values (equivalently, the $\chi$ largest eigenvalues of the reduced density matrix across that bond) and discarding the rest. Because entanglement in a gapped one-dimensional system saturates at a finite value, this truncation introduces only a

small, controllable error, and is the single most important practical operation underlying every tensor-network calculation we have run this session.

## 39.4 Variational ground-state search: the DMRG sweep

Finally, how do we actually find the ground state, rather than just apply $\mathcal{H}$ once? We treat the MPS as a variational ansatz and minimize the energy functional directly. Introducing a Lagrange multiplier $\lambda$ to enforce normalization, we want a stationary point of

$$f[\{M\}] = \langle\Psi[\{M\}]|\,\mathcal{H}\,|\Psi[\{M\}]\rangle - \lambda\,(\langle\Psi[\{M\}]\rangle - 1)\,. \tag{83}$$

The key simplification, and the reason this is tractable at all, is that $|\Psi\rangle$ is *linear* in any single one of its tensors $M_i^{s_i}$ once all the others are held fixed. Setting $\partial f/\partial M_i^{s_i} = 0$ at fixed neighbors, contracting away every tensor we are not currently varying (on both the bra and the ket side, together with the Hamiltonian MPO) collapses the entire, exponentially large problem down to a small effective eigenvalue equation living only on the local physical-times-bond-times-bond space at site $i$,

$$H_{\text{eff}}(i)\,M_i^{s_i} = \lambda\,M_i^{s_i}. \tag{84}$$

We diagonalize this small, dense matrix (its size is set by $\chi$, not by the full $2^L$-dimensional Hilbert space), keep the lowest eigenvector as the new $M_i^{s_i}$, then move to site $i+1$, rebuild its effective Hamiltonian using the freshly-updated tensor at $i$, and repeat. Sweeping all the way to the right end of the chain and back to the left, and iterating this back-and-forth sweep until the energy stops changing, is precisely the **density matrix renormalization group (DMRG)** algorithm: variational MPS optimization and DMRG are, in this sense, one and the same procedure, reached from two different starting points.

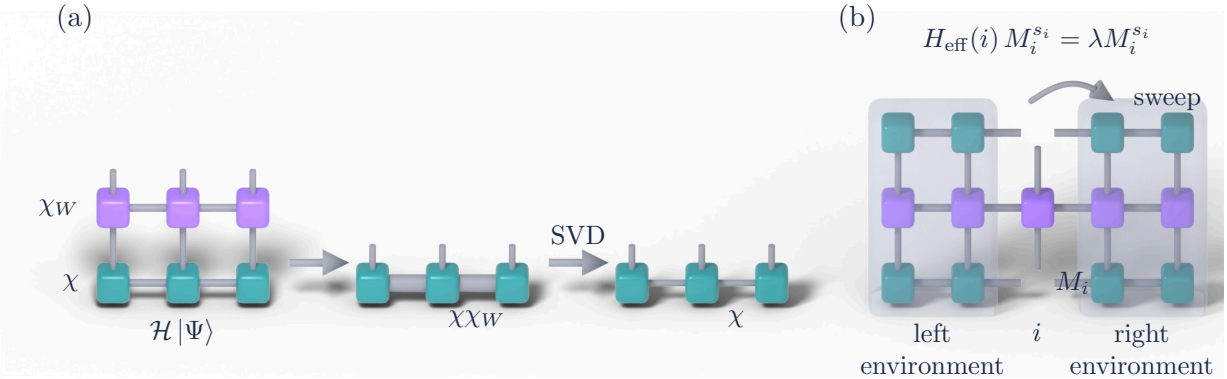



**Figure 25:** The two operations that every calculation in these two sessions is built out of, with the tensors of the state in teal and those of the operator in violet. (a) Acting with $\mathcal{H}$, drawn by joining the matrix product operator's column legs to the physical legs of the state one site at a time, with the state's legs drawn pointing up so that the two chains stack. The two bond legs that now run in parallel, of dimensions $\chi$ and $\chi_W$, merge into a single one of dimension $\chi\chi_W$, drawn thick, so the result is again a matrix product state, just a fatter one, and a singular value decomposition on every bond keeping the largest $\chi$ values compresses it back. (b) The variational ground-state search. The state, the operator and the conjugate state are stacked, everything except site $i$ is contracted into a left and a right environment, the shaded blocks, and the tensors of the state and of its conjugate at $i$ are left out, so what remains is $H_{\text{eff}}(i)$ with the open legs on which $M_i^{s_i}$ sits. This collapses the exponentially large problem to the small eigenvalue problem $H_{\text{eff}}(i)M_i^{s_i} = \lambda M_i^{s_i}$ at that one site. Solving it, stepping one site over and repeating, back and forth along the chain, is the DMRG sweep.

### 39.5 Beyond MPS: PEPS and MERA

Matrix product states are the simplest tensor-network architecture, one tensor per site strung along a line, and that simplicity is exactly why they are so well suited to one dimension: entanglement across any cut of a gapped 1D chain saturates, so a finite bond dimension suffices. The natural two-dimensional generalization, a **projected entangled pair state (PEPS)**, gives every site one bond leg per lattice neighbor instead of just two, letting it capture the larger, area-law entanglement of 2D systems, at the price of losing the cheap, exact contraction we relied on all session, contracting a 2D tensor network exactly is computationally hard in general, and PEPS calculations rely on further approximate contraction schemes. Critical, scale-invariant 1D systems pose a related challenge even within one dimension, since their entanglement grows (logarithmically) with subsystem size rather than saturating, which is exactly the regime the **multiscale entanglement renormalization ansatz (MERA)** was built for, a hierarchical network of alternating disentangling and coarse-graining tensors, layered to mirror the scale invariance of the critical point itself. We will not develop either of these further in this course, but it is worth knowing they exist: MPS is the first, most tractable member of a much larger family of variational tensor architectures, each buying more entanglement capacity at the cost of a harder optimization problem.

## Closing summary

- The Jordan-Wigner mapping identifies the fermionic and spin-$\frac{1}{2}$ local Hilbert spaces site by site, and fixes the resulting anticommutation/commutation mismatch with a string operator $K_i$; nearest-neighbor fermionic hopping maps onto a string-free spin term, which is exactly why 1D nearest-neighbor MPS methods are such a natural fit for fermionic chains.
- Mapping the fermionic Hamiltonian to its Jordan-Wigner spin image and solving it with MPS lets us redo Session 3's spinless CDW transition and Hubbard spin-charge separation at $L \sim 100$, sharp and unambiguous, rather than blurred by the ceiling of exact diagonalization, $L \sim 20$ for spinless fermions and half that for the Hubbard chain; the spinless chain's transition at $V_c \approx 2t$ matches the exact XXZ result.
- Nonlocal particle-particle correlators distinguish gapless (power-law) from gapped (exponential) phases; pairing correlators, a new four-point diagnostic, show that repulsive interactions quench Cooper-pair fluctuations while attractive interactions enhance them, in both the spinless and Hubbard models.
- The correlation matrix built from these correlators has eigenvalues in $[0, 1]$ that are exactly 0 or 1 only for a Slater determinant; the resulting correlation entropy quantifies how far a ground state is from any mean-field description, peaking at the spinless chain's phase transition and dropping smoothly as a staggered field drives the Hubbard model toward its mean-field, symmetry-broken solution.
- The general tensor-network algorithm represents states as MPS and operators as MPOs; applying $\mathcal{H}$ grows the bond dimension, controlled by SVD-based truncation, and the variational ground-state search reduces, tensor by tensor, to a sequence of small effective eigenvalue problems solved by sweeping back and forth along the chain, exactly the DMRG algorithm. PEPS and MERA extend the same idea to two dimensions and to criticality, respectively, beyond the scope of this course.
- Standing back over the whole course: mean-field theory (Session 1) gives us a cheap,

polynomial-cost approximation that can fail qualitatively whenever the true ground state is entangled; exact diagonalization (Sessions 2 and 3) fixes that failure completely but only for small systems, since the Hilbert space grows exponentially; and matrix product states (Sessions 4 and 5) recover the exact many-body answer, entanglement and all, at the large, thermodynamically meaningful system sizes that make the underlying physics, phase transitions, fractionalization, spin-charge separation, unambiguous, at a cost that stays polynomial as long as the entanglement itself stays bounded. That progression, from product states, to unrestricted but tiny Hilbert spaces, to bounded-entanglement states at scale, is the single thread running through this entire course.